Separable Rank of Radial Kernels: Machine-Checked Characterization in Lean 4 and Applications

GitHub Copilot

Source archive (CC0 1.0): https://github.com/OlliNiemitalo/kernel-rank-lean

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Abstract

Filtering an image or three-dimensional wave field directly with a round (radially symmetric) kernel $K(x) = \varphi(\|x\|^2)$ of width W costs W^d multiply–adds per output sample. When a 2-D kernel is a product $h(x_0)v(x_1)$ of a horizontal and a vertical 1-D filter, it can instead be applied as two cheap 1-D passes; the *separable rank* R is the smallest number of such coordinate-product terms needed to sum to K. On a d-dimensional grid, rank R reduces W^d taps to dRW one-dimensional taps per sample (for example, a 9-tap-wide 3-D stencil drops from $9^3 = 729$ taps to $3 \times 4 \times 9 = 108$ taps when $R = 4$). Ordinary bell-shaped Gaussians $e^{-\alpha\|x\|^2}$ have rank 1, but cannot form flat-top camera defocus disks or ringed wave stencils; practical designs therefore sum damped radial waves $e^{-\alpha\|x\|^2}\cos(\omega\|x\|^2 + \theta)$ (*complex-rate Gaussians*), or more generally polynomials in $s = \|x\|^2$ times real or complex exponentials (*exponential polynomials*).

We determine which continuous radial kernels on $\mathbb{R}^d$ ($d \geq 2$) have finite separable rank, and how many product terms each requires. First, $K(x) = \varphi(\|x\|^2)$ has finite rank if and only if $\varphi(s)$ ($s \geq 0$) is a finite sum of exponential polynomials; conversely, decaying complex-rate Gaussians with one shared width and equally spaced frequencies can approximate any continuous radial kernel that fades to zero at large radius. Second, for a pure complex-rate Gaussian ($\deg P = 0$) in d dimensions, expanding real and imaginary parts naively takes $\sum_{2|q} \binom{d}{q} = 2^{d-1}$ real products (4 in 3-D), whereas an alternating angle-sampling identity (Proposition 2.14) achieves the exact minimum of d real products (3 in 3-D); for general polynomial degree $k = \deg P \geq 0$, exact real rank is $(d-1)k+1$ per real exponential (1 at $k = 0$, d at $k = 1$, $2d-1$ at $k = 2$) and $2(d-1)k+d$ per complex-conjugate pair (Proposition 2.15; d at $k = 0$, $3d - 2 = 7$ in 3-D at $k = 1$). Third, allowing sequences of lower-rank kernels that converge uniformly on compact sets (*compact-open border rank*) never lowers rank in 2-D, whereas in $d \geq 3$ dimensions it strictly reduces the cost of polynomial-weighted profiles: for $\|x\|^2\cos(\|x\|^2)$ in 3-D, exact real rank is 7, limits of radial Gaussians take 6 terms, and general real separable limits take 5, at the cost of $O(h^{-1})$–$O(h^{-3})$ cancelling weights in finite precision. All 32 mathematical theorems are verified by the Lean 4 proof assistant, and reproducible Python/Triton case studies examine finite-grid discretization, 16-bit (`float16`) conditioning, and GPU crossovers on an RTX 4070 Ti Super: across 20 HDR bokeh scenes ($R = 12$), orthogonalizing 2-D filter terms cuts max `float16` sRGB error from $7.90 \pm 0.92/255$ ($8.18 \pm 0.92/255$ under magnitude reordering alone) to $0.31 \pm 0.13/255$, and in 3-D acoustics (256^3 voxels), exact 3-term-per-Gaussian stencils cut step time by 24% over naive 4-term expansion (1.51 ± 0.03 ms vs. 1.98 ± 0.02 ms; fourth-order 25-point FDTD takes 0.325 ± 0.0002 ms/step on the same grid, or an estimated $52\times$–$71\times$ more runtime and $40\times$ more memory when grid-refined to match 0.1% dispersion) while keeping directional wave-speed error below 0.2% in spatially varying media where 1-step `cuFFT` does not apply. Code, proofs, and data are public domain (CC0 1.0) at https://github.com/OlliNiemitalo/kernel-rank-lean.

1 Introduction

Convolving a d-dimensional array with a dense $W \times \cdots \times W$ spatial stencil requires W^d multiply–accumulate operations per output sample. When the kernel $K : \mathbb{R}^d \to \mathbb{K}$ admits a rank-R coordinatewise decomposition

$$K(x_0, \ldots, x_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} f_{n,j}(x_j), \tag{1}$$

the d-dimensional convolution reduces to R cascades of d one-dimensional passes, costing dRW operations per sample (or $O(dR)$ operations independent of W when the one-dimensional factors admit recursive causal/anti-causal IIR realizations [19, 77]). Viewing (1) over a commutative semiring $\mathbb{K}$, the *separable rank* $\operatorname{srank}_{\mathbb{K}}(K) \in \mathbb{N} \cup \{\infty\}$ is the least $R \in \mathbb{N}$ for which such a decomposition exists: matrix rank [70] when $d = 2$ and $\mathbb{K}$ is a field, and functional canonical polyadic (CP) or *separation rank* [32, 5, 39, 29] when $d \geq 3$.

For circularly or spherically symmetric (*radial*) kernels in dimension $d \geq 2$,

$$K(x) = \varphi\left(\sum_{j=0}^{d-1} x_j^2\right), \qquad x = (x_0, \ldots, x_{d-1}) \in \mathbb{R}^d,$$

classical characterizations by Maxwell [49] and Kannappan and Sahoo [36] imply that a continuous nonvanishing radial kernel with $\operatorname{srank}_{\mathbb{R}}(K) = 1$ has a Gaussian profile $\varphi(s) = ae^{-\alpha s}$. Real-rate Gaussians ($\alpha > 0$), however, are strictly monotone and convex in the squared radius $s = \|x\|^2$; single Gaussians and nonnegative Gaussian mixtures cannot reproduce flat passbands, sharp cutoffs, or oscillatory diffraction rings [78]. In image processing, real-time graphics, and numerical wave propagation, Niemitalo [54, 57, 58], Garcia [27], and related filter-design studies [44] approximate flat-top and oscillatory radial profiles—such as camera defocus disks, Airy diffraction patterns $(J_1(r)/r)^2$, and ideal circular low-pass filters $\omega_c J_1(\omega_c r)/(2\pi r)$—by sums of *complex-rate* Gaussians $\operatorname{Re}(\beta e^{(-\alpha+i\omega)\|x\|^2})$, whose damped radial oscillations $\cos(\omega\|x\|^2 + \theta)$ synthesize sharp transitions and sidelobe cancellation while maintaining finite separable rank.

1.1 Visual overview: components, rendering terms, and streaming intermediates

Before introducing technical definitions, Figure 1 illustrates in plain visual terms how one-dimensional filters build a circularly symmetric 2-D kernel and why the distinction between *components* and *rendering terms* governs practical GPU execution:

1. **Mathematical "Components" view (Figure 1a):** Replacing a real decay $-\alpha$ by a complex rate $\lambda = -\alpha + i\omega$ preserves the coordinate split $e^{\lambda(x^2+y^2)} = e^{\lambda x^2} e^{\lambda y^2}$ while oscillating radially as $e^{-\alpha r^2}\cos(\omega r^2)$ and $e^{-\alpha r^2}\sin(\omega r^2)$ ($r^2 = x^2 + y^2$). Writing $g(x) = \sqrt{\beta}\, e^{\lambda x^2} = g_{\mathrm{re}}(x) + i g_{\mathrm{im}}(x)$, taking the real part of $g(x)g(y)$ uses two real 1-D×1-D outer products:

$$\operatorname{Re}\big(g(x)g(y)\big) = g_{\mathrm{re}}(x)\, g_{\mathrm{re}}(y) + g_{\mathrm{im}}(x)\, \big(-g_{\mathrm{im}}(y)\big).$$

   Neither outer product is circular on its own—each is a rectangular plaid of positive (blue) and negative (red) stripes—yet adding the pair cancels the grid corners to form a circularly symmetric component (u_1: wide dome; u_2: concentric sharpening rings), whose sum $u_1 + u_2$ approximates a flat-top disc K. (When a component multiplies $e^{\lambda r^2}$ by a polynomial $P(r^2)$ of degree k, expanding $(x^2 + y^2)^q$ naively across monomials would produce $(k+1)(k+2)$ real products, whereas collecting powers of x^2 merges them into $2(k+1)$ real products, or $k+1$ when $\omega = 0$.)

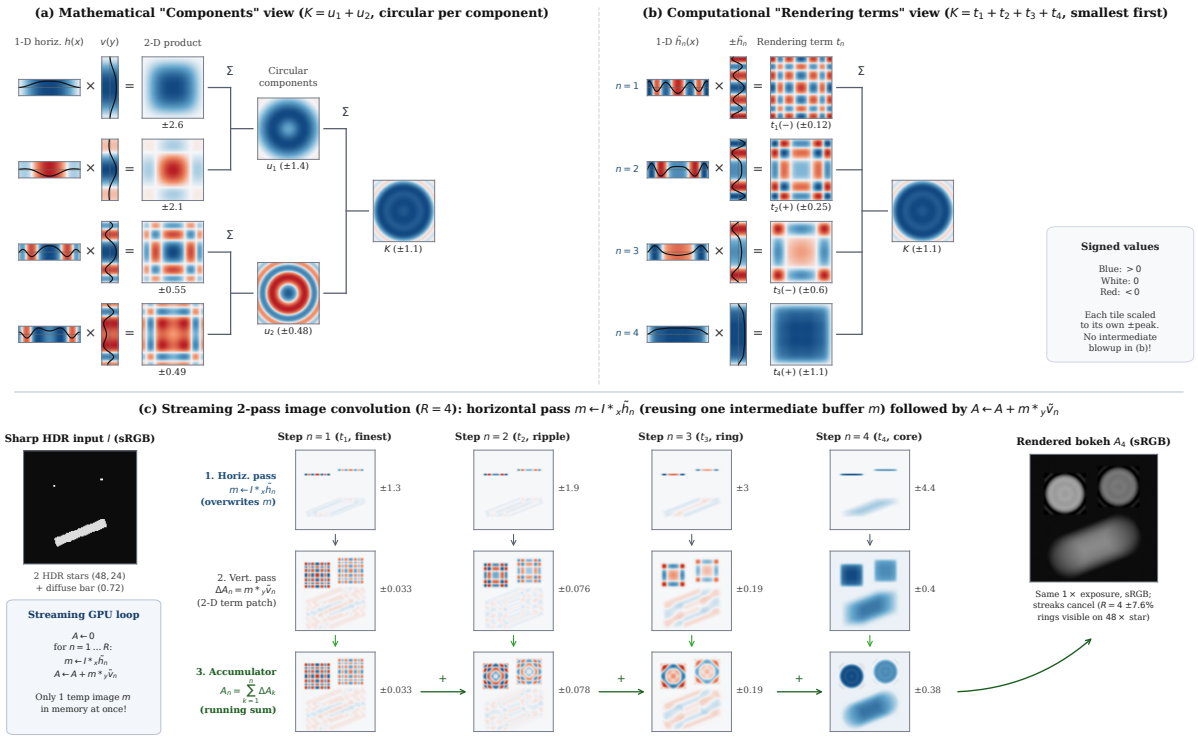


Figure 1: Visual guide to separable radial filtering on a two-complex-Gaussian ($n_{cx} = 2$, exact rank $R = 4$) circular disc-blur kernel [58]: (a) mathematical *Components* view, where pairs of 1-D outer products $h(x)v(y)$ form circularly symmetric concentric-ring components u_1, u_2 that sum to the flat-top disc K; (b) computational *Rendering terms* view, where orthogonalizing rowSpan(K) yields four direct real products $t_n(x, y) = \tilde{h}_n(x) \cdot (\pm\tilde{h}_n(y))$ ordered smallest-first so intermediate amplitudes stay bounded; (c) streaming two-pass linear-light HDR blur using *one* overwritten horizontal intermediate buffer $m \leftarrow I *_x \tilde{h}_n$ followed by vertical accumulation $A \leftarrow A + m *_y \tilde{v}_n$ (with I and A_4 displayed at identical 1× linear exposure through the IEC 61966-2-1 sRGB transfer function; all axis-aligned streaks cancel, while $R = 4$'s ±7.6% pass/stopband ripple remains visible on the 48× HDR star).

2. **Computational "Rendering terms" view (Figure 1b):** A renderer never computes the circular components u_1, u_2 separately—only the total sum $K(x, y) = \sum_{n=1}^{R} \tilde{h}_n(x)\tilde{v}_n(y)$. In high-accuracy approximations (such as $n_{cx} = 6$ complex Gaussians, $R = 12$), individual components u_i grow hundreds of times taller than K and cancel against each other, causing severe precision loss in 16-bit half-float (`float16`) GPU buffers. Orthogonalizing the 1-D filter subspace (`separableChannels` in [58]; Section 5.3) rotates the R filter pairs into orthogonal *rendering terms* $t_n(x, y) = \tilde{h}_n(x) \cdot (\pm\tilde{h}_n(y))$ ordered from smallest to largest magnitude ($n = 1, \ldots, 4$): every term stays comparable to the kernel's own peak (±0.12 to ±1.1 in Figure 1b, vs. ±2.6 in Figure 1a).

3. **Streaming two-pass convolution with one intermediate buffer (Figure 1c):** In the minimal-memory streaming schedule (used, e.g., in fragment-shader pipelines [58]), the GPU initializes an output accumulator $A \leftarrow 0$ and loops over $n = 1, \ldots, R$ (smallest term first). In step n, a 1-D horizontal pass writes $m(x, y) \leftarrow (I *_x \tilde{h}_n)(x, y)$, smearing bright point stars horizontally into 1-D signed streaks; *the same temporary buffer m is overwritten on every step*, keeping DRAM scratch constant at one image regardless of R (in Section 5, `gpu_kernels.py` further tiles the image into L2-cache-resident strips so all R intermediates stay on chip). A 1-D vertical pass then adds $\Delta A_n = m *_y \tilde{v}_n$ into A: steps $n = 1, 2, 3$ build a hollow sharpening ring A_3, and step $n = 4$ adds the soft core so all horizontal and vertical

Table 1: Overview of main theoretical, formalization, and numerical results ($d \geq 2$, degree $k = \deg P$).

Regime	Setting	Main result	Ref.
Finite-rank form	Cont. φ on $\mathbb{R}_{\geq 0}$	$\operatorname{srank}_{\mathbb{R}} < \infty \iff \varphi(s) = \sum_u \operatorname{Re}(P_u(s)e^{\lambda_u s})$ $(s \geq 0)$	Thm. 2.4
Complex rank	$\psi : \mathbb{R} \to \mathbb{C}$, $\mathbb{R}^d$	$(d-1)k+1$ per distinct rate $\mu \in S$	Thm. 2.5
Real rank ($\operatorname{Im}\lambda$=0)	$P \in \mathbb{R}[X]$, $\mathbb{R}^d$	$c_d(\lambda, P) = (d-1)k+1$ $(= 1$ at $k = 0)$	Thm. 2.6
Real rank ($\operatorname{Im}\lambda$>0)	$P \in \mathbb{C}[X]$, $\mathbb{R}^d$	$c_d(\lambda, P) = 2(d-1)k+d$ $(= d$ vs. 2^{d-1} at $k = 0)$	Thm. 2.6, Props. 2.14–2.15
Real–complex gap	Conjugate pair	$c_d(\lambda, P) - 2((d-1)k+1) = d-2$ (all $k \geq 0$)	Rem. 2.19
Border-rank bound	Any list Γ	Multi-Gaussian limit $\leq B_d(\Gamma) = C_d(\Gamma) - (d-2)\sum_u k_u$	Def. 2.3, Thm. 2.9
Exact border rank	$\mathbb{C}$; real λ_u; $\mathbb{R}^3$	$\mathbb{C}$ & real λ_u: sharp; $\mathbb{R}^3$ $\|x\|^2\cos\|x\|^2$ is $5 < B_3$=6	Thm. 2.10
Universal approx.	$\psi \in C_0(\mathbb{R}_{\geq 0})$	Equal-width harmonic Gaussians approx. $\psi(\|x\|^2)$ on $\mathbb{R}^d$	Thm. 2.12
2-D diff. order	$d = 2$, D on $\mathbb{R}_{\geq 0}$	$\operatorname{srank}_{\mathbb{R}} = \deg p_{\min}$ for minimal $p_{\min}(D)\varphi_{\mathbb{C}} = 0$	Thm. 2.7
Stability bounds	$k \geq 0$, limit h	k=0 factor $\leq 2^{d-1}$; border floor $\Theta(\varepsilon_{\text{mach}}^{1/(k+1)})$	Sec. 2.4
Lean 4 proof	`KernelRankLean`	32 theorems (6,935 loc), zero `sorry`, classical axioms	Sec. 4
Case studies	Blur; 3-D wave	Finite-grid, `fp16` ablation, FDTD/`cuFFT` & $c(\mathbf{x})$ studies	Secs. 5–7

streaks cancel in A_4.

1.2 Plain-language summary of main results and paper organization

Table 1 summarizes our results, organized into two formally verified parts—Mathematical Theory (Sections 2–3) and Lean 4 Formalization (Section 4)—followed by an explicitly demarcated part with illustrative numerical and GPU case studies (Sections 5–7, Appendix A):

1. **Part I — Machine-checked mathematical characterization, border rank, universal approximation, and conditioning analysis (Sections 2–3):** Exponential-polynomial profiles in $s = \|x\|^2$ are *universal* for exact finite rank (Theorem 2.4), while equal-width harmonic complex Gaussians uniformly approximate any continuous radial profile vanishing at infinity across $\mathbb{R}^d$ (Theorem 2.12). We establish closed-form exact d-way separable ranks over $\mathbb{C}$ (Theorem 2.5) and over $\mathbb{R}$ (Theorem 2.6), with a plain-language cost guide (Section 2.3), theoretical numerical-stability bounds (Section 2.4), and worked $d = 3$ examples: the alternating angle-sampling ($2d$-th root-of-unity) identity (Proposition 2.14) reduces each degree-0 complex Gaussian from $\sum_{2|q}\binom{d}{q} = 2^{d-1}$ naive real products to d, and its polynomial generalization (Proposition 2.15) reduces each degree-k nonreal component to $2(d-1)k+d$ real products (a constant $d-2$ real-over-complex penalty). After formally defining pointwise and compact-open border rank (Definition 2.3), we prove the multi-Gaussian upper bound $B_d(\Gamma)$ (Theorem 2.9), prove sharpness over $\mathbb{C}$ and for all purely real-rate profiles, and construct an explicit 5-term non-Gaussian limit on $\mathbb{R}^3$ showing that $\|x\|^2\cos(\|x\|^2)$ has exact real border rank $5 < B_3 = 6 < C_3 = 7$ (Theorem 2.10 and Remark 2.11).
2. **Part II — Formalization architecture and axiom audit (Section 4):** All 32 mathematical theorems in Sections 2–3 are machine-checked in Lean 4 using Mathlib's Banach-algebra Bochner integral for the continuous shift semigroup step, Schur-complement and adjugate-trace obstructions for border-rank lower bounds, and periodic Fourier density on `AddCircle` for universal approximation, with every theorem mapped to its Lean identifier and audited for classical axiom dependencies (Tables 6–7).
3. **Part III — Illustrative numerical and GPU case studies (Sections 5–7, Appendix A):** Reproducible Python/Triton experiments (each numerically checked against `float64` references) illustrate finite-grid discretization, 16-bit floating-point stabilization ablations, memory footprints, and hardware crossovers against strong non-separable baselines (Table 8): planar steerable pass-count bounds (Section 5.2), 2-D bokeh `float16` conditioning (comparing $\mathrm{GL}_R(\mathbb{R})$ orthogonalization against magnitude reordering and `float32`

Table 2: Comparison of prior characterizations, tensor decompositions, and formalizations with the present work.

Line of work	Object & domain	Dim. & field	Rank / border-rank result	Formal proof
Maxwell [49]; Kannappan–Sahoo [36]; Schoenberg [65]	Radial $K(\|x\|^2)$ on $\mathbb{R}^d$ (rank 1 or pos.-def.)	$d \geq 2$, $\mathbb{R}$	Rank 1 $\iff$ Gaussian; completely monotone $\iff$ positive Gaussian integral	None
Anselone–Korevaar [3]; Laird [43]; Andersson–Carlsson [2]	2-variable $f(x+y)$ and truncated Hankel operators	$d = 2$, $\mathbb{R}/\mathbb{C}$	Finite 2-way rank $\iff$ exponential polynomial ($n = \dim \mathcal{W}(f)$); 2-D rank closed	None
Beylkin–Mohlenkamp [5]; Oseledets [59]; Rahimi–Recht [61]	Gaussian-sum CP, TT/Tucker, and pairwise RFF $k(x-y)$	$d \geq 2$, $\mathbb{R}/\mathbb{C}$	Constructive approximations; radial TT/Tucker rank $= n$ (n^d or $n \times n$ cores); no CP converse	None
Winograd [75]; Alder–Strassen [1]; Bläser et al. [10]	Multiplication tensor of finite-dim. algebra $\mathcal{A}$	$d = 3$ or $d \geq 3$, field $\mathbb{K}$	Exact rank $(d-1)\dim \mathcal{A} - (d-2)m_{\mathrm{irr}}$ on $\mathbb{K}[X]/(p)$ ($\|\mathbb{K}\|$ large); discrete border bounds [79, 16]	None
Bentkamp et al. [4]; Jain [35]; Yang et al. [76]	Discrete circuit CP rank; 8^3 octonion; 3×3 $\mathbb{F}_2$ matmul	Finite tensors, $\mathbb{R}/\mathbb{F}_2$	Matricization bounds; fixed-size lower bounds (≥ 18, ≥ 21)	Isabelle/HOL; Lean 4
This work	Continuous radial $K(x) = \varphi(\|x\|^2)$ on $\mathbb{R}^d$	All $d \geq 2$, $\mathbb{R}$ and $\mathbb{C}$	Finite-rank iff + exact $\mathbb{C}/\mathbb{R}$ CP rank + angle-sampling syntheses + border gap ($5 < 6 < 7$ on $\mathbb{R}^3$) + harmonic Gaussian density	**Lean 4** (32 thms, 0 `sorry`)

outer accumulation), discrete-to-continuous convergence, and warm-starting (Section 5.3), and homogeneous and spatially varying $c(\mathbf{x})$ 3-D acoustic wave propagation compared against four explicit FDTD schemes and `cuFFT` (Section 5.4). Sections 6–7 examine practical limitations, open questions, and conclusions, with explicit coefficients in Appendix A.

1.3 Related work

Table 2 contrasts our contributions against five adjacent literatures:

Classical radial characterizations and 2-variable Hankel equations. Maxwell [49] and Kannappan and Sahoo [36] proved that continuous rotation-invariant rank-1 functions are Gaussians, while Schoenberg [65] characterized completely monotone radial profiles as positive Gaussian mixtures. Under $s_j = x_j^2$, radial separability on $\mathbb{R}^d$ reduces to the d-variable additive Hankel equation $\varphi(s_0 + \cdots + s_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} g_{n,j}(s_j)$ on $[0,\infty)^d$, studied for $d = 2$ by Levi-Civita [46], Stone [66], Anselone and Korevaar [3], Laird [43], and Laczkovich [42], and via L^2 Hankel/convolution operators by Rochberg [64], Andersson and Carlsson [2], and Mourrain [51]. Unlike L^2 formulations requiring integrability or bounded domains, our pointwise formulation on $[0,\infty)^d$ characterizes all continuous profiles on $[0,\infty)$—including growing profiles such as $\|x\|^2 \cos(\|x\|^2)$—without assuming differentiability or decay.

Separation rank, TT/Tucker formats, random features, and exponential sums. Beylkin and Mohlenkamp [5], Kolda and Bader [39], Oseledets [59], Hackbusch [29], and Khoromskij [37] develop CP, Tucker, and tensor-train (TT) approximations of radial operators via exponential sums [12, 28, 13, 6, 47] and cross approximations [71, 74, 69], while Rahimi and Recht [61] approximate shift-invariant positive-definite kernels via random Fourier features (RFF). Three distinctions separate $\mathrm{srank}_{\mathbb{K}}$ (Definition 2.1) from these settings: (i) RFF factors $K_0(x-y)$ across *sample pairs* $(x,y) \in \mathbb{R}^d \times \mathbb{R}^d$ (where even $e^{-\alpha\|x-y\|^2}$ has infinite exact two-argument rank via $e^{2\alpha x \cdot y}$, and $e^{i\omega \cdot z}$ is non-decaying and linear in z_j rather than radial in $\|z\|^2$), whereas across Cartesian coordinates $(z_0, \ldots, z_{d-1}) \in \mathbb{R}^d$ a single Gaussian has exact d-way rank 1 and

damped complex Gaussians localize spatially (Theorem 2.12); (ii) every prefix/mode unfolding of $f(s_0+\cdots+s_{d-1})$ depends only on partial sums, so TT [59] and Tucker [39, 29] unfolding ranks equal $n = \dim_{\mathbb{K}} \mathcal{W}(f)$ independent of d at the cost of an n^d core or $n \times n$ inter-axis transfer contractions, whereas uncoupled CP cascades (1) pay only $\operatorname{srank}_{\mathbb{K}} = (d-1)n - (d-2)m_{\mathrm{irr}}$; and (iii) we establish the exact converse ($\operatorname{srank}_{\mathbb{R}} < \infty \iff$ exponential polynomial) across all signed and polynomial-oscillatory profiles.

Algebra complexity, Waring/border rank, applied filters, and formalization. Identifying $\mathcal{W}(f)$ with $\mathcal{A} = \mathbb{K}[X]/(p_{\min,\mathbb{K}})$ (m_{irr} primary factors) reduces our multilinear core to $(d-1)$-fold multiplication in $\mathcal{A}$, whose exact complexity $(d-1)\dim_{\mathbb{K}} \mathcal{A} - (d-2)m_{\mathrm{irr}}$ was established at $d = 3$ by Winograd [75], Fiduccia–Zalcstein [23], and Alder–Strassen [1, 9, 15] and for $d \geq 4$ by Bläser, Mayer, and Shringi [10]. Theorems 2.5–2.10 lift this invariant to continuous radial kernels on $\mathbb{R}^d$, replace real Vandermonde inversion on nonreal factors by closed-form angle-sampling syntheses (Propositions 2.14–2.15), and resolve real border-rank non-sharpness on $\mathbb{R}^3$ ($5 < B_3 = 6 < C_3 = 7$), complementing Hankel/Waring rank [17, 34, 41, 53], real–complex rank gaps [18, 20, 63, 11], and border-rank degenerations [8, 45, 14, 79, 16]. Our case studies build on steerable and complex-Gaussian filters [70, 26, 38, 60, 7, 62, 72, 44, 54, 27, 55, 56, 57, 58] and wideband acoustic propagators [68, 67, 40, 73, 30, 31, 25], while our Lean 4 formalization [52, 22, 50, 48, 21] extends machine-checked tensor-rank work from discrete circuits and fixed-size tensors [4, 35, 76] to infinite-dimensional function spaces on $\mathbb{R}^d$ for all $d \geq 2$ and $k \geq 0$.

2 Problem Formulation, Harmonized Notation, and Main Theorems

2.1 Definitions: separable rank, canonical representations, and border rank

Throughout the paper, we adhere to one uniform profile convention: $\varphi : \mathbb{R} \to \mathbb{R}$ denotes a *real-valued* radial profile, $\varphi_{\mathbb{C}}(s) = (\varphi(s) : \mathbb{C})$ denotes its complexification, $\psi : \mathbb{R} \to \mathbb{C}$ denotes a general *complex-valued* radial profile, and $f : \mathbb{R} \to \mathbb{K}$ denotes a profile taking values in an arbitrary commutative semiring or field $\mathbb{K}$. Similarly, for any kernel $K : X \to \mathbb{R}$, we write $K_{\mathbb{C}}(x) = (K(x) : \mathbb{C})$ for its scalar extension to $\mathbb{C}$.

Definition 2.1 (Coordinatewise separable rank; Lean: `IsSumOfProducts`, `separableRank`)**.** Let ι be a finite index set, $(X_i)_{i\in\iota}$ a family of sets, and $\mathbb{K}$ a commutative semiring. A kernel $K : (\prod_{i\in\iota} X_i) \to \mathbb{K}$ is a *sum of R products* ($R \in \mathbb{N}$) if there exist univariate factors $f_{n,i} : X_i \to \mathbb{K}$ ($0 \leq n < R$, $i \in \iota$) such that $K(x) = \sum_{n=0}^{R-1} \prod_{i\in\iota} f_{n,i}(x_i)$ for all $x \in \prod_{i\in\iota} X_i$ (taking the empty sum 0 when $R = 0$). Its *separable rank* over $\mathbb{K}$ is the extended natural number

$$\operatorname{srank}_{\mathbb{K}}(K) := \inf\{R \in \mathbb{N} : K \text{ is a sum of } R \text{ products over } \mathbb{K}\} \in \mathbb{N} \cup \{\infty\},$$

where $\inf \emptyset = \infty$, and an equation $\operatorname{srank}_{\mathbb{K}}(K) = R$ with $R \in \mathbb{N}$ denotes equality in $\mathbb{N} \cup \{\infty\}$ under the canonical inclusion $\mathbb{N} \hookrightarrow \mathbb{N} \cup \{\infty\}$.

Separable rank is subadditive under finite sums and cannot increase when we substitute coordinate maps $y_i \mapsto g_i(y_i)$ into an existing decomposition $\sum_n \prod_i f_{n,i}(g_i(y_i))$. When every $g_i : Y_i \to X_i$ ($i \in \iota$) is *onto* (surjective), the reverse inequality holds as well: picking for each $x_i \in X_i$ at least one preimage $h_i(x_i) \in Y_i$ with $g_i(h_i(x_i)) = x_i$ turns any decomposition $\sum_n \prod_i \tilde{f}_{n,i}(y_i)$ of $K(g_0(y_0), \dots)$ back into $\sum_n \prod_i \tilde{f}_{n,i}(h_i(x_i)) = K(x)$ with the same number of terms:

$$\operatorname{srank}_{\mathbb{K}}\big(y \mapsto K(i \mapsto g_i(y_i))\big) = \operatorname{srank}_{\mathbb{K}}(K). \tag{2}$$

For $x_j \in \mathbb{R}$, the squaring map $x_j \mapsto x_j^2$ covers $\mathbb{R}_{\geq 0} = [0, \infty)$ (with right inverse $s_j \mapsto \sqrt{s_j}$). Applying (2) therefore equates the d-way separable rank of any radial kernel on $\mathbb{R}^d$ with that of the d-way additive Hankel kernel on $\mathbb{R}^d_{\geq 0}$:

$$\operatorname{srank}_{\mathbb{K}}\left(x \in \mathbb{R}^d \mapsto f\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = \operatorname{srank}_{\mathbb{K}}\left(s \in \mathbb{R}^d_{\geq 0} \mapsto f\left(\sum_{j=0}^{d-1} s_j\right)\right). \tag{3}$$

Notice two immediate consequences of (3): (i) because $\|x\|^2 \geq 0$ never visits negative arguments, $K(x) = \varphi(\|x\|^2)$ depends only on $\varphi|_{[0,\infty)}$ and leaves $\varphi(s)$ for $s < 0$ completely unconstrained; and (ii) replacing $x_j \mapsto x_j^2$ by $x_j \mapsto x_j^2/a_j^2$ ($a_j \neq 0$) or $x_j \mapsto |x_j|^p$ ($p > 0$) is still onto $[0, \infty)$, so every theorem below holds verbatim for axis-aligned ellipsoidal kernels $f(\sum_j x_j^2/a_j^2)$ and ℓ_p superellipsoidal kernels $f(\sum_j |x_j|^p)$. Finally, for $d \geq 2$, setting $x_2 = \cdots = x_{d-1} = 0$ turns each d-way product $\prod_{j=0}^{d-1} f_{n,j}(x_j)$ into the 2-way product $(c_n f_{n,0}(x_0)) f_{n,1}(x_1)$ with constant $c_n = \prod_{j=2}^{d-1} f_{n,j}(0)$, giving the 2-D slice lower bound:

$$\operatorname{srank}_{\mathbb{K}}\big(y \in \mathbb{R}^2 \mapsto f(y_0^2 + y_1^2)\big) \leq \operatorname{srank}_{\mathbb{K}}\left(x \in \mathbb{R}^d \mapsto f\left(\sum_{j=0}^{d-1} x_j^2\right)\right). \tag{4}$$

Definition 2.2 (Canonical real representations and cost functions)**.** A *component* (`Component`) is a pair $(\lambda, P) \in \mathbb{C} \times \mathbb{C}[X]$, evaluating at $s \in \mathbb{R}$ as $\operatorname{Re}(P(s)e^{\lambda s})$ of degree $k = \deg P$ (multiplicity $k+1$). It is *well-formed* if $P \neq 0$, $\operatorname{Im}\lambda \geq 0$, and $\operatorname{Im}\lambda = 0 \implies P \in \mathbb{R}[X]$. A finite list $\Gamma = [(\lambda_1, P_1), \ldots, (\lambda_m, P_m)]$ is a *representation* (`IsRepresentation`, written `cs : List Component` in Lean and denoted Γ here to stay distinct from the rank R and the field $\mathbb{R}$) if every (λ_u, P_u) is well-formed and the rates $\lambda_1, \ldots, \lambda_m$ are pairwise distinct; its radial profile is $\operatorname{profile}(\Gamma, s) = \sum_{u=1}^m \operatorname{Re}(P_u(s)e^{\lambda_u s})$.

To motivate the two integer cost functions assigned to each component (λ, P) in dimension $d \geq 2$: $c_d(\lambda, P)$ counts the *exact* minimum number of real d-way product terms $\prod_{j=0}^{d-1} f_{n,j}(x_j)$ needed to represent $x \mapsto \operatorname{Re}(P(\|x\|^2)e^{\lambda\|x\|^2})$ on $\mathbb{R}^d$ (1 product for a pure real Gaussian at $\operatorname{Im}\lambda = 0, \deg P = 0$ and d real products for a pure complex Gaussian at $\operatorname{Im}\lambda > 0, \deg P = 0$, plus $(d-1)\deg P$ or $2(d-1)\deg P$ for polynomial powers of $\|x\|^2$), whereas $b_d(\lambda, P)$ counts the number of real product terms when $P(\|x\|^2)e^{\lambda\|x\|^2}$ is instead approximated on bounded sets by finite differences of $\deg P + 1$ pure Gaussians (1 real product per real Gaussian, d per complex Gaussian):

$$c_d(\lambda, P) = \begin{cases} (d-1)\deg P + 1, & \operatorname{Im}\lambda = 0, \\ 2(d-1)\deg P + d, & \operatorname{Im}\lambda > 0, \end{cases} \qquad b_d(\lambda, P) = \begin{cases} \deg P + 1, & \operatorname{Im}\lambda = 0, \\ d(\deg P + 1), & \operatorname{Im}\lambda > 0, \end{cases} \tag{5}$$

with total costs $C_d(\Gamma) = \sum_{u=1}^m c_d(\lambda_u, P_u)$ (`multiTotalCost`) and $B_d(\Gamma) = \sum_{u=1}^m b_d(\lambda_u, P_u)$ (`borderTotalCost`). When every $\deg P_u = 0$ (*multi-Gaussian* representation with n_{re} real and n_{cx} nonreal rates), $C_d(\Gamma) = B_d(\Gamma) = n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$.

Formally, a representation Γ specifies $m = n_{\mathrm{re}} + n_{\mathrm{cx}}$ radially symmetric *components* $U_u(x) = \operatorname{Re}(P_u(\|x\|^2)e^{\lambda_u\|x\|^2})$ (Figure 1a), whereas real Lagrange interpolation ($\operatorname{Im}\lambda_u = 0$) and Propositions 2.14–2.15 ($\operatorname{Im}\lambda_u > 0$) expand each U_u into $c_d(\lambda_u, P_u)$ rank-1 coordinate-product (*rendering*) terms $\prod_{j=0}^{d-1} f_{n,j}(x_j)$ totaling $R = C_d(\Gamma)$ terms; in $d = 2$, writing $K = HV^\top$ relates all minimal R-term rendering bases via $(HT, VT^{-\top})$ for $T \in \mathrm{GL}_R(\mathbb{R})$ (Figure 1b).

Definition 2.3 (Compact-open topology and border rank)**.** Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ and $d \geq 1$. The *compact-open topology* τ_{co} on $C(\mathbb{R}^d, \mathbb{K})$ is the locally convex topology generated by the uniform seminorms $\|F\|_{E,\infty} := \sup_{x \in E} |F(x)|$ over all compact subsets $E \subseteq \mathbb{R}^d$ (metrizable via $\mathbb{R}^d = \bigcup_{M=1}^\infty [-M, M]^d$): $F_\iota \to K$ in τ_{co} iff $\|F_\iota - K\|_{E,\infty} \to 0$ for every compact $E \subseteq \mathbb{R}^d$. Writing

Table 3: Harmonized symbol glossary and cost specializations ($d \geq 2$, $k = \deg P$).

Mathematical symbol	Lean 4 identifier	Meaning / specialization
$\operatorname{srank}_{\mathbb{K}}(K) \in \mathbb{N} \cup \{\infty\}$	`separableRank K (ENat)`	Coordinatewise separable rank over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$
$\underline{\operatorname{srank}}^{\mathrm{co}}_{\mathbb{K}}(K)$, $\underline{\operatorname{srank}}^{\mathrm{pt}}_{\mathbb{K}}(K)$	`TendstoUniformlyOn` / `Tendsto`	Compact-open and pointwise border rank (Def. 2.3)
φ, $\varphi_{\mathbb{C}}$, ψ, f	φ : $\mathbb{R} \to \mathbb{R}$, ψ : $\mathbb{R} \to \mathbb{C}$	Real profile, complexification, complex profile, $\mathbb{K}$-profile
$W = 2L + 1$; $s = \|x\|^2$	—; $\sum_j$ `x j ^ 2`	1-D discrete tap width on $[-L, L]$; squared radius on $\mathbb{R}^d$
$\mathcal{W}(f)$	`translateSpan f`	Half-line translate span $\operatorname{span}_{\mathbb{K}}\{u \mapsto f(u+t) : t \geq 0\}$
$\Gamma = [(\lambda_1, P_1), \ldots, (\lambda_m, P_m)]$	`cs : List Component`	Canonical real representation (λ_u distinct, $\operatorname{Im} \lambda_u \geq 0$)
n_{re}, n_{cx}; m_{irr}	—	Real and nonreal component counts; $m_{\mathrm{irr}} = n_{\mathrm{re}} + n_{\mathrm{cx}}$
$C_d(\Gamma) = \sum_u c_d(\lambda_u, P_u)$	`multiTotalCost d cs`	Exact d-D real radial rank (Theorem 2.6)
$C_2(\Gamma)$; $n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$	`totalCost`; `gaussianTotalCost`	Specializations at $d = 2$ and at $\deg P_u = 0$
$B_d(\Gamma) = \sum_u b_d(\lambda_u, P_u)$	`borderTotalCost d cs`	Multi-Gaussian border bound $C_d(\Gamma) - (d-2)\sum_u \deg P_u$
$H_P(s_0; X)$; $E_k(s_j; X)$	`revHassePoly`; `truncExpPoly`	Reversed Hasse poly.; truncated exp $\sum_{q=0}^{k} s_j^q X^q / q!$
$M^{\mathbb{C}}_{d,k}$, $M^{\mathbb{R}}_{d,k}$	$(d-1)k+1$, $2(d-1)k+d$	Complex and nonreal root-of-unity node counts

$\mathcal{F} \neq \bot$ for a nontrivial filter, the *compact-open* and *pointwise border ranks* of $K : \mathbb{R}^d \to \mathbb{K}$ in $\mathbb{N} \cup \{\infty\}$ are

$$\underline{\operatorname{srank}}^{\mathrm{co}}_{\mathbb{K}}(K) := \inf\big\{r \in \mathbb{N} : \exists (F_n)_{n\in\mathbb{N}},\ \operatorname{srank}_{\mathbb{K}}(F_n) \leq r,\ \|F_n - K\|_{E,\infty} \to 0\ (\forall \text{ compact } E)\big\},$$
$$\underline{\operatorname{srank}}^{\mathrm{pt}}_{\mathbb{K}}(K) := \inf\big\{r \in \mathbb{N} : \exists \mathcal{F} \neq \bot,\ (F_\iota),\ \operatorname{srank}_{\mathbb{K}}(F_\iota) \leq r \text{ along } \mathcal{F},\ F_\iota(x) \xrightarrow{\mathcal{F}} K(x)\ (\forall x)\big\},$$

satisfying $\underline{\operatorname{srank}}^{\mathrm{pt}}_{\mathbb{K}}(K) \leq \underline{\operatorname{srank}}^{\mathrm{co}}_{\mathbb{K}}(K) \leq \operatorname{srank}_{\mathbb{K}}(K)$. In Lean 4 / Mathlib (`Approximation.lean`), upper bounds prove `TendstoUniformlyOn (fun h x => F h x) K (nhdsWithin 0 {0}ᶜ) E` for every `IsCompact E` along the punctured real filter $h \to 0$ ($h \neq 0$, with $\operatorname{srank}_{\mathbb{K}}(F_h) \leq r$ for all h, using $0^{-1} = 0$ at $h = 0$), whereas lower bounds hold for arbitrary nontrivial filters `(l : Filter ι) [l.NeBot]` under pointwise convergence ∀ `x, Tendsto (fun a => F a x) l (nhds (K x))` by passing finite evaluation determinants and traces to the limit on `[T2Space` $\mathbb{K}$`]` without requiring individual factors $f^{(a)}_{n,j}$ to converge.

Why are the conditions $\operatorname{Im} \lambda \geq 0$ and ($\operatorname{Im} \lambda = 0 \implies P \in \mathbb{R}[X]$) canonical? Because $\operatorname{Re}(\overline{P}(s)e^{\bar{\lambda}s}) = \operatorname{Re}(P(s)e^{\lambda s})$ and $\operatorname{Re}(iae^{\mu s}) = 0$ for $a, \mu \in \mathbb{R}$, unrestricted pairs over-parameterize real profiles. Under Definition 2.2, each real rate ($\operatorname{Im} \lambda = 0$) contributes a single term $P(s)e^{\lambda s}$ over $\mathbb{C}$, while each upper-half-plane rate ($\operatorname{Im} \lambda > 0$) expands via $\operatorname{Re} z = \frac{1}{2}(z + \bar{z})$ to the conjugate pair $\frac{1}{2}P(s)e^{\lambda s} + \frac{1}{2}\overline{P}(s)e^{\bar{\lambda}s}$ at distinct rates $\lambda \neq \bar{\lambda}$. Since exponential polynomials at distinct complex rates are $\mathbb{C}$-linearly independent on $(0, \infty)$, every continuous real profile of finite rank determines its unordered set of well-formed components uniquely. Table 3 summarizes all symbols and their Lean 4 counterparts.

2.2 Statement of main theorems and algebraic identities

Theorem 2.4 (d-D Finite-Rank Classification; Lean: `separableRank_radial_lt_top_iff_exists`)**.** *Let $d \geq 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$. Then $\operatorname{srank}_{\mathbb{R}}\big(x \in \mathbb{R}^d \mapsto \varphi(\sum_{j=0}^{d-1} x_j^2)\big) < \infty$ if and only if there exists a representation $\Gamma = [(\lambda_1, P_1), \ldots, (\lambda_m, P_m)]$ such that $\varphi(s) = \sum_{u=1}^{m} \operatorname{Re}(P_u(s)e^{\lambda_u s})$ for all $s \geq 0$.*

Theorem 2.5 (d-D Complex Radial Rank; Lean: `separableRank_radial_complex_eq_iff`)**.** *Let $d \geq 2$ and let $\psi : \mathbb{R} \to \mathbb{C}$ be continuous on $[0, \infty)$. For every $R \in \mathbb{N}$,*

$$\operatorname{srank}_{\mathbb{C}}\left(x \in \mathbb{R}^d \mapsto \psi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = R$$

if and only if there exist a finite set $S \subset \mathbb{C}$ and nonzero polynomials $(P_\mu)_{\mu\in S} \subset \mathbb{C}[X]$ such that $\psi(s) = \sum_{\mu\in S} P_\mu(s)e^{\mu s}$ for all $s \geq 0$ and $\sum_{\mu\in S}\big((d-1)\deg P_\mu + 1\big) = R$.

Theorem 2.6 (Exact d-D Real Radial Rank; Lean: `separableRank_radial_eq_iff`)**.** *Let $d \geq 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$. For every $R \in \mathbb{N}$,*

$$\operatorname{srank}_{\mathbb{R}} \left(x \in \mathbb{R}^d \mapsto \varphi \left(\sum_{j=0}^{d-1} x_j^2 \right) \right) = R$$

if and only if there exists a canonical representation $\Gamma = [(\lambda_1, P_1), \ldots, (\lambda_m, P_m)]$ (distinct rates with $\operatorname{Im} \lambda_u \geq 0$, nonzero P_u, and $P_u \in \mathbb{R}[X]$ when $\operatorname{Im} \lambda_u = 0$) such that $\varphi(s) = \sum_{u=1}^{m} \operatorname{Re}(P_u(s) e^{\lambda_u s})$ for all $s \geq 0$ and

$$C_d(\Gamma) = \sum_{\operatorname{Im} \lambda_u = 0} \big((d-1) \deg P_u + 1 \big) + \sum_{\operatorname{Im} \lambda_u > 0} \big(2(d-1) \deg P_u + d \big) = R.$$

In particular, for a multi-Gaussian representation ($\deg P_u = 0$) with n_{re} real and n_{cx} nonreal rates, $\operatorname{srank}_{\mathbb{R}} = n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$, and for $d = 2$, $\operatorname{srank}_{\mathbb{R}} = C_2(\Gamma) = \operatorname{srank}_{\mathbb{C}}(x \mapsto \varphi_{\mathbb{C}}(x_0^2 + x_1^2))$.

Theorem 2.7 (2-D Minimal Differential Annihilator; Lean: `exists_minimal_differential_annihilator`)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$ with $\operatorname{srank}_{\mathbb{R}}(x \in \mathbb{R}^2 \mapsto \varphi(x_0^2 + x_1^2)) = R < \infty$. For any $q(X) = \sum_{k=0}^{\deg q} q_k X^k \in \mathbb{C}[X]$ and $g : \mathbb{R} \to \mathbb{C}$, define the half-line differential operator (`differentialOperator`)*

$$\big(q(D)g\big)(t) := \sum_{k=0}^{\deg q} q_k \, D^k_{[0,\infty)} g(t),$$

where $D^0_{[0,\infty)} g = g$ and $D^k_{[0,\infty)} g(t)$ (`iteratedDerivWithin k g (Set.Ici 0) t`, $k \geq 1$) is Mathlib's totalized k-th derivative within $[0, \infty)$ (returning 0 where a within-derivative fails to exist, and agreeing for $g \in C^\infty([0, \infty))$ with the classical interior derivative at $t > 0$ and the right-hand limit $\lim_{h \to 0^+} h^{-1}(D^{k-1}_{[0,\infty)} g(h) - D^{k-1}_{[0,\infty)} g(0))$ at $t = 0$). Then $\varphi_{\mathbb{C}}|_{[0,\infty)} \in C^\infty([0, \infty))$ (so every $D^k_{[0,\infty)} \varphi_{\mathbb{C}}(t)$ is a genuine classical derivative), and there exists a unique monic polynomial $p_{\min} \in \mathbb{C}[X]$ of degree R such that for every $q \in \mathbb{C}[X]$, $\big(q(D)\varphi_{\mathbb{C}}\big)(t) = 0$ for all $t \geq 0$ if and only if $p_{\min} \mid q$.

Remark 2.8 (Automatic smoothness, domain of D, and identity with the generator's characteristic polynomial)**.** Theorem 2.7 assumes *only* continuity of φ on $[0, \infty)$ and $\operatorname{srank}_{\mathbb{R}} = R < \infty$, not differentiability: Section 3.1 shows by integral smoothing of the shift semigroup that $\varphi_{\mathbb{C}}$ agrees on $[0, \infty)$ with an analytic exponential polynomial $\sum_{\mu \in S} P_\mu(t) e^{\mu t}$, so every classical right-hand derivative $D^k_{[0,\infty)} \varphi_{\mathbb{C}}(t)$ ($t \geq 0$, $k \geq 0$) exists automatically! (Conversely, any continuous profile on $[0, \infty)$ that fails to be C^∞—such as $|s - 1|$—has $\operatorname{srank}_{\mathbb{R}} = \infty$. In Lean, `derivWithin` is totalized to return 0 where a derivative does not exist, so restricting Theorem 2.7 to finite-rank continuous profiles ensures that $D^k_{[0,\infty)} \varphi_{\mathbb{C}}$ always coincides with the classical within-derivative.) Uniqueness of $p_{\min}$ is immediate because any two monic polynomials dividing each other are equal; moreover, in terms of the shift generator $A \in \mathcal{L}(\mathcal{W}(\varphi_{\mathbb{C}}))$ of Section 3.1, $\deg p_{\min} = R = \dim_{\mathbb{C}} \mathcal{W}(\varphi_{\mathbb{C}}) = \deg \chi_A$ together with $p_{\min} \mid \chi_A$ implies $p_{\min}(X) = \chi_A(X) = \mu_A(X)$ ($\varphi_{\mathbb{C}}|_{\mathbb{R}_{\geq 0}}$ is a cyclic vector for A).

By Definition 2.3, $\underline{\operatorname{srank}}^{\mathrm{pt}}_{\mathbb{K}}(K) = \underline{\operatorname{srank}}^{\mathrm{co}}_{\mathbb{K}}(K) = \operatorname{srank}_{\mathbb{K}}(K)$ in $d = 2$ because $(r+1) \times (r+1)$ minors are continuous (cf. Laird [43]), whereas in $d \geq 3$ dimensions $\underline{\operatorname{srank}}^{\mathrm{co}}_{\mathbb{R}}(K)$ can drop strictly below $\operatorname{srank}_{\mathbb{R}}(K)$ via $O(1/h)$ cancelling weights—just as $s_0 + s_1 + s_2 = \lim_{h \to 0} h^{-1}\big((1 + hs_0)(1 + hs_1)(1 + hs_2) - 1\big)$ approximates a rank-3 sum by two 3-way products [8, 45]:

Theorem 2.9 (General d-D Multi-Gaussian Border-Rank Bound and Gap)**.** *Let $d \geq 2$ and let $\Gamma = [(\lambda_1, P_1), \ldots, (\lambda_m, P_m)]$ be any finite list of well-formed components. Formalized as `multiTotalCost_eq_borderTotalCost_add` and `exists_rank_le_borderTotalCost_tendstoUniformlyOn_radial_profile`:*

1. Exact gap identity: *Both real and nonreal components satisfy* $c_d(\lambda_u, P_u) = b_d(\lambda_u, P_u) + (d-2)\deg P_u$, *giving*

$$C_d(\Gamma) = B_d(\Gamma) + (d-2)\sum_{u=1}^{m} \deg P_u. \tag{6}$$

2. Uniform multi-Gaussian approximation on compact sets: *Replacing* $s = \|x\|^2$ *inside each* $P_u(s)$ *by the difference quotient* $u_h(s) = (e^{hs}-1)/h$ *yields an explicit family of multi-Gaussian radial kernels* $F_h : \mathbb{R}^d \to \mathbb{R}$ *(*$h \in \mathbb{R}\setminus\{0\}$*, extended to* $h = 0$ *in Lean via* $0^{-1} = 0$*) satisfying* $\operatorname{srank}_{\mathbb{R}}(F_h) \le B_d(\Gamma)$ *for all* h *and* $F_h(x) \to \operatorname{profile}(\Gamma, \|x\|^2)$ *uniformly on every compact subset* $C \subset \mathbb{R}^d$ *as* $h \to 0$ *(*$h \neq 0$*). Consequently, whenever* Γ *is a representation in dimension* $d \ge 3$ *with* $\sum_u \deg P_u \ge 1$*, its compact-open real border rank is strictly less than its exact real rank* $C_d(\Gamma)$ *by at least* $(d-2)\sum_u \deg P_u \ge d-2$*. In particular, for* $K_d(x) = \|x\|^2\cos(\|x\|^2)$ *(*$\Gamma = [(i, X)]$*),* $\operatorname{srank}_{\mathbb{C}}((K_d)_{\mathbb{C}}) = 2d$*,* $\operatorname{srank}_{\mathbb{R}}(K_d) = 3d-2$*, and* $\operatorname{srank}_{\mathbb{R}}(F_h) \le 2d$*.*

Is the multi-Gaussian radial bound $B_d(\Gamma)$ equal to the exact compact-open border rank when arbitrary non-Gaussian separable approximants (1) are allowed? Over $\mathbb{C}$ (for all profiles) and over $\mathbb{R}$ whenever all rates are real (or for a single complex Gaussian in $d = 3$), the bound is exact, *but for* $\deg P_u \ge 1$ *with nonreal rates over* $\mathbb{R}$ *in* $d \ge 3$*,* $B_d(\Gamma)$ *is strictly non-sharp*:

Theorem 2.10 (Exact Border-Rank Sharpness Regimes and Strict Non-Sharpness on $\mathbb{R}^3$)**.** *Formalized in* `KernelRankLean/Approximation.lean`*:*

1. Pointwise closure in $d = 2$, exact d-D complex border rank, and all-degree real-rate sharpness*: Over any Hausdorff topological field* $\mathbb{K}$*, if* $F_\iota : X_0 \times X_1 \to \mathbb{K}$ *eventually satisfies* $\operatorname{srank}_{\mathbb{K}}(F_\iota) \le r$ *along a nontrivial filter and converges pointwise to* K*, then* $\operatorname{srank}_{\mathbb{K}}(K) \le r$ *(*`separableRank_fin2_le_of_tendsto`*). Padding two coordinates by zeros implies that for every* $d \ge 2$*, any representation* Γ *has pointwise and compact-open real border rank bounded between* $C_2(\Gamma)$ *and* $B_d(\Gamma)$ *(*`totalCost_le_of_tendsto_radial_profile`*). Consequently: (i) over* $\mathbb{C}$*, every radial exponential polynomial* $\sum_{\mu\in S} P_\mu(\|x\|^2)e^{\mu\|x\|^2}$ *(*$P_\mu \neq 0$*) has exact pointwise and compact-open complex border rank* $\sum_{\mu\in S}(\deg P_\mu + 1)$ *(*`sum_natDegree_add_one_le_of_tendsto_radial_complex`*,* `exists_rank_le_sum_natDegree_add_one_tendstoUniformlyOn_radial_complex`*); and (ii) over* $\mathbb{R}$*, whenever every rate of* Γ *is real (*$\operatorname{Im}\lambda_u = 0$*, arbitrary polynomial degrees* $\deg P_u \ge 0$*),* $B_d(\Gamma) = C_2(\Gamma) = \sum_{u=1}^m(\deg P_u + 1)$ *is the exact pointwise and compact-open real border rank in all dimensions* $d \ge 2$ *(*`borderTotalCost_le_of_tendsto_radial_real_profile`*).*

2. Nonreal $k = 0$ sharpness and strict separation $4 < 5 < 6 < 7$ on $\|x\|^2\cos(\|x\|^2)$ in $d = 3$*: On* $\mathbb{R}^3$*, no family of real kernels of separable rank* ≤ 2 *converges pointwise along a nontrivial filter to* $\cos(\|x\|^2)$ *(*`not_exists_rank_le_two_tendsto_radial_cos`*), so its pointwise and compact-open real border rank equals* $3 = B_3$*. For* $K_3(x) = \|x\|^2\cos(\|x\|^2)$ *on* $\mathbb{R}^3$*, no family of real kernels of separable rank* ≤ 4 *converges pointwise to* K_3 *(*`not_exists_rank_le_four_tendsto_radial_sq_mul_cos`*), while the explicit* 5*-term non-Gaussian real family* (7) *below has* $\operatorname{srank}_{\mathbb{R}}(F_h) \le 5$ *and converges to* K_3 *uniformly on every compact subset* $C \subset \mathbb{R}^3$ *(*`exists_rank_le_five_tendstoUniformlyOn_radial_sq_mul_cos`*). Thus* K_3 *strictly separates complex border rank* (4)*, real border rank* (5)*, radial multi-Gaussian border rank and exact complex rank* (6)*, and exact real rank* (7*; see Table 14 in Section 7*)*.*

Remark 2.11 (Explicit rank-5 construction, sharpness classification, and numerical conditioning implications)**.** To see explicitly how non-Gaussian separable approximants drop below $B_3([(i, X)]) = 6$ in Theorem 2.10(2) (formalized in `Approximation.lean`, `exists_rank_le_five_tendstoUniformlyOn_radial_sq_mul_cos`, and verified algebraically in Section 3.4),

Table 4: Classification of when the multi-Gaussian border-rank bound ($B_d(\Gamma)$ over $\mathbb{R}$, $\sum_{\mu\in S}(\deg P_\mu + 1)$ over $\mathbb{C}$) equals the exact compact-open (and pointwise) border rank.

Field & dim.	Representation / profile class	Exact border rank	Sharp?	Formalized theorem
$\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, d=2	All representations $(\forall \lambda_u, \deg P_u)$	$C_2(\Gamma) = B_2(\Gamma) = \mathrm{srank}_{\mathbb{K}}$	**Yes**	`separableRank_fin2_le_of_tendsto`
$\mathbb{C}$, any $d \geq 2$	All complex exp.-polynomials	$\sum_{\mu\in S}(\deg P_\mu + 1)$	**Yes**	`sum_natDegree_add_one_le_...`
$\mathbb{R}$, any $d \geq 2$	All real rates $(\mathrm{Im}\,\lambda_u = 0, \forall \deg P_u)$	$\sum_u(\deg P_u + 1) = B_d(\Gamma)$	**Yes**	`borderTotalCost_le_of_tendsto_...`
$\mathbb{R}$, $d = 3$	Single nonreal Gaussian $\cos(\|x\|^2)$	$3 = B_3 = C_3$	**Yes**	`not_exists_rank_le_two_...`
$\mathbb{R}$, $d = 3$	Degree-1 nonreal $\|x\|^2\cos(\|x\|^2)$	$5 < B_3 = 6 < C_3 = 7$	**No**	`exists_rank_le_five_...`
$\mathbb{R}$, $d \geq 3$	General $\mathrm{Im}\,\lambda_u > 0$ ($d \geq 4$ or $k_u \geq 2$)	In $[C_2(\Gamma), B_d(\Gamma)]$	Open	`totalCost_le_...` & Thm. 2.9

write $c_j = \cos(x_j^2)$, $s_j = \sin(x_j^2)$, $p_{1,j} = \frac{1}{2}c_j - \frac{\sqrt{3}}{2}s_j$, $p_{2,j} = -\frac{1}{2}c_j - \frac{\sqrt{3}}{2}s_j$, and $q_{m,j} = x_j^2 p_{m,j}$ for $j \in \{0,1,2\}$. For any $h \in \mathbb{R}\setminus\{0\}$, the 5-product real kernel

$$F_h(x) = \frac{1}{3h}\prod_{j=0}^{2}(c_j + 4hx_j^2 c_j) + \frac{1}{h^3}\left(\prod_{j=0}^{2}(s_j + h^2 c_j) - \prod_{j=0}^{2} s_j\right) - \frac{4}{3h}\prod_{j=0}^{2}(p_{1,j} + hq_{1,j}) + \frac{4}{3h}\prod_{j=0}^{2}(p_{2,j} + hq_{2,j}) \tag{7}$$

breaks radial symmetry at finite $h \neq 0$ yet satisfies $F_h(x) - \|x\|^2\cos(\|x\|^2) = O(h)$ uniformly on every compact set $C \subset \mathbb{R}^3$. Table 4 summarizes all regimes in which the multi-Gaussian bound is proved sharp versus strictly non-sharp. Two practical lessons follow directly from these theorems:

1. *Convergence topology:* All border-rank *lower* bounds in Theorem 2.10 hold under bare pointwise convergence on $\mathbb{R}^d$, whereas the *upper* bounds hold uniformly on compact sets $C \subset \mathbb{R}^d$ (for profiles with $\mathrm{Re}\,\lambda_u \geq 0$ such as $\|x\|^2\cos\|x\|^2$, $|u_h(s) - s|$ grows as $s \to \infty$, so uniform convergence on all of $\mathbb{R}^d$ is neither claimed nor possible, in contrast to decaying profiles in Theorem 2.12).
2. *Floating-point cancellation penalty:* Saving $(d-2)\sum_u \deg P_u$ terms via Theorem 2.9 requires $O(h^{-\deg P_u})$ weights, and saving one further term in (7) requires $O(h^{-3})$ weights: achieving 10^{-3} truncation accuracy in (7) inflates intermediate products by $h^{-3} \approx 10^9$, wiping out IEEE `float16` (3.3 decimal digits) and `float32` (7.2 digits) precision. Consequently, border-rank degenerations are primarily structural obstructions explaining near-coalescing rate ill-conditioning ($1c_6$ in Table 11), whereas practical filter and stencil designs should use well-separated pure Gaussians ($k = 0$, where $B_d = C_d$ and weights remain $O(1)$).

Finally, whereas Theorem 2.4 shows that non-exponential-polynomial targets have $\mathrm{srank}_{\mathbb{R}} = \infty$, pure decaying complex Gaussians sharing a *single* width parameter and equally spaced harmonic frequencies are dense in the uniform norm over *all* of $\mathbb{R}^d$ for radial profiles vanishing at infinity:

Theorem 2.12 (Equal-Width Harmonic Complex-Gaussian Universal Approximation on $\mathbb{R}^d$; Lean: `exists_harmonic_gaussian_uniform_approx`)**.** *Let $d \geq 2$, let $\psi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0,\infty)$ with $\lim_{s\to\infty}\psi(s) = 0$, and let $\alpha > 0$ and $\varepsilon > 0$. There exist a fundamental frequency $\omega_0 > 0$, $M \in \mathbb{N}$, $a_0 \in \mathbb{R}$, and $a_1, \ldots, a_M \in \mathbb{C}$ such that the equal-width harmonic multi-Gaussian kernel*

$$K_M(x) := a_0 e^{-\alpha\|x\|^2} + \sum_{m=1}^{M}\mathrm{Re}\Big(a_m e^{(-\alpha+im\omega_0)\|x\|^2}\Big) \tag{8}$$

satisfies $\operatorname{srank}_{\mathbb{R}}(K_M) \le 1 + dM$ *and* $|K_M(x) - \psi(\|x\|^2)| < \varepsilon$ *for all* $x \in \mathbb{R}^d$.

Remark 2.13 (Envelope compensation vs. near-cancelling nonlinear rates)**.** In (8), factoring out the shared Gaussian envelope $e^{-\alpha s}$ ($s = \|x\|^2$) leaves a T-periodic trigonometric polynomial $\operatorname{Re}(a_0 + \sum_{m=1}^{M} a_m e^{im\omega_0 s})$ of period $T = 2\pi/\omega_0$. If $\psi(s)$ vanishes (or is below $\varepsilon/4$) beyond S_0, taking $T > S_0 + 2$ avoids wrap-around inside $[0, S_0]$, compensating $g(u) = e^{+\alpha u}\psi(u)$ on $[0, S_0]$ recovers any continuous radial shape once $e^{-\alpha u} g(u) = \psi(u)$, and choosing αT large enough damps every periodic replica at $s = kT + u$ ($k \ge 1, u \in [0, S_0]$) because $e^{-\alpha(kT+u)} e^{+\alpha u}\psi(u) = e^{-k\alpha T}\psi(u)$ cancels the $+\alpha u$ ramp identically. At the same time, fixing equal widths and harmonic spacing $\omega_m = m\omega_0$ explains why (8) is *not* optimal at small rank R: keeping $\alpha S_0 \ll 1$ (a wide envelope needing little compensation) forces $T \gg S_0$ so resolving features of width Δs requires $M \propto T/\Delta s$ harmonics, whereas keeping $T \sim 2S_0$ forces $\alpha S_0 \gg 1$ and amplifies $|g(u)|$ by $e^{+\alpha S_0}$ near the rim. Freeing (α_u, ω_u) allows unequal-width, non-harmonic (or nearly coalescing, Theorem 2.9) components to nearly cancel and sculpt sharp transitions at small R [12, 58].

The constructive upper bounds for nonreal components in Theorems 2.6, 2.9, and 2.12 rest on two alternating angle-sampling ($2d$-th and $M^{\mathbb{R}}_{d,k}$-th root-of-unity) identities, proved in Section 3.2 (item 3) and machine-checked in `KernelRankLean/MultiRealRank.lean` (throughout, $[X^k]F(X)$ denotes the coefficient of X^k in $F \in \mathbb{C}[X]$):

Proposition 2.14 (Alternating angle-sampling ($2d$-th root-of-unity) identity; Lean: `re_prod_eq_sum_re`)**.** *For every integer* $d \ge 1$ *and all* $z_0, \dots, z_{d-1} \in \mathbb{C}$,

$$\operatorname{Re}\left(\prod_{j=0}^{d-1} z_j\right) = \frac{2^{d-1}}{d} \sum_{r=0}^{d-1} (-1)^r \prod_{j=0}^{d-1} \operatorname{Re}\left(z_j e^{-i\pi r/d}\right). \tag{9}$$

Proposition 2.15 (Polynomial alternating angle-sampling identity; Lean: `re_coeff_prod_eq_sum_re`)**.** *Let* $d \ge 2$, $k \ge 0$, *and* $M^{\mathbb{R}}_{d,k} = 2(d-1)k + d$. *For any polynomials* $U_0, \dots, U_{d-1} \in \mathbb{C}[X]$ *with* $\deg U_j \le k$,

$$\operatorname{Re}\left([X^k] \prod_{j=0}^{d-1} U_j(X)\right) = \frac{2^{d-1}}{M^{\mathbb{R}}_{d,k}} \sum_{r=0}^{M^{\mathbb{R}}_{d,k}-1} (-1)^r \prod_{j=0}^{d-1} \operatorname{Re}\left(U_j\left(e^{\frac{2\pi i r}{M^{\mathbb{R}}_{d,k}}}\right) e^{-\frac{i\pi r(2k+1)}{M^{\mathbb{R}}_{d,k}}}\right). \tag{10}$$

When $k = 0$, $M^{\mathbb{R}}_{d,0} = d$, *each* $U_j(X) = z_j$ *is a constant polynomial,* $[X^0]\prod_j z_j = \prod_j z_j$, *and the phase* $e^{-i\pi r(2\cdot 0+1)/d} = e^{-i\pi r/d}$ *reduces* (10) *identically to* (9).

Remark 2.16 (Reconciling the product counts 2^{d-1}, d, $(d-1)k+1$, and $2(d-1)k+d$)**.** To keep the distinct real-product counts separate across degrees $k = \deg P$:

- *Degree* $k = 0$, *naive expansion* (2^{d-1} *products*) *vs. Proposition 2.14* (d *products*)*:* Writing $z_j = a_j + ib_j$, expanding $\operatorname{Re}(\prod_{j=0}^{d-1}(a_j + ib_j))$ sums $(-1)^{|S|/2}(\prod_{j\notin S} a_j)(\prod_{j\in S} b_j)$ over all even-cardinality subsets $S \subseteq \{0, \dots, d-1\}$, which has $\sum_{2|q} \binom{d}{q} = \frac{1}{2}((1+1)^d + (1-1)^d) = 2^{d-1}$ real d-way products ($2, 4, 8$ for $d = 2, 3, 4$). By contrast, averaging over all $2d$ angles $r = 0, \dots, 2d-1$ folds in antipodal pairs $r \leftrightarrow r + d$ because $\operatorname{Re}(z_j e^{-i\pi(r+d)/d}) = -\operatorname{Re}(z_j e^{-i\pi r/d})$ and $(-1)^{r+d}(-1)^d = (-1)^r$, leaving just d real products in (9) ($2, 3, 4$ for $d = 2, 3, 4$, with scalar prefactor $2^{d-1}/d$).
- *Degree* $k \ge 1$*:* A *real* rate ($\operatorname{Im}\lambda = 0$) takes $c_d(\lambda, P) = M^{\mathbb{C}}_{d,k} = (d-1)k+1$ real products (d at $k = 1$, $2d-1$ at $k = 2$). A *nonreal* rate ($\operatorname{Im}\lambda > 0$) takes $c_d(\lambda, P) = M^{\mathbb{R}}_{d,k} = 2(d-1)k+d$ real products via (10) ($3d-2 = 7$ at $d = 3, k = 1$), compared with $M^{\mathbb{C}}_{d,k} \cdot 2^{d-1} = ((d-1)k+1)2^{d-1}$ products (12 at $d = 3, k = 1$; Table 5) if each complex root-of-unity product is expanded into 2^{d-1} real products, or $d^2 = 9$ at $d = 3, k = 1$ if $s_0 + s_1 + s_2$ is split into 3 complex Gaussians evaluated by (9).

Example 2.17 (Concrete $d = 3, k = 0$ complex Gaussian: 3 real products vs. 4 naive products)**.** Let $d = 3$ and write $z_j = a_j + ib_j \in \mathbb{C}$ $(a_j, b_j \in \mathbb{R})$ for $z_j(x_j) = \beta^{1/3} e^{(-\alpha+i\omega)x_j^2}$. Expanding $\mathrm{Re}(z_0 z_1 z_2)$ directly into real and imaginary parts requires $2^{3-1} = 4$ real 3-way products:

$$\mathrm{Re}(z_0 z_1 z_2) = a_0 a_1 a_2 - a_0 b_1 b_2 - b_0 a_1 b_2 - b_0 b_1 a_2.$$

Instead, Proposition 2.14 at $d = 3$ evaluates $u_{j,r} = \mathrm{Re}(z_j e^{-i\pi r/3}) = a_j \cos(\frac{\pi r}{3}) + b_j \sin(\frac{\pi r}{3})$ at the three angles $0, \frac{\pi}{3}, \frac{2\pi}{3}$ $(u_{j,0} = a_j,\ u_{j,1} = \frac{1}{2}a_j + \frac{\sqrt{3}}{2}b_j,\ u_{j,2} = -\frac{1}{2}a_j + \frac{\sqrt{3}}{2}b_j)$:

$$\mathrm{Re}(z_0 z_1 z_2) = \frac{4}{3}\Big(u_{0,0}u_{1,0}u_{2,0} - u_{0,1}u_{1,1}u_{2,1} + u_{0,2}u_{1,2}u_{2,2}\Big).$$

Expanding the right-hand side confirms that all $a_0 a_1 b_2$ and $b_0 b_1 b_2$ terms cancel via $-\frac{\sqrt{3}}{8} + \frac{\sqrt{3}}{8} = 0$ and $-\frac{3\sqrt{3}}{8} + \frac{3\sqrt{3}}{8} = 0$, while $a_0 a_1 a_2$ keeps $1 - \frac{1}{8} - \frac{1}{8} = \frac{3}{4}$, leaving $\frac{4}{3}(\frac{3}{4}a_0 a_1 a_2 - \frac{3}{4}(a_0 b_1 b_2 + b_0 a_1 b_2 + b_0 b_1 a_2)) = \mathrm{Re}(z_0 z_1 z_2)$ using *three* symmetric real 3-way products rather than four, and Theorem 2.6 proves that two never suffice when $\omega > 0$. Algebraically, pairing $(z_0, z_1) \mapsto z_0 z_1$ against $\bar{z}_2 \mapsto \mathrm{Re}((z_0 z_1) z_2)$ identifies this $2 \times 2 \times 2$ real tensor with bilinear multiplication in $\mathbb{C} \cong \mathbb{R}[X]/(X^2 + 1)$, whose real rank is 3 by Gauss/Karatsuba complex multiplication [75].

Example 2.18 (Concrete $d = 3, k = 1$ polynomial oscillation: 7 real products via $M_{3,1}^{\mathbb{R}} = 7$ vs. 9 or 12 naive)**.** Consider $K_3(x) = (x_0^2 + x_1^2 + x_2^2)\cos(x_0^2 + x_1^2 + x_2^2)$ $(\lambda = i,\ P(s) = s,\ k = 1)$. Splitting $x_0^2 \cos\|x\|^2 + x_1^2 \cos\|x\|^2 + x_2^2 \cos\|x\|^2$ into three shifted complex Gaussians costs $3 \times 4 = 12$ real products under direct a_j, b_j expansion (Table 5) or $3 \times 3 = 9$ real products using Example 2.17 on each summand. Instead, put $U_j(X) = e^{ix_j^2}(1 + x_j^2 X)$ $(j = 0, 1, 2)$, so $[X^1](U_0 U_1 U_2) = (x_0^2 + x_1^2 + x_2^2)e^{i\|x\|^2}$. Here $M_{3,1}^{\mathbb{R}} = 2(3-1)(1) + 3 = 7$ and $2k + 1 = 3$; evaluating $\mathrm{Re}\big(U_j(e^{2\pi i r/7})e^{-3\pi i r/7}\big)$ gives the seven real univariate functions

$$K_3(x) = \frac{4}{7}\sum_{r=0}^{6}(-1)^r f_r(x_0)\, f_r(x_1)\, f_r(x_2), \qquad f_r(u) := \cos\left(u^2 - \frac{3\pi r}{7}\right) + u^2 \cos\left(u^2 - \frac{\pi r}{7}\right).$$

Why must the node count be 7 with alternating signs $(-1)^r = e^{7\pi i r/7}$? Writing $\cos\theta = \frac{1}{2}(e^{-i\theta} + e^{i\theta})$ in $f_r(x_0) f_r(x_1) f_r(x_2)$ produces harmonics $e^{2\pi i r K/7}$ where $m \in \{0, 1, 2, 3\}$ factors pick the $-i$ branch and $q_j \in \{0, 1\}$ picks 1 vs. x_j^2, giving $K = 3m + \sum_{j \notin S} q_j - \sum_{j \in S} q_j - 1$. For mixed branches $m = 1$ $(K \in \{1, \ldots, 4\})$ and $m = 2$ $(K \in \{3, \ldots, 6\})$, 7 is the smallest modulus keeping every mixed harmonic inside $\{1, \ldots, M-1\}$ so $\sum_{r=0}^{6} e^{2\pi i r K/7} = 0$, while on pure branches $m = 0$ $(K = q_0 + q_1 + q_2 - 1)$ and $m = 3$ $(K = 8 - q_0 - q_1 - q_2)$ only $q_0 + q_1 + q_2 = 1$ is divisible by 7, isolating $K_3(x)$ in 7 real products (proved optimal in Section 3.3).

2.3 Intuitive guide: what $c_d(\lambda, P)$ counts and why the $d - 2$ penalty arises

For readers primarily interested in filter design or scientific computing rather than commutative algebra, four plain-language rules explain every number in Theorems 2.4–2.9 (illustrated in Table 5):

1. **What $c_d(\lambda, P)$ counts computationally:** A single component (λ, P) is one radially symmetric building block (such as u_1 or u_2 in Figure 1a), whereas $c_d(\lambda, P)$ counts how many rank-1 real product channels $\prod_{j=0}^{d-1} f_{n,j}(x_j)$ (the rendering terms t_n in Figure 1b) must be summed to evaluate that component over $\mathbb{R}$. On a d-dimensional grid of width W, rank $R = C_d(\Gamma) = \sum_u c_d(\lambda_u, P_u)$ executes dR one-dimensional passes (dRW multiply–adds per voxel).

Table 5: Component-cost comparison across $d \in \{2,3,4\}$ and degrees $k = \deg P \in \{0,1\}$: exact complex rank, exact real rank $c_d(\lambda, P)$ (each term costing d one-dimensional passes), naive real expansion (grouping p_0 into mode 0 to form d complex products at $k=1$ and expanding each $\mathrm{Re}(z_0 \cdots z_{d-1})$ into 2^{d-1} real products, giving $d\,2^{d-1}$), and multi-Gaussian border-rank bound $b_d(\lambda, P)$.

Single component type	Profile $\varphi(s)$ ($s = \|x\|^2$)	Dim. d	Complex rank	Exact real c_d	Naive real	Border bound b_d
Real Gaussian ($k=0$)	$a\,e^{-\alpha s}$	$2,3,4$	$1,1,1$	$1,1,1$	$1,1,1$	$1,1,1$
Real Laguerre ($k=1$)	$(p_0 + p_1 s)e^{-\alpha s}$	$2,3,4$	$2,3,4$	$2,3,4$	$2,3,4$	$2,2,2$
Complex Gaussian ($k=0$)	$\mathrm{Re}(\beta\, e^{(-\alpha+i\omega)s})$ ($\omega > 0$)	$2,3,4$	$2,2,2$	$2,3,4$	$2,4,8$	$2,3,4$
Complex poly. ($k=1$)	$\mathrm{Re}((p_0 + p_1 s)e^{(-\alpha+i\omega)s})$	$2,3,4$	$4,6,8$	$4,7,10$	$4,12,32$	$4,6,8$

2. **Why a real rate costs $(d-1)k+1$:** Expanding $P(s_0 + \cdots + s_{d-1})$ of degree k places a degree-k polynomial on each of the last $d-1$ coordinates; their product has degree $(d-1)k$, which has $(d-1)k+1$ independent coefficients (1 when $k=0$; d when $k=1$, matching $s_0 + \cdots + s_{d-1}$) and can be reconstructed from $(d-1)k+1$ real evaluation points.

3. **Where the constant $d-2$ real-over-complex penalty comes from:** Over $\mathbb{C}$, a nonreal component $\mathrm{Re}(P(s)e^{\lambda s}) = \frac{1}{2}P(s)e^{\lambda s} + \frac{1}{2}\overline{P}(s)e^{\bar{\lambda} s}$ splits into two independent complex rates of cost $(d-1)k+1$ each, totaling $2(d-1)k+2$ (2 complex products when $k=0$, in *every* dimension d). Over $\mathbb{R}$, however, even at $k=0$ ($\mathrm{Re}(z_0 \cdots z_{d-1})$ with $z_j \in \mathbb{C} \cong \mathbb{R}^2$), every real linear functional $\ell : \mathbb{C} \to \mathbb{R}$ has a 1-dimensional nullspace: choosing $z_j \neq 0$ in that nullspace kills one real product term per coordinate $j = 0, \ldots, d-2$ while leaving the remaining product nonzero, forcing at least d real products (2 in 2-D, 3 in 3-D, 4 in 4-D)—a base penalty of $d-2$ over the 2 complex products! Each polynomial degree increment $k \mapsto k+1$ then adds $2(d-1)$ real degrees of freedom on modes $0, \ldots, d-2$ over *both* $\mathbb{C}$ and $\mathbb{R}$, so $c_d(\lambda, P) = d + 2(d-1)k$ and the complex pair cost $2 + 2(d-1)k$ grow at the same rate $2(d-1)k$, keeping their difference locked at $d-2$ for every $k \geq 0$.

4. **Why the multi-Gaussian border-rank bound drops by $(d-2)k$:** Approximating $se^{\lambda s} \approx \frac{e^{(\lambda+h)s} - e^{\lambda s}}{h}$ replaces one degree-1 polynomial factor by a difference of *two* pure Gaussians ($k=0$): in $d \geq 3$, two complex Gaussians cost $2d$ real products whereas exact $k=1$ costs $2(d-1)+d = 3d-2$, lowering the multi-Gaussian border-rank upper bound $B_d(\Gamma)$ by $d-2$ products per polynomial degree at the price of $O(1/h)$ cancelling weights as $h \to 0$.

Remark 2.19 (Unified commutative-algebra invariant and comparison with symmetric Waring rank)**.** Write $\mathcal{A}_\varphi = \mathbb{R}[X]/(p_{\min,\mathbb{R}}(X))$ for the finite-dimensional commutative polynomial quotient algebra associated to the minimal real differential annihilator $p_{\min,\mathbb{R}}(D)\varphi = 0$ (isomorphic to $\mathcal{W}(\varphi)$ as an $\mathbb{R}[D]$-module, with multiplication induced by composition of D rather than pointwise product of profiles). Over $\mathbb{R}$, $p_{\min,\mathbb{R}}(X)$ factors into $m_{\mathrm{irr}} = n_{\mathrm{re}} + n_{\mathrm{cx}}$ distinct powers of irreducible real polynomials (n_{re} linear factors $(X-\lambda_u)^{k_u+1}$ and n_{cx} irreducible real quadratics $((X+\alpha_u)^2+\omega_u^2)^{k_u+1}$; equivalently, m_{irr} maximal ideals of $\mathcal{A}_\varphi$), while $\dim_{\mathbb{R}} \mathcal{A}_\varphi = \deg p_{\min,\mathbb{R}} = C_2(\Gamma)$. Checking $(d-1)(k+1) - (d-2)\cdot 1 = (d-1)k+1$ and $(d-1)\cdot 2(k+1) - (d-2)\cdot 1 = 2(d-1)k + d$ shows that Theorem 2.6 takes one unified form across all $d \geq 2$:

$$\mathrm{srank}_{\mathbb{R}}(K) = C_d(\Gamma) = (d-1)\dim_{\mathbb{R}} \mathcal{A}_\varphi - (d-2)\, m_{\mathrm{irr}} \qquad (d \geq 2), \tag{11}$$

and the identical formula holds over $\mathbb{C}$ with $m_{\mathrm{irr}} = |S|$ linear factors. At $d=2$ (1-fold identity pairing), (11) reduces to $\dim_{\mathbb{K}} \mathcal{A}$; at $d=3$ (2-fold bilinear multiplication), it coincides with the Winograd–Fiduccia–Zalcstein–Alder–Strassen complexity $2\dim_{\mathbb{K}} \mathcal{A} - m_{\mathrm{irr}}$ [75, 23, 1, 15]; and for $d \geq 4$ ($(d-1)$-fold multiplication), it matches the multilinear Alder–Strassen lower bound of

Bläser, Mayer, and Shringi [10] with equality. Notice the contrast with symmetric Waring rank: for a real binary form $F(u,v) = \mathrm{Re}((u+iv)^D)$ ($D \geq 2$), complex Waring rank is 2 whereas real symmetric Waring rank equals D [18, 63, 11]—a gap $D-2$ that grows with degree. In (11), complexifying $\mathbb{R} \hookrightarrow \mathbb{C}$ splits each irreducible real quadratic into two conjugate linear factors (m_{irr} increases by 1 per nonreal pair while $\dim \mathcal{A}$ is unchanged), producing a *degree-independent* real-over-complex penalty of $(d-2)(2-1) = d-2$ per conjugate pair because coordinatewise decompositions (1) break mode symmetry by placing $H_P(x_0^2; X)$ on mode 0 and $E_k(x_j^2; X)$ on modes $j \geq 1$.

2.4 Theoretical numerical stability and practical scope of the decompositions

Because Theorems 2.4–2.10 assume exact real or complex arithmetic, assessing when each rank formula yields a numerically stable algorithm in floating-point arithmetic with unit roundoff $\varepsilon_{\mathrm{mach}}$ (`float16`: $2^{-11} \approx 4.9 \times 10^{-4}$; `float32`: $2^{-24} \approx 6.0 \times 10^{-8}$) requires bounding the intermediate term amplification $\sum_{n=0}^{R-1} \prod_{j=0}^{d-1} |f_{n,j}(x_j)|$ across our three synthesis mechanisms:

1. **Degree-0 angle sampling (Proposition 2.14) vs. inter-component cancellation:** For a single complex Gaussian $z_j = \beta^{1/d} e^{(-\alpha+i\omega)x_j^2}$, $|\mathrm{Re}(z_j e^{-i\pi r/d})| \leq |z_j|$ gives

$$\frac{2^{d-1}}{d} \sum_{r=0}^{d-1} \prod_{j=0}^{d-1} \left|\mathrm{Re}(z_j e^{-i\pi r/d})\right| \leq 2^{d-1} \prod_{j=0}^{d-1} |z_j| = 2^{d-1} |\beta|\, e^{-\alpha \|x\|^2}. \tag{12}$$

   For $d \in \{2,3,4\}$ ($2^{d-1} = 2, 4, 8$), single-component angle sampling is well-conditioned; only for high dimensions $d \gg 1$ does 2^{d-1} grow exponentially. Instead, the severe ill-conditioning seen in multi-component flat-top filters ($\kappa_\infty^{\mathrm{raw}} = 595$ on 6c, Table 9) stems from *inter-component cancellation* $\sum_{u=1}^{m} |\beta_u| e^{-\alpha_u s} \gg |\varphi(s)|$ when distinct rates (λ_u) cluster. In $d = 2$, transitive $\mathrm{GL}_R(\mathbb{R})$ orthogonalization on $\mathrm{rowSpan}(K)$ reduces $\sum_{n=1}^{R} \|\tilde{h}_n\|_2 \|\tilde{v}_n\|_2$ all the way to the nuclear norm $\|K\|_{S_1}$ (Section 5.3), whereas in $d \geq 3$ minimal CP decompositions are generically rigid [41] and require keeping $\sum_u |\beta_u|$ modest during rate optimization.

2. **High-degree polynomials ($k = \deg P \geq 1$) and large d:** In Section 3.2, scaling the indeterminate $X \mapsto \rho X$ ($\rho > 0$) extracts $P(s) = \rho^{-k}[X^k]\big(H_P(s_0; \rho X) \prod_{j=1}^{d-1} E_k(s_j; \rho X)\big)$. For the model monomial $P(s) = s^k/k!$ at $s = \|x\|^2 > 0$, $|E_k(s_j; \rho e^{i\theta})| \leq e^{\rho s_j}$ bounds the absolute sum of the $M_{d,k}^{\mathbb{R}}$ terms in (10) (or $M_{d,k}^{\mathbb{C}}$ terms over $\mathbb{C}$, without the 2^{d-1} factor) by $2^{d-1} \rho^{-k} e^{\rho s} |e^{\lambda s}|$. Relative to $(s^k/k!)|e^{\lambda s}|$, choosing the saddle-point radius $\rho_\star = k/s$ gives Stirling amplification

$$\frac{2^{d-1} \rho_\star^{-k} e^{\rho_\star s}}{s^k/k!} = 2^{d-1}\, k! \left(\frac{e}{k}\right)^k \sim 2^{d-1} \sqrt{2\pi k}, \tag{13}$$

   whereas an unscaled evaluation $\rho = 1$ blows up as $k!\, e^s/s^k$ for large $s = \|x\|^2$, and real Lagrange interpolation on $M = (d-1)k+1$ *equispaced* nodes $0, \ldots, M-1$ suffers exponential Lebesgue-constant growth $\Lambda_M \sim \frac{2^M}{eM \log M}$ unless evaluated on Chebyshev nodes in $[-1,1]$ ($\Lambda_M = \frac{2}{\pi} \log M + O(1)$).

3. **Finite-h border-rank approximations (Theorem 2.9 and Eq. (7)):** Replacing $s^k e^{\lambda s}$ by $u_h(s)^k e^{\lambda s} = h^{-k} \sum_{m=0}^{k} (-1)^{k-m} \binom{k}{m} e^{(\lambda+mh)s}$ trades $O(k s^{k+1} |h|)$ truncation bias against $O(2^k |h|^{-k} \varepsilon_{\mathrm{mach}})$ catastrophic cancellation. Optimizing over $h \neq 0$ sets $|h_\star| \asymp \varepsilon_{\mathrm{mach}}^{1/(k+1)}$ and imposes a fundamental relative-error floor of $\Theta\big(\varepsilon_{\mathrm{mach}}^{1/(k+1)}\big)$ (already $\sim 2.2\%$ at $k = 1$ and $\sim 8\%$ at $k = 2$ in `float16`, and $\Theta(\varepsilon_{\mathrm{mach}}^{1/4})$—$\sim 15\%$ in `float16`, $\sim 1.6\%$ in `float32`—for the $O(h^{-3})$ 5-term $\mathbb{R}^3$ limit (7)).

Practical scope: Exact rank decompositions are directly practical for $d \in \{2,3,4\}$ with well-separated pure Gaussians ($k = 0$) and for scaled $k = 1$ components in `float32`/`float64`; high-degree ($k \geq 2$) or high-dimensional ($d \geq 6$) syntheses require radius scaling $\rho \approx k/S^2$, Chebyshev nodes, and higher precision; and $d \geq 3$ border-rank degenerations ($B_d < C_d$) serve primarily as complexity lower bounds and models of coalescing-rate ill-conditioning rather than finite-precision numerical schemes.

3 Mathematical Proofs and Core Constructions

Radial $K(x) = f(\|x\|^2)$ on $\mathbb{R}^d$ $\xleftarrow[\text{surjection } s_j = x_j^2]{\text{Eq. (3)}}$ Hankel $(s_0, \dots, s_{d-1}) \mapsto f\left(\sum_{j=0}^{d-1} s_j\right)$ on $\mathbb{R}^d_{\geq 0}$

Constructive upper bounds ($\leq R$)
- $\mathbb{C}$: $M^{\mathbb{C}}_{d,k} = (d-1)k+1$ roots of unity (Sec. 3.2)
- $\mathbb{R}$, $\operatorname{Im}\lambda{=}0$: Lagrange at $0, \dots, M^{\mathbb{C}}_{d,k}-1$ (Sec. 3.2)
- $\mathbb{R}$, $\operatorname{Im}\lambda{>}0$: $M^{\mathbb{R}}_{d,k} = 2(d-1)k+d$ roots (Prop. 2.15)

Necessity & multilinear lower bounds ($\geq R$)
- $d{=}2$ slice $\Rightarrow \dim_{\mathbb{C}} \mathcal{W}(\psi) < \infty \Rightarrow T_t = e^{tA}$ (Sec. 3.1)
- Dual span of $v(\mathbb{R}_{\geq 0}) \Rightarrow$ linear $\ell_{n,j} \in V^*$ (Sec. 3.3)
- Alder–Strassen peeling + $+1$ shift (Sec. 3.3)

3.1 Necessity and the two-dimensional core

In two coordinates over a field $\mathbb{K}$, let $\operatorname{rowSpan}(K) \leq (X \to \mathbb{K})$ be the $\mathbb{K}$-linear span of the row sections $\{x \mapsto K(x,y) : y \in X\}$. Then $\operatorname{srank}_{\mathbb{K}}(K) = \dim_{\mathbb{K}} \operatorname{rowSpan}(K) \in \mathbb{N} \cup \{\infty\}$: if $K(x,y) = \sum_{n=0}^{R-1} f_n(x) g_n(y)$, every row $x \mapsto K(x,y)$ is a linear combination of $f_0, \dots, f_{R-1}$ so $\dim \operatorname{rowSpan}(K) \leq R$; conversely, if $b_0(x), \dots, b_{R-1}(x)$ is a basis of $\operatorname{rowSpan}(K)$, expanding each row $x \mapsto K(x,y) = \sum_{n=0}^{R-1} c_n(y) b_n(x)$ directly gives a rank-R sum of products. For the two-variable Hankel kernel $(s_0, s_1) \mapsto f(s_0 + s_1)$ on $\mathbb{R}^2_{\geq 0}$, the row span is the *translate span*

$$\mathcal{W}(f) = \operatorname{span}_{\mathbb{K}}\{u \in \mathbb{R}_{\geq 0} \mapsto f(u+t) : t \in \mathbb{R}_{\geq 0}\} \leq (\mathbb{R}_{\geq 0} \to \mathbb{K}).$$

Scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ satisfies $\operatorname{srank}_{\mathbb{C}}(K_{\mathbb{C}}) \leq \operatorname{srank}_{\mathbb{R}}(K)$ in every dimension d. In $d = 2$, equality $\operatorname{srank}_{\mathbb{C}}(K_{\mathbb{C}}) = \operatorname{srank}_{\mathbb{R}}(K)$ always holds: if real functions $b_0, \dots, b_{R-1} : X \to \mathbb{R}$ are $\mathbb{R}$-linearly independent, any complex relation $\sum_n (\alpha_n + i\beta_n) b_n(x) = 0$ forces $\sum_n \alpha_n b_n(x) = 0$ and $\sum_n \beta_n b_n(x) = 0$ separately, so $\dim_{\mathbb{C}} \operatorname{rowSpan}(K_{\mathbb{C}}) = \dim_{\mathbb{R}} \operatorname{rowSpan}(K)$.

For any $d \geq 2$, if $\operatorname{srank}_{\mathbb{R}}(x \in \mathbb{R}^d \mapsto \varphi(\|x\|^2)) < \infty$, the slice inequality (4) and 2-D complexification give $\dim_{\mathbb{C}} \mathcal{W}(\varphi_{\mathbb{C}}) < \infty$. Necessity and the 2-D annihilator theorem then follow from a single basis-free operator construction on $\mathcal{W}(\psi)$ for any continuous $\psi : \mathbb{R} \to \mathbb{C}$ with $\dim_{\mathbb{C}} \mathcal{W}(\psi) = n < \infty$:

1. Equip the algebraic subspace $\mathcal{W}(\psi) \leq (\mathbb{R}_{\geq 0} \to \mathbb{C})$—not a space of bounded continuous functions, since ψ may grow polynomially or exponentially on $[0, \infty)$—with the Hausdorff product topology of pointwise convergence inherited from $\mathbb{R}_{\geq 0} \to \mathbb{C}$. Because $\dim_{\mathbb{C}} \mathcal{W}(\psi) = n < \infty$ and all Hausdorff vector topologies on an n-dimensional $\mathbb{C}$-vector space coincide (`isNormableSpace_of_finiteDimensional`), $\mathcal{W}(\psi)$ admits a compatible norm whose topology equals pointwise convergence and is automatically complete (`FiniteDimensional.complete`), making $\mathcal{W}(\psi)$ a complex Banach space without any growth or smoothness hypothesis on ψ.

2. Non-negative shifts $(T_t g)(u) = g(u+t)$ $(t \geq 0)$ map $\mathcal{W}(\psi)$ into itself because shifting a translate $u \mapsto \psi(u+s)$ gives $u \mapsto \psi(u + (s+t)) \in \mathcal{W}(\psi)$, requiring no derivatives of ψ. Continuity of ψ on $[0, \infty)$ implies that $t \mapsto (T_t g)(u)$ is continuous at each point $u \geq 0$ on every spanning translate, hence norm-continuous on $\mathcal{W}(\psi)$, and therefore $t \mapsto T_t \in \mathcal{L}(\mathcal{W}(\psi))$ is continuous in operator norm on $[0, \infty)$ (evaluating a linear map on a finite basis of $\mathcal{W}(\psi)$).

3. Following Nagumo [52] (see Engel and Nagel [22, Thm. I.3.7]; cf. Anselone and Korevaar's matrix-semigroup proof on $\mathbb{R}$ [3, p. 750]), in any real Banach algebra $\mathcal{A}$ (here $\mathcal{L}(\mathcal{W}(\psi))$), continuity of $t \mapsto T_t$ at 0 with $T_0 = I$ yields $\delta > 0$ such that the Bochner integral $I_\delta = \int_0^\delta T_u \, du$

satisfies $\|I_\delta - \delta I\| < \delta$, making I_δ invertible in $\mathcal{A}$ by Neumann series (`Semigroup.lean`). For $t \geq 0$, changing variables in $T_t I_\delta = \int_t^{t+\delta} T_u\, du$ shows by the fundamental theorem of calculus that $t \mapsto T_t$ is right-differentiable on $[0,\infty)$ with generator $A = (T_\delta - I)I_\delta^{-1}$ commuting with T_t; differentiating $t \mapsto T_t \exp(-tA)$ gives 0, so $T_t = \exp(tA)$ for all $t \geq 0$ and $\psi(t) = (T_t\psi|_{\mathbb{R}_{\geq 0}})(0) = \mathrm{eval}_0(\exp(tA)\psi|_{\mathbb{R}_{\geq 0}})$ is automatically C^∞ on $[0,\infty)$.

4. Over the algebraically closed field $\mathbb{C}$, $\mathcal{W}(\psi) = \bigoplus_{\mu\in S} \ker((A-\mu I)^n)$ decomposes into generalized eigenspaces. On a summand where $(A-\mu I)^n v = 0$, writing $tA = \mu t I + t(A-\mu I)$ as a sum of commuting operators gives $\exp(tA)v = e^{\mu t}\sum_{q=0}^{n-1} \frac{t^q}{q!}(A-\mu I)^q v$, because all powers $q \geq n$ vanish. Evaluating at 0 turns each vector $(A-\mu I)^q v$ into a complex scalar, yielding $\psi(t) = \sum_{\mu\in S} P_\mu(t)e^{\mu t}$ on $[0,\infty)$. When $\psi = \varphi_{\mathbb{C}}$ for a real profile φ, averaging $\varphi(t) = \frac{1}{2}(\varphi_{\mathbb{C}}(t) + \overline{\varphi_{\mathbb{C}}(t)})$ and grouping conjugate rates $(\mu, \bar\mu)$ produces a canonical real representation Γ.

Finally, differentiating $t \mapsto \exp(tA)\psi|_{\mathbb{R}_{\geq 0}}$ q times on $[0,\infty)$ brings down A^q, so $q(D)\psi(t) = \mathrm{eval}_0\big(q(A)\exp(tA)\psi|_{\mathbb{R}_{\geq 0}}\big)$ for every polynomial $q \in \mathbb{C}[X]$. In $d = 2$, the Cayley–Hamilton theorem ($q = \chi_A \implies \chi_A(A) = 0$) immediately implies that the monic characteristic polynomial $\chi_A(X)$ of degree $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ satisfies $\chi_A(D)\psi = 0$ on $[0,\infty)$. Conversely, applying $(D-\mu)^{\deg P_\mu + 1}$ kills $P_\mu(t)e^{\mu t}$ while shifting every other rate $\nu \neq \mu$ by an invertible factor $(\nu-\mu)^{\deg P_\mu+1}$ on leading terms, so $q(D)\big(\sum_{\mu\in S} P_\mu(t)e^{\mu t}\big) = 0$ on $[0,\infty)$ ($P_\mu \neq 0$) if and only if $p_{\min}(X) = \prod_{\mu\in S}(X-\mu)^{\deg P_\mu+1}$ divides $q(X)$. Hence $p_{\min} \mid \chi_A$, giving $\sum_{\mu\in S}(\deg P_\mu + 1) = \deg p_{\min} \leq \deg \chi_A = \dim_{\mathbb{C}} \mathcal{W}(\psi)$, matched by Taylor expansion of $P_\mu(s_0+s_1)e^{\mu(s_0+s_1)}$ to prove Theorem 2.7.

3.2 Constructive multi-way upper bounds via Hasse polynomials and roots of unity

Warm-up ($d = 3, k = 1$): extracting a degree-1 radial polynomial as an $[X^1]$ coefficient. Consider $P(s) = p_0 + p_1 s$ on $s = s_0 + s_1 + s_2$. Notice that if we put $U_0(X) = p_1 + (p_0 + p_1 s_0)X$, $U_1(X) = 1 + s_1 X$, and $U_2(X) = 1 + s_2 X$, then the coefficient of X^1 in $U_0(X)U_1(X)U_2(X)$ is

$$[X^1]\big((p_1+(p_0+p_1s_0)X)(1+s_1X)(1+s_2X)\big) = (p_0+p_1s_0)\cdot 1\cdot 1 + p_1 s_1 \cdot 1 + p_1 \cdot 1 \cdot s_2 = P(s_0+s_1+s_2),$$

a product of three univariate polynomials in X whose j-th factor depends *only on* s_j! Since $\deg(U_0U_1U_2) \leq 3$, over $\mathbb{C}$ averaging $\zeta^{-r}U_0(\zeta^r)U_1(\zeta^r)U_2(\zeta^r)$ over the $M_{3,1}^{\mathbb{C}} = 3$ cube roots of unity ($\zeta = e^{2\pi i/3}$) isolates the X^1 term (because X^0, X^2, X^3 shift to X^{-1}, X^1, X^2, none of which is a multiple of 3!), whereas over $\mathbb{R}$ for a nonreal rate $e^{\lambda(s_0+s_1+s_2)}$ ($\mathrm{Im}\,\lambda > 0$), Proposition 2.15 extracts $\mathrm{Re}([X^1]\prod_{j=0}^2 e^{\lambda s_j}U_j(X))$ using $M_{3,1}^{\mathbb{R}} = 2(3-1)(1)+3 = 7$ real 3-way products.

General degree $k = \deg P$ ($d \geq 2$). Using the q-th Hasse derivative $D^{[q]}P(s_0) = \frac{1}{q!}P^{(q)}(s_0)$—simply the q-th Taylor coefficient of P around s_0, defined algebraically without division by $P(s_0+t) = \sum_{q=0}^k (D^{[q]}P)(s_0)\, t^q$—define the *reversed Hasse polynomial* $H_P(s_0;X)$ on mode 0 and the *truncated exponential polynomials* $E_k(s_j;X)$ on modes $j = 1,\dots,d-1$:

$$H_P(s_0;X) = \sum_{q=0}^{k} q!\,(D^{[q]}P)(s_0)X^{k-q},\ E_k(s_j;X) = \sum_{q=0}^{k}\frac{s_j^q}{q!}X^q,\ Q(X) = H_P(s_0;X)\prod_{j=1}^{d-1}E_k(s_j;X). \tag{14}$$

Binomial convolution gives $[X^q]\prod_{j=1}^{d-1}E_k(s_j;X) = (s_1+\cdots+s_{d-1})^q/q!$ for all $0 \leq q \leq k$, so $[X^k]Q(X) = P(s_0+\cdots+s_{d-1})$ with $\deg Q \leq dk$. Three coefficient-extraction schemes then yield all three upper bounds:

1. *Complex rates over* $\mathbb{C}$*:* Put $M_{d,k}^{\mathbb{C}} = (d-1)k+1$ and $\zeta = e^{2\pi i/M_{d,k}^{\mathbb{C}}}$. Because $d \geq 2$, every monomial index $n \in \{0,\dots,dk\}$ satisfies $-M_{d,k}^{\mathbb{C}} < -k \leq n-k \leq (d-1)k < M_{d,k}^{\mathbb{C}}$, so $M_{d,k}^{\mathbb{C}} \mid (n-k) \iff n = k$. Discrete Fourier averaging $[X^k]Q(X) = \frac{1}{M_{d,k}^{\mathbb{C}}} \sum_{r=0}^{M_{d,k}^{\mathbb{C}}-1} \zeta^{-rk} Q(\zeta^r)$ multiplied by $\prod_{j=0}^{d-1} e^{\mu s_j}$ gives $M_{d,k}^{\mathbb{C}} = (d-1)k+1$ complex products.
2. *Real rates over* $\mathbb{R}$ *(*$\operatorname{Im}\lambda = 0$*,* $P \in \mathbb{R}[X]$*):* On modes $1,\dots,d-1$, the real polynomial $A(X) = \prod_{j=1}^{d-1} E_k(s_j;X) \in \mathbb{R}[X]$ has $\deg A \leq (d-1)k < M_{d,k}^{\mathbb{C}}$. Real Lagrange interpolation of $A(X)$ at the $M_{d,k}^{\mathbb{C}}$ distinct real nodes $r = 0,1,\dots,M_{d,k}^{\mathbb{C}}-1 \in \mathbb{R}$ expresses each $[X^q]A(X)$ $(0 \leq q \leq k)$, and hence $P(\sum_j s_j)e^{\lambda \sum_j s_j}$, as a sum of $M_{d,k}^{\mathbb{C}} = (d-1)k+1$ real d-way products.
3. *Nonreal rates over* $\mathbb{R}$ *(*$\operatorname{Im}\lambda > 0$*, Proof of Propositions 2.14 and 2.15):* Put $M = M_{d,k}^{\mathbb{R}} = 2(d-1)k+d = (2k+1)d-2k$ (note $M = d$ when $k = 0$). Writing $\operatorname{Re} w = \frac{1}{2}(\bar{w}+w)$ in (10) and expanding over subsets $S \subseteq \{0,\dots,d-1\}$ of cardinality $m = |S|$ and degree tuples $q \in \{0,\dots,k\}^d$, the identity $(-1)^r = e^{i\pi rM/M}$ produces the geometric phase $\exp(2\pi i r K(S,q)/M)$ with integer frequency

$$K(S,q) = (2k+1)m + \sum_{j\notin S} q_j - \sum_{j\in S} q_j - k \in \mathbb{Z}.$$

Because $0 \leq q_j \leq k$ and $d \geq 2$: if $m = 0$ $(S = \emptyset)$, $-M < -k \leq K(\emptyset,q) = \sum_j q_j - k \leq (d-1)k < M$, so $M \mid K \iff \sum_j q_j = k$; if $m = d$ $(S = \{0,\dots,d-1\})$, $K = M + (k - \sum_j q_j)$, so $M \mid K \iff \sum_j q_j = k$; and if $1 \leq m \leq d-1$,

$$1 \leq (k+1)m - k \leq K(S,q) \leq (k+1)m + (d-1)k \leq 2(d-1)k + d - 1 = M - 1,$$

so $M \nmid K(S,q)$ and $\sum_{r=0}^{M-1} e^{2\pi i r K(S,q)/M} = 0$. Applying (10) to $U_0(X) = e^{\lambda s_0} H_P(s_0;X)$ and $U_j(X) = e^{\lambda s_j} E_k(s_j;X)$ $(j \geq 1)$ yields $M_{d,k}^{\mathbb{R}} = 2(d-1)k+d$ real d-way products.

3.3 Multilinear lower bounds via Alder–Strassen degree peeling

Step 1: From set-theoretic 1-D functions to multilinear dual functionals. Let V be a finite-dimensional $\mathbb{K}$-vector space $(\mathbb{K} \in \{\mathbb{R},\mathbb{C}\})$, let $v : \mathbb{R}_{\geq 0} \to V$ be an evaluation curve whose image spans V $(\operatorname{span}_{\mathbb{K}}(v(\mathbb{R}_{\geq 0})) = V$, verified by checking $\Lambda \circ v = 0 \implies \Lambda = 0$ for all $\Lambda \in V^*)$, and let $\mathcal{M} : V^d \to \mathbb{K}$ be d-multilinear. If the Hankel kernel $(s_0,\dots,s_{d-1}) \mapsto \mathcal{M}(v(s_0),\dots,v(s_{d-1}))$ admits a rank-R decomposition (1) with arbitrary functions $f_{n,j} : \mathbb{R}_{\geq 0} \to \mathbb{K}$, choosing a basis $\{v(t_b)\}_{b=1}^{\dim V}$ of V and defining unique linear functionals $\ell_{n,j} \in V^*$ by $\ell_{n,j}(v(t_b)) = f_{n,j}(t_b)$ yields

$$\mathcal{M}(u^{(0)},\dots,u^{(d-1)}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} \ell_{n,j}(u^{(j)}) \qquad \text{for all } u^{(0)},\dots,u^{(d-1)} \in V, \tag{15}$$

because both sides are d-multilinear maps agreeing on every d-tuple of basis vectors.

Step 2: Geometric intuition on $d = 3, k = 1$ **(**$\|x\|^2\cos(\|x\|^2)$ **requires 7 real terms).** Before stating the general induction, consider $K_3(x) = (x_0^2+x_1^2+x_2^2)\cos(x_0^2+x_1^2+x_2^2)$, where $v(s) = (e^{is}, se^{is}) \in \mathbb{C}^2$ spans $V = \mathbb{C} \oplus \mathbb{C}\varepsilon \cong \mathbb{C}[T]/(T^2)$ $(\varepsilon^2 = 0)$ over $\mathbb{R}$ and (15) represents the real 3-multilinear multiplication map $\mathcal{M}((u_0,u_1),(v_0,v_1),(w_0,w_1)) = \operatorname{Re}(u_1v_0w_0 + u_0v_1w_0 + u_0v_0w_1)$. Why must $R \geq 7$, even though difference quotients of two complex Gaussians have rank 6 for every $h \neq 0$?

- On the nilpotent subspace $E = 0 \oplus \mathbb{C}\varepsilon \cong \mathbb{R}^2$, $\mathcal{M}((0,z_1),(v_0,0),(w_0,0)) = \mathrm{Re}(z_1 v_0 w_0)$ vanishes only if $z_1 = 0$, so the restrictions $\{\ell_{n,0}|_E : 0 \le n < R\}$ must span E^* ($\dim_{\mathbb{R}} E^* = 2$). Picking two functionals (say $n = 0, 1$) forming a basis of E^*, there is a unique linear map $\gamma_0 : \mathbb{C} \to \mathbb{C}$ such that $\psi_0(u_0) := (u_0, \gamma_0(u_0)) \in \ker \ell_{0,0} \cap \ker \ell_{1,0}$ for all $u_0 \in \mathbb{C}$, *killing two terms* ($n = 0, 1$).
- Crucially, because $\varepsilon^2 = 0$ in $\mathbb{C}[T]/(T^2)$, substituting $\psi_0(u_0)$ on mode 0 and testing $(0, v_1) \in E$ on mode 1 gives $\mathcal{M}(\psi_0(u_0),(0,v_1),(w_0,0)) = \mathrm{Re}(u_0 v_1 w_0)$—*completely unaffected* by $\gamma_0(u_0)$! Thus the remaining functionals $\{\ell_{n,1}|_E : n \ge 2\}$ on mode 1 must still span E^*, letting a second map $\psi_1(v_0) = (v_0, \gamma_1(v_0))$ kill *two more terms* ($n = 2, 3$).
- Finally, evaluating $(u_0, v_0, w_1) \mapsto \mathcal{M}(\psi_0(u_0), \psi_1(v_0), (0, w_1)) = \mathrm{Re}(u_0 v_0 w_1)$ uses only the surviving $R - 4$ terms ($n \ge 4$). Since every $\mathbb{R}$-linear functional $\Lambda : \mathbb{C} \to \mathbb{R}$ has a nonzero kernel vector ($\dim_{\mathbb{R}} \mathbb{C} = 2 > 1$), induction on d shows $\mathrm{Re}(u_0 v_0 w_1)$ requires at least 3 real terms, forcing $R - 4 \ge 3$, i.e. $R \ge 7$.

Step 3: General multi-mode degree peeling over $\mathbb{C}$ and over $\mathbb{R}$. Over $\mathbb{C}$, let $V = \prod_{i\in\iota} \mathbb{C}^{\tau_i^0}$ with $\tau_i^0 = \deg P_i + 1$ and $v(s)_{i,q} = \frac{s^q}{q!} e^{\mu_i s}$. Exponential-polynomial uniqueness implies $\mathrm{span}_{\mathbb{C}}(v(\mathbb{R}_{\ge 0})) = V$, mapping the Hankel kernel to the truncated-convolution form with cutoffs $\tau_i \le \tau_i^0$:

$$\mathcal{T}_\tau(u^{(0)}, \dots, u^{(d-1)}) = \sum_{i\in\iota} \sum_{q_0+\cdots+q_{d-1}<\tau_i} w_i(q_0 + \cdots + q_{d-1}) \prod_{j=0}^{d-1} u_{i,q_j}^{(j)}, \qquad w_i(\tau_i - 1) \ne 0 \ (\tau_i > 0).$$

Strong induction on $\sum_i \tau_i$ peels one block i_0 at a time:

- When $\tau_{i_0} = 1$, testing $(e_{i_0,0}, \dots, e_{i_0,0})$ finds $\ell_{n_0,0}(e_{i_0,0}) \ne 0$; projecting block i_0 to 0 on modes $j \ge 1$ and subtracting $\frac{\ell_{n_0,0}(\pi_{\ne i_0}(x))}{\ell_{n_0,0}(e_{i_0,0})} e_{i_0,0}$ on mode 0 kills 1 term and reduces τ_{i_0} to 0.
- When $\tau_{i_0} \ge 2$, since $2(\tau_{i_0} - 1) \ge \tau_{i_0}$, rank-1 perturbations along $e_{i_0,\tau_{i_0}-1}$ on modes $j = 0, \dots, d-2$ do not cross-interact when another coordinate equals $e_{i_0,\tau_{i_0}-1}$, killing $d - 1$ distinct terms; truncating modes $0, \dots, d-2$ to $q < \tau_{i_0} - 1$ and shifting mode $d-1$ up by $+1$ ($e_{i_0,q} \mapsto e_{i_0,q+1}$) reduces $\tau_{i_0} \mapsto \tau_{i_0} - 1$ with shifted top weight $w'_{i_0}(\tau_{i_0} - 2) = w_{i_0}(\tau_{i_0} - 1) \ne 0$.

Over $\mathbb{R}$, the same induction on $G = (\prod_{i\in\iota_{\mathbb{R}}} \mathbb{R}^{\tau^0_{\mathbb{R},i}}) \times (\prod_{u\in\iota_{\mathbb{C}}} \mathbb{C}^{\tau^0_{\mathbb{C},u}})$ peels real blocks identically (1 term at $\tau_{\mathbb{R},i_0} = 1$, $d - 1$ terms at $\tau_{\mathbb{R},i_0} \ge 2$), peels nonreal $\tau_{\mathbb{C},u_0} = 1$ blocks by d terms (using $\dim_{\mathbb{R}} \ker(\Lambda : \mathbb{C} \to \mathbb{R}) \ge 1$ to kill 2 terms on mode 0 and 1 term on each mode $1, \dots, d-2$), and peels nonreal $\tau_{\mathbb{C},u_0} \ge 2$ blocks by $2(d-1)$ terms (2 independent real functionals on $0 \oplus \mathbb{C}e_{u_0,\tau_{\mathbb{C},u_0}-1} \cong \mathbb{R}^2$ per mode $j = 0, \dots, d-2$, followed by $+1$ index shifting on mode $d - 1$), completing Theorem 2.6.

3.4 Proofs of border-rank bounds, adjugate obstruction, and universal approximation (Theorems 2.9–2.12)

Identity (6) is immediate from (5): for $d \ge 2$, $(k+1) + (d-2)k = (d-1)k + 1$ and $d(k+1) + (d-2)k = 2(d-1)k + d$. For a single well-formed component (λ, P) of degree $k = \deg P$ and any $h \in \mathbb{R} \setminus \{0\}$, replace $s = \sum_{j=0}^{d-1} x_j^2$ inside $P(s)$ by the exponential difference quotient $u_h(s) = \frac{e^{hs}-1}{h}$ (setting $u_0(s) := 0$ via $0^{-1} = 0$ in Lean):

$$F_h^{(\lambda,P)}(x) := \mathrm{Re}\Big(P\big(u_h(\|x\|^2)\big)\, e^{\lambda\|x\|^2}\Big).$$

Because $u_h(s) = L_h(e^{hs})$ is a degree-1 polynomial $L_h(Y) = -h^{-1} + h^{-1}Y$ in $Y = e^{hs}$, the composed polynomial $Q_h(Y) = (P \circ L_h)(Y)$ still has degree at most k:

$$P(u_h(s))\, e^{\lambda s} = \sum_{m=0}^{k} ([Y^m]Q_h)\, (e^{hs})^m e^{\lambda s} = \sum_{m=0}^{k} ([Y^m]Q_h) \prod_{j=0}^{d-1} e^{(\lambda+mh)x_j^2}.$$

When $\operatorname{Im}\lambda = 0$ (so $P, Q_h \in \mathbb{R}[X]$), this is a sum of $k+1 = b_d(\lambda, P)$ real Gaussian products; when $\operatorname{Im}\lambda > 0$, it is a sum of $k+1$ complex Gaussian products, whose real part has real d-way rank at most $d(k+1) = b_d(\lambda, P)$ by Proposition 2.14. (At $k = 1$ and $\lambda = 0$, $u_h(s) = h^{-1}(\prod_{j=0}^{d-1} e^{hx_j^2} - 1)$ realizes the classical W-tensor degeneration $[X^1]\prod_{j=0}^{d-1}(1 + s_j X)$ of exact rank d and border rank 2 [8, 14, 79] directly by smooth radial Gaussians.) Summing over $(\lambda_u, P_u) \in \Gamma$ gives $\operatorname{srank}_{\mathbb{R}}(F_h) \le B_d(\Gamma)$ for all $h \in \mathbb{R}$. For any compact $C \subset \mathbb{R}^d$, let $B = \sup_{x\in C}\|x\|^2 < \infty$; twice applying the mean-value inequality gives $|e^y - 1 - y| \le y^2 e^{|y|}$, so $|u_h(s) - s| \le |h|(B+1)^2 e^{B+1} \to 0$ uniformly on $[0, B]$, and uniform continuity of $(u, s) \mapsto \operatorname{Re}(P(u)e^{\lambda s})$ on $[-1, B+1]\times[0, B]$ yields uniform convergence on C (and likewise over $\mathbb{C}$ with rank $\le \sum_{\mu\in S}(\deg P_\mu + 1)$).

For Theorem 2.10(1), if $\operatorname{srank}_{\mathbb{K}}(K) > r$ on $X_0 \times X_1$, induction on $k \le r+1$ via the Schur complement $\det\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) = (\det A)(D - CA^{-1}B)$ constructs an $(r+1)\times(r+1)$ evaluation submatrix $M(K)$ with $\det M(K) \neq 0$, whereas $\det M(F_\iota) = 0$ whenever $\operatorname{srank}_{\mathbb{K}}(F_\iota) \le r$ since $M(F_\iota) = UV$ factors through $\mathbb{K}^r$; when every $\operatorname{Im}\lambda_u = 0$, $b_d(\lambda_u, P_u) = \deg P_u + 1 = c_2(\lambda_u, P_u)$ gives $B_d(\Gamma) = C_2(\Gamma)$ immediately. For Theorem 2.10(2), if $G : X^3 \to \mathbb{R}$ has $\operatorname{srank}_{\mathbb{R}}(G) \le n$, then for any $u_0, u_1 \in X$ and $p, q \in X^n$, its mode-0 slices $S_0 = V\operatorname{diag}(d_0)W$ and $S_1 = V\operatorname{diag}(d_1)W$ satisfy the global polynomial sum-of-squares identity (requiring no invertibility assumptions)

$$\operatorname{tr}\Big(\big(\operatorname{adj}(S_0)\, S_1\big)^2\Big) = (\det V\ \det W)^2 \sum_{k=0}^{n-1}\Big(d_{1,k}\prod_{j\neq k} d_{0,j}\Big)^2 \ge 0.$$

Evaluating $\cos(\|x\|^2)$ at $p = q = (0, \sqrt{\pi/2})$ ($u_0 = 0, u_1 = \sqrt{\pi/2}$) gives $\operatorname{tr}((\operatorname{adj}(S_0)S_1)^2) = -2 < 0$, ruling out rank ≤ 2; evaluating $\|x\|^2\cos(\|x\|^2)$ at $p = q = (0, \sqrt{\pi/2}, \sqrt{\pi}, \sqrt{3\pi/2})$ ($u_0 = 0, u_1 = \sqrt{\pi/2}$) gives $\operatorname{tr}((\operatorname{adj}(S_0)S_1)^2) = -1024(\pi/2)^8 < 0$, ruling out rank ≤ 4 (compare real eigenvalue obstructions on $n\times n\times n$ slices in [20, 45]). Conversely, expanding the 5-term real family $F_h(x)$ of (7) in powers of h: the $-\frac{4}{3h}\prod_j(p_{1,j} + hq_{1,j}) + \frac{4}{3h}\prod_j(p_{2,j} + hq_{2,j})$ pair supplies two of the three modes of Example 2.17 applied to $\frac{1}{h}\operatorname{Re}\prod_j((c_j + is_j)(1 + hx_j^2))$, leaving at $O(h^{-1})$ only $-\frac{1}{3h}c_0c_1c_2 + \frac{1}{h}(c_0s_1s_2 + s_0c_1s_2 + s_0s_1c_2)$, which are cancelled respectively by $\frac{1}{3h}\prod_j(c_j + 4hx_j^2c_j)$ (contributing $+\frac{1}{3h}c_0c_1c_2 + \frac{4}{3}(x_0^2 + x_1^2 + x_2^2)c_0c_1c_2 + O(h)$) and by the 2-term W-degeneration $\frac{1}{h^3}(\prod_j(s_j + h^2c_j) - \prod_j s_j) = \frac{1}{h}(c_0s_1s_2 + s_0c_1s_2 + s_0s_1c_2) + O(h)$, leaving $F_h(x) - \|x\|^2\cos(\|x\|^2) = h\operatorname{Rem}(x, h)$ with continuous Rem, hence uniform convergence on compact sets.

Finally, for Theorem 2.12, extend ψ by $\psi(\max(s, 0))$, pick $S_0 \ge 1$ so $|\psi(s)| < \varepsilon/4$ for $s \ge S_0$, set $M_{\mathrm{b}} = \max(\sup_{u\in[0,S_0+1]} e^{\alpha u}|\psi(u)|, |\psi(0)|) + 1$, and choose $T > S_0 + 3$ so $e^{-\alpha(T-1)}M_{\mathrm{b}} < \varepsilon/4$. The continuous clamped function $g(u) = \max(0, \min(1, S_0 + 1 - u))\, e^{\alpha u}\psi(u) + \max(0, \min(1, u - (T-1)))\,\psi(0)$ satisfies $g(0) = g(T) = \psi(0)$, lifting to $G \in C(\mathbb{R}/T\mathbb{Z}, \mathbb{C})$. By Fourier density on $\mathbb{R}/T\mathbb{Z}$ (`span_fourier_closure_eq_top`), a trigonometric polynomial P_{trig} approximates G within $\varepsilon/4$ uniformly; folding $\pm m$ harmonics via $\operatorname{Re}(c_{-m}e^{-im\omega_0 s}) = \operatorname{Re}(\bar{c}_{-m}e^{im\omega_0 s})$ ($\omega_0 = 2\pi/T$) and multiplying by $e^{-\alpha s} \le 1$ yields (8) with $\operatorname{srank}_{\mathbb{R}}(K_M) \le 1 + dM$ by Proposition 2.14 and global error $< \varepsilon/4 + 2(\varepsilon/4) < \varepsilon$ across all $x \in \mathbb{R}^d$.

4 Formalization Architecture and Verification Audit

Table 6 breaks down the `KernelRankLean` library in Lean 4 [50, 48] (6,935 physical lines, 6,586 nonblank lines; earlier commits comprised 5,750 lines at `94a5b4a`, 5,578 lines at `1094bf1`, 5,324

Table 6: Module architecture and verified source line counts of the `KernelRankLean` library.

Module(s)	Mathematical content	Phys.	Nonbl.
`Defs`, `KernelRankLean`	Multi-way rank, surjections, slices, exact & border costs	254	216
`RowSpan`, `Complexification`	2-way row-span rank; $\mathbb{R} \to \mathbb{C}$ invariance	221	189
`Translations`	d-D radial-to-Hankel reduction; `translateSpan`	92	80
`ExpPoly`, `RealExpPoly`	Exponential-polynomial uniqueness	198	170
`Semigroup`, `TranslationSemigroup`	Banach semigroups; half-line shift semigroup	262	219
`Necessity`, `Representation`	Generalized eigenvectors; canonical conjugate pairing	222	198
`Differential`, `ComplexRank`	Half-line operators $q(D)$; 2-D complex rank	318	284
`Annihilator`	Minimal constant-coefficient differential annihilator	69	62
`MultiComplexRank`	Exact d-way complex rank $\sum_\mu((d-1)\deg P_\mu + 1)$	1,015	978
`MultiRealRank`	Exact d-way real rank (`multiTotalCost`)	2,562	2,521
`Main`	Unified d-D/2-D theorems & multi-Gaussian border bound	538	516
`Approximation`	Exact complex/real border ranks & harmonic approx.	1,184	1,153
Total (`KernelRankLean`)		**6,935**	**6,586**

lines at `9722ff5`, 3,822 lines at `6aabb4d` for d-D complex and multi-Gaussian ranks, 1,587 lines at `cae13d6` for 2-D plus differential annihilators, and 1,340 lines at `240dea0` for 2-D classification alone), and Table 7 maps every numbered mathematical statement to its exported Lean 4 identifier. Whereas prior formalized tensor-rank developments establish deep-network CP-rank bounds via generalized matricization in Isabelle/HOL [4] or finite-dimensional octonion and $\mathbb{F}_2$ matrix-multiplication rank bounds in Lean 4 [35, 76], our library couples continuous-domain Banach-algebra analysis with exact arbitrary-dimension real and complex multilinear rank formulas. Four proof-engineering challenges and solutions proved decisive:

1. **Dependent multi-way definitions with homogeneous two-coordinate specialization:** Defining `IsSumOfProducts` and `separableRank` over a dependent family $\prod_{i\in\iota} X_i$ unifies the 2-D and d-D theorems, while specializing `rowSpan` in `RowSpan.lean` to homogeneous $\operatorname{Fin} 2 \to \alpha$ (via a pattern-matched helper `fin2Vec`) avoids 200,000-heartbeat higher-order unification timeouts on $\operatorname{Fin} 2 \to \mathbb{R}_{\geq 0}$.

2. **Explicit uniformity selection on `translateSpan`:** Transferring a norm from $\mathbb{C}^n$ onto $\mathcal{W}(\psi) \leq (\mathbb{R}_{\geq 0} \to \mathbb{C})$ creates a second `UniformSpace` instance alongside the subtype uniformity; explicitly installing the norm-induced uniformity as the local instance prevents typeclass diamond timeouts while letting finite-dimensional linear maps inherit automatic continuity.

3. **Cayley–Hamilton divisibility in place of cyclic vectors:** Bounding $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ from below via divisibility of $\chi_A(X)$ by $\prod_\mu (X-\mu)^{\deg P_\mu + 1}$ eliminated coefficient-synthesis operators and cyclic-vector constructions, shrinking the 2-D proof plus differential-annihilator extension by 93 lines while yielding Theorem 2.7 directly.

4. **Decidable equality hygiene and basis-free adjugate obstructions:** In d-way product manipulations (`MultiRealRank.lean`), invoking local `let := Classical.decEq` rather than global `classical` avoids shadowing $\operatorname{Fin} d$'s canonical decidable equality inside `Finset.mul_prod_erase`; in `Approximation.lean`, formulating 3-D border-rank lower bounds via the everywhere-defined adjugate trace polynomial $\operatorname{tr}((\operatorname{adj}(S_0)S_1)^2) \geq 0$ avoids generic slice-invertibility hypotheses along limiting sequences.

Comparison with formalized tensor-rank developments and reusable components. Recent formalizations of tensor rank address complementary settings: Bentkamp, Blanchette, and Klakow [4] formalize finite-dimensional CP-rank and generalized matricization bounds over $\mathbb{R}$ in Isabelle/HOL to separate shallow and deep convolutional arithmetic circuits; Jain [35]

Table 7: Audit map from paper theorems and propositions to exported Lean 4 identifiers in `KernelRankLean` (all 32 exported theorems checked for standard classical axioms in `scripts/check`).

Paper result	Exported Lean 4 theorem identifier(s) (`KernelRank.*`)	File(s)	Loc
Eq. (2)–(4)	`separableRank_comp_surjective`, `separableRank_padTwo_le`	`Defs.lean`	239
Sec. 3.1 (2-D core)	`separableRank_eq_finrank_rowSpan`, `separableRank_ofReal`	`RowSpan`, `Complexification`	221
Sec. 3.1 (semigroup)	`exists_eq_exp_of_semigroup`, `exists_translation_generator`	`Semigroup`, `TranslationSemigroup`	262
Thm. 2.4	`separableRank_radial_lt_top_iff_exists`	`Main.lean`	538
Thm. 2.5	`separableRank_radial_complex_eq_iff`, `separableRank_sum_expPoly`	`Main`, `MultiComplexRank`	1,015
Thm. 2.6	`separableRank_radial_eq_iff`, `separableRank_radial_two_eq_iff`, `separableRank_radial_gaussian_eq_iff`	`Main`, `MultiRealRank`	2,562
Thm. 2.7	`exists_minimal_differential_annihilator`	`Annihilator.lean`	69
Thm. 2.9	`multiTotalCost_eq_borderTotalCost_add`, `exists_rank_le_borderTotalCost_tendstoUniformlyOn_radial_profile`	`Defs`, `Main`	538
Thm. 2.10(1, $\mathbb{R}$)	`separableRank_fin2_le_of_tendsto`, `totalCost_le_of_tendsto_radial_profile`, `borderTotalCost_le_of_tendsto_radial_real_profile`	`Approximation.lean`	1,184
Thm. 2.10(1, $\mathbb{C}$)	`sum_natDegree_add_one_le_of_tendsto_radial_complex`, `exists_rank_le_sum_natDegree_add_one_tendstoUniformlyOn_radial_complex`	`Approximation.lean`	1,184
Thm. 2.10(2, $\mathbb{R}^3$)	`not_exists_rank_le_two_tendsto_radial_cos`, `not_exists_rank_le_four_tendsto_radial_sq_mul_cos`, `exists_rank_le_five_tendstoUniformlyOn_radial_sq_mul_cos`	`Approximation.lean`	1,184
Thm. 2.12	`exists_harmonic_gaussian_uniform_approx`	`Approximation.lean`	1,184
Props. 2.14–2.15	`re_prod_eq_sum_re`, `re_coeff_prod_eq_sum_re`	`MultiRealRank.lean`	2,562

formalizes slice-peeling lower bounds (≥ 18, via a two-slice real pencil bound analogous to the $d = 3, k = 0$ base case of `cplxTruncForm_rank_ge`) and the Krawczyk contraction theorem in Lean 4 (checking the concrete rational certificate inequalities for rank ≤ 25 via external scripts) on the $8 \times 8 \times 8$ real octonion multiplication tensor; and Yang et al. [76] verify $R_{\mathbb{F}_2} \geq 21$ for 3×3 matrix multiplication over $\mathbb{F}_2$ via finite quotient-rank certificates in Lean 4. By contrast, `KernelRankLean` formalizes *functional* CP rank on infinite-dimensional domains $\mathbb{R}^d$ and proves exact parametric formulas for all dimensions $d \geq 2$ and degrees $k \geq 0$. Roughly two-thirds of the codebase ($\approx$ 4,400 of 6,935 lines) is general-purpose infrastructure reusable outside radial kernel theory:

- *Reusable general-purpose modules:* `Defs`, `RowSpan`, `Complexification`, and the lower-semicontinuity / adjugate-trace sections of `Approximation` (coordinatewise `separableRank` on arbitrary finite Pi types, surjection invariance, 2-way row-span rank and pointwise border-rank closure, and 3-way polynomial adjugate border-rank obstructions); `Semigroup` ($M(t) = \exp(tA)$ in any real Banach algebra, 96 lines via Mathlib's Bochner integral and semilinear analysis [21]); and the algebraic cores of `MultiComplexRank` and `MultiRealRank` (Propositions 2.14–2.15, the dual-span function-to-multilinear bridge `exists_linear_factors_of_span_eq_top`, and Alder–Strassen truncated-convolution / local-algebra rank lower bounds over $\mathbb{C}$ and $\mathbb{R}$).
- *Problem-specific radial-kernel glue:* `Translations`, `TranslationSemigroup`, `ExpPoly`, `RealExpPoly`, `Necessity`, `Representation`, `Differential`, `Annihilator`, `Main`, and the radial specializations of `Approximation` (the $s_j = x_j^2$ reduction, half-line shift semigroup on `translateSpan`, canonical upper-half-plane conjugate pairing, 5-term and multi-Gaussian border limits, and harmonic Gaussian universal approximation).

Classical axioms, constructive witnesses, and trustworthiness audit. In Lean 4, `separableRank` is marked `noncomputable` because taking the infimum in Definition 2.1 over all univariate factor families $X_i \to \mathbb{K}$ has no general decision procedure, and Mathlib's real numbers are classical Cauchy-sequence quotients. Nevertheless, all upper-bound and border-rank proofs in `MultiComplexRank.lean`, `MultiRealRank.lean`, `Main.lean`, and `Approximation.lean` construct *explicit closed-form* univariate factor formulas (Propositions 2.14–2.15, Remark 2.11, and Section 3.4), evaluated directly in `Applications/acoustics_3d.py`. Across all 32 audited top-level theorems, `#print axioms` reports *exclusively* Lean 4 and Mathlib's three standard classical axioms:

1. `propext` (*propositional extensionality*, $(P \leftrightarrow Q) \to P = Q$);
2. `Quot.sound` (*quotient soundness*, $r\,a\,b \to \text{Quot.mk}\,r\,a = \text{Quot.mk}\,r\,b$, underlying Cauchy-completion $\mathbb{R}$ and quotient spaces);
3. `Classical.choice` (*Hilbert's choice operator*, $\text{Nonempty}\,\alpha \to \alpha$, which together with `propext` and `Quot.sound` implies the law of the excluded middle $P \vee \neg P$ via Diaconescu's theorem and enters through $\mathbb{R}/\mathbb{C}$ arithmetic, coordinate surjection right inverses (2), and finite-dimensional basis extraction in $\mathcal{W}(\psi)$ and V).

Calculus of Inductive Constructions extended by {`propext`, `Quot.sound`, `Classical.choice`} is equiconsistent with ZFC set theory with countably many inaccessible cardinals [50, 48]. To audit trustworthiness mechanically, `scripts/check` (wired as the Git pre-commit hook) builds `KernelRankLean` and `Experiments` from source, rejects `sorry` (`sorryAx`), and checks `#print axioms` on all 32 theorems—guaranteeing zero custom `axiom` declarations, zero compiler-trust reflection axioms (`Lean.ofReduceBool` / `native_decide`), and zero `unsafe` blocks—while concrete positive and negative test theorems (`separableRank_radial_sq_mul_cos` $= 3d - 2$ and `not_exists_rank_le_four_...`) confirm that the 239-line specification in `Defs.lean` is non-vacuous. Alongside `Semigroup.lean`, the repository (`Experiments/` and `Isabelle/Semigroup_NSA.thy`) also includes four self-contained finite-dimensional proofs of $M(t) = \exp(tA)$ (206–399 lines) via finite differences and Euler's infinitesimal step in Lean 4 and Isabelle/HOL [24, 33] (audited with zero oracles via `Thm_Deps.all_oracles`).

5 Illustrative Numerical and GPU Case Studies

5.1 Discretization regimes, error metrics, and hardware protocol

Theorems 2.4–2.12 are already established with mathematical certainty by the Lean 4 proofs of Sections 2–4 and require no empirical validation. Rather, because practical filters and PDE solvers operate on finite $[-L, L]^d$ integer grids, in 16- or 32-bit floating-point arithmetic, and alongside competing non-separable algorithms (`cuFFT` and compact FDTD), Sections 5.2–5.4 present reproducible numerical and GPU case studies (outside the Lean 4 formalization, with every GPU kernel output checked against a `float64` reference) illustrating finite-grid discretization effects, finite-precision conditioning, memory footprints, and hardware crossovers, as mapped in Table 8.

Hardware platform, reproducibility, and explicit error-metric definitions. All scripts, auto-tuned Triton/CUDA kernels (`Applications/gpu_kernels.py`), package requirements (`Applications/requirements.txt`), and JSON result archives live in `Applications/` (explicit coefficients in Appendix A). Measurements were collected on Linux `7.0.0-43-generic` (`x86_64`) with an Intel Core Ultra 5 125H CPU (14 cores / 18 threads, 18 MB L3 cache, 61 GiB DDR5) and an external NVIDIA GeForce RTX 4070 Ti Super eGPU (16,376 MiB GDDR6X, 48 MB L2 cache, 672 GB/s peak VRAM bandwidth, connected via Thunderbolt / PCIe

Table 8: Summary of how the numerical and GPU case studies (Sections 5.2–5.4) connect to the continuous-domain theory (Sections 2–3).

Theoretical result	Finite-grid / finite-precision question	Observed behavior in reproducible case studies	Section / ref.
Prop. 2.14 & Thm. 2.6 ($d = 2$)	Discrete SVD rank vs. $C_2(\Gamma)$; steerable pass bound $2(m+1)R$	Sampled SVD rank equals $C_2(\Gamma)$ across all 18 blur families ($20\times$–$40\times$ tap reduction on 161^2 gradient pair)	Secs. 5.2, 5.3; Tab. 9
Sec. 2.4 & $\mathrm{GL}_R(\mathbb{R})$	Does magnitude reordering or `fp32` outer sum suffice vs. $\mathrm{GL}_R(\mathbb{R})$?	Reordering (8.18/255) and `fp32` outer sum (6.76/255) leave intra-channel `fp16` rounding intact on $6c$; $\mathrm{GL}_R(\mathbb{R})$ cuts error to 0.30–0.31/255	Sec. 5.3; Tab. 11, Fig. 2c
Thm. 2.4 (srank $< \infty$ iff exp.-poly)	When do discrete rank-R stencils approach continuous profiles?	Converges onto $1c/2c/4c/6c$ once $R_{\mathrm{px}} \geq 16$ (warm-starts beat cold SVD by $1.3\times$–$2.2\times$); discrete error hits 0 when $L{+}1 \leq R$	Sec. 5.3; Tab. 10, Figs. 3–4
Props. 2.14–2.15 ($d = 3$: $4 \to 3$, $9 \to 7$)	Wall-clock and memory trade-offs vs. naive split, FDTD, and `cuFFT`	$< 10^{-15}$ stencil match; 24.0% GPU ($1.98 \to 1.51$ ms, +28 MiB scratch) and 21.2% CPU savings; $52\times$–$71\times$ accuracy-matched gain vs. 25-pt FDTD	Sec. 5.4; Tab. 13, Fig. 5
Thm. 2.6 (C_d linear in weights)	Variable $c(\mathbf{x})$ via shared rates and local weights $\gamma_u(\nu(\mathbf{x}))$	$\leq 0.185\%$ dispersion across $\nu \in [0.35, 0.80]$ and 7.8% 3-D lens error where 1-step `cuFFT` is inapplicable (40% at singular caustic)	Sec. 5.4; Fig. 6
Complexity dRW vs. W^d & FFT	Crossover regimes against tuned W^d Triton, FDTD, and `cuFFT`	Wins at $R{=}4$, $R_{\mathrm{px}} \leq 16$ in 2-D and in `fp16` / $c(\mathbf{x})$ 3-D wave; `cuFFT` and 1-pass FDTD win outside those regimes	Secs. 5.3–5.4; Tabs. 12, 13

Gen3 $\times 4$, driver `615.71.09`), running Python `3.14.4`, NumPy `2.5.4`, SciPy `1.18.1` (HiGHS LP), PyTorch `2.15.0` (CUDA `13.2`), and Triton `3.9.0`. All GPU timings use on-device `torch.cuda.Event` barriers around VRAM-resident tensors (excluding host-to-eGPU PCIe $\times 4$ transfers). Six quantitative error metrics govern the evaluations below:

1. *2-D Chebyshev pass/stopband ripple:* $\delta(\varphi) := \max_{r \in [0,1] \cup [1+\Delta r,\, \sqrt{2}(1+\Delta r)]} |\varphi(r^2) - \mathbf{1}_{r \leq 1}|$ ($\Delta r = 0.2$, evaluated on 2,000 radial nodes in continuous LP design and on all discrete grid points $\|\mathbf{m}\|_2 / R_{\mathrm{px}}$ of $\{-L, \ldots, L\}^2$, $L = \lceil 1.2\, R_{\mathrm{px}} \rceil$).
2. *Equal-radius discrete shell anisotropy:* $\Delta_{\mathrm{sh}}(K) := \max_{N \leq 2L^2,\, |S_N| > 1} \big(\max_{\mathbf{m} \in S_N} K[\mathbf{m}] - \min_{\mathbf{m} \in S_N} K[\mathbf{m}]\big)$ over Euclidean lattice shells $S_N = \{\mathbf{m} \in \{-L, \ldots, L\}^2 : m_x^2 + m_y^2 = N\}$.
3. *IEEE `float16` 8-bit sRGB and linear-light bokeh errors:* Given a unit-DC kernel ($\sum_{\mathbf{m}} K[\mathbf{m}] = 1$) on 512×512 scene-linear HDR images $I \geq 0$, we compare the `float16`-intermediate/accumulator output $\hat{I}_{16}$ against the exact `float64` output I_{64} via $E_{\mathrm{sRGB}} = 255 \max_{x,y} |\mathrm{sRGB}(\hat{I}_{16}^{[0,1]}) - \mathrm{sRGB}(I_{64}^{[0,1]})|$ (clamping to $[0, 1]$ before the exact piecewise IEC 61966-2-1 transfer function) and $E_{\mathrm{lin}} = 255 \max_{x,y} |\hat{I}_{16} - I_{64}|$, across 20 random glint scenes (`seed=0..19`) and glint peaks $8\times, 48\times, 256\times$.
4. *3-D Brillouin-octant phase-velocity error and effective bandwidth:* For discrete symbol $\widehat{M}(\mathbf{k}) = \sum_{\mathbf{m} \in [-L,L]^3} M[\mathbf{m}] e^{-i\mathbf{k}\cdot\mathbf{m}}$ at Courant number $\nu = c_0 \Delta t / \Delta x$, $|\delta c / c_0| := \max_{0 < \|\mathbf{k}\|_2 \leq k_c} \big|\arccos(\widehat{M}(\mathbf{k})/2)/(\nu \|\mathbf{k}\|_2) - 1\big|$, with effective τ-bandwidth $k_\tau := \sup\{\kappa \in (0, \pi] : \max_{\|\mathbf{k}\|_2 \leq \kappa} |c_{\mathrm{num}}(\mathbf{k})/c_0 - 1| \leq \tau\}$ ($\tau \in \{0.1\%, 0.5\%\}$) and von Neumann stability $|\widehat{M}(\mathbf{k})| \leq 2$ audited across 232,620 symmetry-reduced octant/shell wavevectors plus multi-start Nelder–Mead refinement.
5. *3-D spherical-pulse relative L^2 waveform error:* $\|p_{\mathrm{num}}(\mathbf{x}_r, \cdot) - p_{\mathrm{ref}}(\mathbf{x}_r, \cdot)\|_{\ell^2} / \|p_{\mathrm{ref}}(\mathbf{x}_r, \cdot)\|_{\ell^2}$ (where $\|g\|_{\ell^2} = (\sum_n |g(t_n)|^2)^{1/2}$ over all time steps) across axial $(48, 0, 0)$, face-diagonal $(34, 34, 0)$, and body-diagonal $(28, 28, 28)$ receivers on 256^3 against the exact analytical 3-D D'Alembert solution (and on 128^3 lens refraction against a 256^3 fourth-order spectral Lax–Wendroff PDE reference).
6. *Implementation verification and run-to-run timing variability:* Every GPU benchmark run asserts numerical agreement of its output tensor against a `float64` reference with identical

boundary rules (`err_max_*`, `err_rms_*`: max `float32` error $\le 6.1 \times 10^{-6}$ in 2-D and $\le 6.6 \times 10^{-6}$ in 3-D) and reports mean $\pm$ std across 10 CUDA-Event batches (6 batches on `DirectStep3D`). Startup autotuning and VRAM buffer alignment shift means by up to $\approx 6\%$ in 2-D and $\approx 3\%$ in 3-D between full runs, so few-percent differences are within empirical noise.

Before turning to discrete designs, note a continuous-domain Fourier duality: for $\operatorname{Re}\lambda < 0$, differentiating $\mathcal{F}_{\mathbb{R}^d}[e^{\lambda\|x\|^2}](\xi) = (\pi/(-\lambda))^{d/2}\exp(\|\xi\|^2/(4\lambda))$ k times with respect to λ maps $P(\|x\|^2)e^{\lambda\|x\|^2}$ $(\deg P = k)$ to $\widetilde{P}(\|\xi\|^2)e^{\|\xi\|^2/(4\lambda)}$ with $\deg\widetilde{P} = k$ and $\operatorname{Re}(1/(4\lambda)) < 0$. By Theorems 2.5 and 2.6, the d-dimensional Fourier transform preserves exact complex and real d-way separable rank across all integrable radial exponential-polynomial kernels.

5.2 Harmonic "beach-ball" differential operators and steerable kernels

Complex partial-derivative operators $(\partial_x + i\partial_y)^m$ provide a canonical rotation-equivariant basis for 2-D symmetry and steerable filters [26, 38, 7, 62]. Specializing Proposition 2.14 to equal symbolic factors $z_0 = \cdots = z_{m-1} = \partial_x + i\partial_y$ (writing $m \ge 1$ for the harmonic order and $D_\theta = \cos\theta\,\partial_x + \sin\theta\,\partial_y$ for the directional derivative along angle θ) gives, for *every* integer $m \ge 1$ (both odd and even),

$$\operatorname{Re}\big((\partial_x + i\partial_y)^m\big) = \frac{2^{m-1}}{m}\sum_{r=0}^{m-1}(-1)^r D^m_{\pi r/m}. \tag{16}$$

On a regular $2m$-point ring $\theta_n = \pi n/m$ $(n = 0,\dots,2m-1)$ of radius $h \to 0$, antipodal points satisfy $D_{\theta_{r+m}} = -D_{\theta_r}$ while Fourier orthogonality of $(-1)^r = e^{i\pi r}$ annihilates all Taylor orders below m. For $f \in C^m$ near (x,y), (16) recovers—by elementary geometric series rather than Cauchy residues and Chebyshev partial fractions—both the odd-order $(m = 2N+1)$ hexagonal/decagonal "beach-ball" operators of Niemitalo [56] and the even-order operators of Perona [60] in a single formula:

$$\lim_{h\to 0}\frac{1}{h^m}\sum_{n=0}^{2m-1}(-1)^n f\Big(x + h\cos\frac{\pi n}{m},\, y + h\sin\frac{\pi n}{m}\Big) = \frac{1}{(m-1)!\,2^{m-2}}\operatorname{Re}\big((\partial_x + i\partial_y)^m\big) f(x,y),$$

with a half-sector ring rotation $+\frac{\pi}{2m}$ yielding $+\operatorname{Im}((\partial_x + i\partial_y)^m)$. Recognizing $\partial_{\bar z} = \frac{1}{2}(\partial_x + i\partial_y)$ as the Cauchy–Riemann operator also explains why applying $\partial^m_{\bar z}$ to *any* smooth radial kernel $\varphi(x^2+y^2) = \varphi(z\bar z)$ produces no lower-order radial derivatives [7, 62]: since $\partial_{\bar z}(z) = 0$, induction gives

$$\left(\frac{\partial_x + i\partial_y}{2}\right)^m \varphi(x^2+y^2) = (x+iy)^m\,\varphi^{(m)}(x^2+y^2) = r^m e^{im\theta}\,\varphi^{(m)}(r^2). \tag{17}$$

When $\varphi(s)$ is a well-formed multi-Gaussian profile with n_{re} real and n_{cx} nonreal rates (all $\lambda_u \ne 0$, approximating, e.g., an ideal circular low-pass filter [57]), $\varphi^{(m)}(s)$ is *again* a well-formed multi-Gaussian profile at the same rates (weighted by λ_u^m) of continuous 2-way rank $R = n_{\mathrm{re}} + 2n_{\mathrm{cx}}$. Analytically differentiating the continuous-space kernel on $\mathbb{R}^2$—avoiding discrete finite-difference distortion up to Nyquist [55]—absorbs the monomials of $\operatorname{Re}((x+iy)^m)$ and $\operatorname{Im}((x+iy)^m)$ into the 1-D Gaussian factors, giving exact *continuous-domain* 2-way real separable ranks on $\mathbb{R}^2$:

$$\big(\lfloor m/2\rfloor + 1\big)(n_{\mathrm{re}} + 2n_{\mathrm{cx}}) \quad \text{for } \cos(m\theta), \qquad \lceil m/2\rceil(n_{\mathrm{re}} + 2n_{\mathrm{cx}}) \quad \text{for } \sin(m\theta).$$

Sampling those 1-D factors on a finite grid yields a discrete separable realization with at most that many terms. On the 161×161 ($\omega_c = \pi/64$) orientation-estimation filter of [55], replacing direct $2 \times 161^2 = 51{,}842$ taps/pixel by a two-complex-Gaussian ($R = 4$) analytical gradient pair reduces scalar multiply–adds to 2,576 (20×, or 40× with even/odd tap-parity folding).

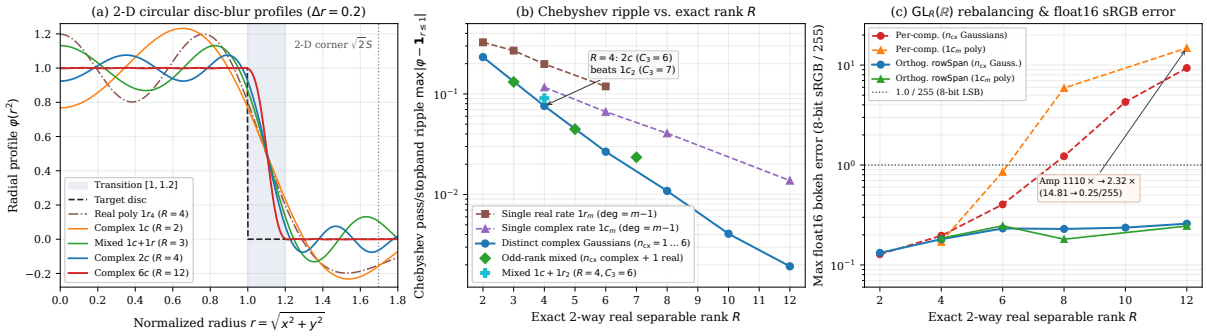


Figure 2: Two-dimensional circular disc blur ($\Delta r = 0.2$, `Applications/blur_2d.py`): (a) minimax radial profiles $\varphi(r^2)$ against the unit disc; (b) minimax pass/stopband ripple across exact 2-way real separable ranks $R \in [2, 12]$ for distinct complex Gaussians, odd-rank mixed Gaussians, and repeated-rate polynomial-Gaussians; (c) maximum IEEE `float16` bokeh error in 8-bit sRGB units (/255, via IEC 61966-2-1 sRGB encoding on a 512×512 linear-light HDR scene with night shadows 0.001 and 3×3 glints up to 48.0, $R_{\text{px}} = 25$, $W = 61$) before and after nuclear-norm-optimal $\mathrm{GL}_R(\mathbb{R})$ rebalancing on rowSpan(K).

Continuous-domain pass-count bounds for simultaneous steerable pairs. Can sharing intermediate 1-D continuous convolutions reduce simultaneous evaluation of $(C_m * I, S_m * I)$ below separate evaluation $2(\lfloor m/2 \rfloor + 1)R + 2\lceil m/2 \rceil R = 2(m+1)R$? No: if any two-stage 1-D schedule on $\mathbb{R}^2$ splits $(C_m, S_m) = (A_c + B_c, A_s + B_s)$ into P_y shared vertical passes followed by $\mathrm{rank}(A_c) + \mathrm{rank}(A_s)$ horizontal passes plus P_x shared horizontal passes followed by $\mathrm{rank}(B_c) + \mathrm{rank}(B_s)$ vertical passes, then

$$\dim \mathrm{rowSpan}_x(C_m, S_m) \le \mathrm{rank}(A_c) + \mathrm{rank}(A_s) + P_x,$$
$$\dim \mathrm{colSpan}_y(C_m, S_m) \le P_y + \mathrm{rank}(B_c) + \mathrm{rank}(B_s).$$

Because C_m and S_m on $\mathbb{R}^2$ span all powers $\{x^0, \dots, x^m\} g_n(x)$ in x and $\{y^0, \dots, y^m\} g_n(y)$ in y, adding the two inequalities shows that *every* two-stage continuous 1-D schedule requires at least $2(m+1)R$ passes ($4R$ passes for (h_x, h_y)). Savings instead come from: (i) definite even/odd parity of every 1-D factor $u^q g_n(u)$, halving multiplications per sampled pass via $I[y+\ell] \pm I[y-\ell]$ folding; (ii) sharing $\{I *_y (y^q g_n)\}_{q=0}^{M}$ across *multiple* harmonic orders ($5R$ instead of $6R$ for $m = 0, 1$; $10R$ instead of $12R$ for $m = 1, 3$); and (iii) for the d-D gradient $(h_{x_0}, \dots, h_{x_{d-1}})$ of a multi-Gaussian radial kernel ($d \ge 3$, base rank $R_d = n_{\text{re}} + d\, n_{\text{cx}}$), recursively splitting $\{0, \dots, d-1\}$ into halves to reduce $d^2 R_d$ passes to $T(d) R_d$ with $T(1) = 1$ and $T(d) = T(\lfloor d/2 \rfloor) + T(\lceil d/2 \rceil) + d \le d(\lceil \log_2 d \rceil + 1)$ ($8R_3$ vs. $9R_3$ in $d = 3$; $12R_4$ vs. $16R_4$ in $d = 4$).

5.3 Worked application I: 2-D circular lens blur and $\mathrm{GL}_R(\mathbb{R})$ conditioning

In depth-of-field bokeh rendering [54, 27, 58], the ideal defocus point-spread function is the discontinuous unit disc $\mathbf{1}_{r \le 1}$ ($r = \sqrt{x^2 + y^2}$), which has infinite separable rank: for any $0 < r_1 < \cdots < r_M < 1$, evaluating $\mathbf{1}_{x_i^2 + y_j^2 \le 1}$ at $x_i = r_i$, $y_j = \sqrt{1 - r_j^2}$ produces an invertible $M \times M$ upper-triangular binary matrix. Introducing a transition band $[1, S]$ with $S = 1 + \Delta r$ ($\Delta r = 0.2$) and truncating each 1-D pass to $|x|, |y| \le S$ restricts the 2-D separable support $[-S, S]^2$ to radii $r \le \sqrt{2}\, S \approx 1.697$. For any fixed rates (α_u, ω_u), minimizing the sampled Chebyshev ripple $\max_{r \in [0,1] \cup [S, \sqrt{2}S]} |\varphi(r^2) - \mathbf{1}_{r \le 1}|$ (2,000 radial nodes) over all polynomial weights is a linear program, solved with HiGHS inside differential-evolution/Nelder–Mead rate search in `Applications/blur_2d.py`.

Figure 2 and Table 9 highlight four structural findings:

Table 9: Representative 2-D circular disc-blur designs ($\Delta r = 0.2$, $R_{\text{px}} = 25$, $W = 61$; explicit parameters in Table 15). Exact 2-D rank $R = C_2(\Gamma)$ and 3-D ball rank $C_3(\Gamma)$ follow Theorem 2.6; SVD is the empirical singular-value count ($\sigma_j > 10^{-10}\sigma_1$) of the 61×61 kernel; $\kappa_\infty = (\sum_n \|h_n\|_\infty \|v_n\|_\infty)/\|K\|_\infty$ measures peak channel amplification before (raw) and after (orth) rowSpan rebalancing, alongside max `float16` HDR bokeh error in 8-bit sRGB units (/255, IEC 61966-2-1).

Structure	Degrees	R (d=2)	SVD	C_3 (d=3)	Ripple $\pm\delta$	$\kappa_\infty^{\text{raw}}$	$\kappa_\infty^{\text{orth}}$	sRGB raw	sRGB orth
$1c$ (Gaussian)	0	2	2	3	0.2324	1.65	1.30	0.13	0.13
$1c + 1r$ (mixed)	0, 0	3	3	4	0.1315	3.45	1.65	0.16	0.20
$2c$ (Gaussians)	0, 0	4	4	6	0.0759	5.28	1.95	0.20	0.18
$1c + 1r_2$ (mixed poly)	0, 1	4	4	6	0.0905	4.50	1.92	0.12	0.23
$1c_2$ (complex poly)	1	4	4	7	0.1165	5.68	1.88	0.17	0.19
$1r_4$ (real Laguerre)	3	4	4	7	0.1982	4.74	1.56	0.21	0.23
$2c + 1r$ (mixed)	0, 0, 0	5	5	7	0.0446	8.73	2.08	0.26	0.20
$4c$ (Gaussians)	0^4	8	8	12	0.0109	70.06	2.31	1.22	0.23
$6c$ (Gaussians)	0^6	12	12	18	0.0019	595.01	2.49	9.35	0.26
$1c_6$ (complex poly)	5	12	12	23	0.0138	1,109.75	2.32	14.81	0.25

1. **Role of nonreal rates, odd budgets, and 3-D lifting:** Across all 18 tested families, empirical SVD rank matches $C_2(\Gamma)$ identically. Distinct complex Gaussians achieve geometric ripple decay with $R = 2n_{\text{cx}}$ ($0.2324 \to 0.0019$ from $1c$ to $6c$, or 0.0014 under global Remez search [58]), while adding a single real Gaussian ($1c+1r, 2c+1r$) interpolates cleanly at odd ranks $R = 3, 5$. Free real-only rates coalesce into repeated-rate Laguerre–Gaussians ($1r_m$). At $R = 4$, four distinct structures tie in 2-D cost, but lifting to a 3-D spherical ball splits their exact 3-way ranks: $2c$ gives both the lowest ripple (± 0.0759) and lower 3-D rank ($C_3 = 6$ vs. $C_3 = 7$ for $1c_2$ and $1r_4$).

2. **Transitive $\mathrm{GL}_R(\mathbb{R})$ freedom and six-stage `separableChannels` ablation against non-orthogonalized stabilization (Table 11):** While $R = \dim \text{rowSpan}(K)$ is fixed, in $d = 2$ writing the $W \times W$ symmetric kernel matrix as $K = HV^\top$ ($H, V \in \mathbb{R}^{W \times R}$) allows replacing (H, V) by $(HT, VT^{-\top})$ for *any* invertible matrix $T \in \mathrm{GL}_R(\mathbb{R})$ [70]. To remove inter-component cancellation ($\kappa_\infty^{\text{raw}} = 595$ on $6c$ and 1,110 on $1c_6$) without a full $W \times W$ SVD, compute a thin QR factorization $H = QR_H$ ($Q^\top Q = I_R$), project $K = QMQ^\top$ onto the $R \times R$ symmetric matrix $M = R_H V^\top Q$, and diagonalize $M = U \,\mathrm{diag}(e_1, \ldots, e_R) U^\top$. Setting $G = QU = [g_1, \ldots, g_R]$ yields $K(x, y) = \sum_{n=1}^R e_n g_n(x) g_n(y)$, where $\tilde{h}_n = \sqrt{|e_n|}\, g_n$ and $\tilde{v}_n = \mathrm{sign}(e_n)\tilde{h}_n$ achieve the nuclear-norm minimum $\sum_{n=1}^R \|\tilde{h}_n\|_2 \|\tilde{v}_n\|_2 = \|K\|_{S_1}$ ($\kappa_\infty \le 2.50$). Table 11 compares $\mathrm{GL}_R(\mathbb{R})$ orthogonalization across $N = 20$ random HDR glint scenes (`seed=0..19`) against three non-orthogonalized baselines: (i) diagonal pair balancing $\|c_n h_n\|_2 = \|v_n / c_n\|_2$ (Stage 1 $\to$ Stage 2) prevents `float16` intermediate overflow on $1c_6$ (20/20 overflow $\to$ $15.17 \pm 1.39/255$) but leaves $\kappa_\infty^{\text{raw}}$ unchanged ($7.90 \pm 0.92/255$ on $6c$); (ii) *reordering non-orthogonalized channels from smallest to largest* $\|h_n\|_2 \|v_n\|_2$ (Stage 2b) gives *no improvement* ($8.18 \pm 0.92/255$ on $6c$, $15.59 \pm 1.84/255$ on $1c_6$), and even accumulating the R `float16`-filtered channels in `float32` without orthogonalization (Stage 2c, equivalent to exact/compensated outer summation) still leaves $6.76 \pm 1.10/255$ on $6c$ and $13.33 \pm 1.30/255$ on $1c_6$, because $\sim 600\times$–$1{,}100\times$ amplitude inflation already rounds each 1-D intermediate $I *_x h_n$ at a coarse `float16` ULP *before* the outer sum; (iii) cross-component $\mathrm{GL}_R(\mathbb{R})$ orthogonalization (Stage 3) collapses κ_∞ to ≤ 2.49, cutting 20-scene sRGB max error to 0.25 ± 0.07 ($4c$), 0.30 ± 0.06 ($6c$, -96.2%), and $0.28 \pm 0.05/255$ ($1c_6$, -98.2%); and (iv) summing orthogonalized terms smallest-$|e_n|$-first (Stage 3 $\to$ Stage 4) further cuts *linear-light* highlight error by 1.5×–2.9× ($0.40 \to 0.15/255$ at 48× and $1.41 \to 0.60/255$ at 256× on $6c$, and $0.23 \to 0.08/255$ in sRGB at unclipped 8×). In $d \ge 3$, minimal d-way CP decompositions lack transitive $\mathrm{GL}_R(\mathbb{R})$ rotations [41], making well-separated $k = 0$ rates

essential (Section 2.4).

3. **GPU and CPU throughput vs. tuned direct W^2 stencils and global `cuFFT` (Table 12):** In `Applications/gpu_kernels.py`, all three 1920×1080 RGBA GPU paths are tuned around the 48 MB L2 cache: `SeparableBlur2D` evaluates all R `rowSpan`-orthogonalized terms per pass ($2c/4c/6c$ for $R = 4, 8, 12$) using K-output register sliding windows over transposed L2-resident row strips; `DirectBlur2D` is a dense $W \times W$ Triton kernel sharing loaded row segments across outputs; and `FFTBlur2D` packs RGBA channel pairs into two complex planes for complex 2-D `cuFFT` (`fft2`/`ifft2` with in-place real spectral multiply) at a tuned 7-smooth period ($\geq \text{size} + L$ for zero padding, $\geq \text{size} + 2L$ for clamped borders). Notably, direct $W \times W$ and `cuFFT` runtimes depend *only* on (R_{px}, W)—they evaluate any symmetric kernel on $[-L, L]^2$ (whether $2c$, $6c$, or an unapproximated disc) in the same time, whereas separable FIR cost grows with R. At $R_{\text{px}} = 64$, the separable cascade beats the tuned direct $W \times W$ kernel by 28.0× (`fp32`) / 29.5× (`fp16`) at $R = 4$, 13.9× / 14.6× at $R = 8$, and 9.2× / 9.4× at $R = 12$ (whereas at $R_{\text{px}} = 8$, direct 0.52 ms beats $R \geq 8$). Against `cuFFT` (0.451–0.555 ms across all radii), the separable cascade wins only at $R = 4$ for $R_{\text{px}} \in \{8, 16\}$ in both precisions (0.331 / 0.417 ms vs. 0.457 ms; 0.354 / 0.396 ms vs. 0.454 ms; at $R_{\text{px}} = 32, R = 4$, `fp16` 0.496 ms and clamp `cuFFT` 0.481 ms are within run-to-run variation), while `cuFFT` is both faster and free of finite-rank approximation ripple for every $R \geq 8$ and for $R = 4$ at $R_{\text{px}} = 64$. Because strip intermediates stay in L2 rather than DRAM, `float16` is no longer uniformly faster than `float32` (`fp16`/`fp32` time ratios span 0.84× to 1.26× due to 16-bit gather/store overhead, though Table 11's conditioning remains essential whenever textures or APIs mandate `float16`). On the 14-core CPU (`float32`), by contrast, unfused $2R$-pass PyTorch `conv2d` writes 33 MB frame buffers to DDR5 RAM (169.1, 286.7, 444.0 ms at $R = 4$ vs. `PocketFFT` `rfft2` at 32.6, 40.7, 68.1 ms and direct `conv2d` at 68.2, 208.2, 676.6 ms for $R_{\text{px}} = 8, 16, 32$). Separable decompositions at large radii are thus most compelling with $O(1)$ recursive IIR Gaussians [19, 77], spatially varying depth-of-field radii $R_{\text{px}}(x, y)$ [27], or L2/line-buffered pipelines without spectral padding buffers.

4. **Direct discrete rank-R optimization, convergence, and warm-starting (Figures 3–4, Table 10):** Because any finite-tap separable cascade on $\{-L, \ldots, L\}^d$ is rectangular-truncated and even in every coordinate ($h_n[-m] = h_n[m]$), its 1-D factors lie in an $(L+1)$-dimensional even subspace, capping 2-D discrete rank at $\min(R, L+1)$ (for instance, sampling $6c$ of continuous rank 12 gives exact discrete SVD rank $4, 6, 11, 12, 12, 12$ at $R_{\text{px}} = 2, 4, 8, 16, 25, 64$, with σ_{12}/σ_1 locking onto 4.37–4.34 $\times 10^{-4}$ for $R_{\text{px}} \geq 16$). Table 10 and Figures 3–4 compare sampled continuous Gaussians against (i) direct Eckart–Young ℓ_2 SVD truncation of a raised-cosine smooth disc on $[1, 1.2]$, (ii) unconstrained D_4-symmetric discrete rank-R minimax optimization (alternating pass/stopband clamping with SVD rank-R projection), and (iii) shell-isotropic discrete rank-R minimax (equalizing lattice points of equal Euclidean radius $m_x^2 + m_y^2 = N$, with shell anisotropy $\Delta_{\text{sh}} = \max_N(\max_{\|\mathbf{m}\|^2=N} K - \min_{\|\mathbf{m}\|^2=N} K)$ reported to six decimals in Figure 3 to distinguish exact machine-zero continuous symmetry $\Delta_{\text{sh}} = 0.000000$ from discrete rank-4 SVD projection residuals 0.000001 at $R_{\text{px}} = 6$ and 0.000178 at $R_{\text{px}} = 25$):

   - *Small stencils and lattice chord coincidences ($R_{\text{px}} \leq 12$):* Once $R \geq L + 1$, every even matrix has rank $\leq R$ and discrete error is identically 0 (∇ in Figure 4a). Even slightly above $R = L + 1$, small integer lattices permit discrete gains (± 0.00476 shell-isotropic vs. ± 0.07593 continuous at $R_{\text{px}} = 6, R = 4$, or $\leq 1.4 \times 10^{-5}$ at isolated radii such as $R_{\text{px}} \in \{4, 5, 8\}$ for $R = 2$ and $R_{\text{px}} = 12$ for $R = 4$ where the $[1, 1.2]$ transition gap partitions passband/stopband circle chords into $\leq R$ staircase blocks; at $R_{\text{px}} = 6$, only $1^2 + 7^2 = 5^2 + 5^2 = 50$ shares a squared radius in $\{0, \ldots, 7\}^2$, leaving 0°-vs.-45° freedom in Figure 3).

Table 10: Sampled continuous complex Gaussians vs. direct discrete rank-R optimization across stencil radii $R_{\text{px}} \in \{3, 6, 12, 25, 64\}$ ($L + 1 = \lceil 1.2\, R_{\text{px}} \rceil + 1$ even 1-D degrees of freedom). Each entry lists pass/stopband Chebyshev ripple $\pm\delta$, with equal-radius shell anisotropy Δ_{sh} in parentheses.

R_{px}	W	$L{+}1$	R	Cont. Gaussian	Direct ℓ_2 SVD (Δ_{sh})	Disc. unconstr. (Δ_{sh})	Disc. shell-iso. (Δ_{sh})
3	9	5	4	0.07593 (0)	0.00000 (0)	0.00000 (0)	0.00000 (0)
6	17	9	4	0.07593 (0)	0.02182 (0.0111)	0.00479 (0.1285)	0.00476 ($1{\times}10^{-6}$)
12	31	16	4	0.07593 (0)	0.09496 (0.1066)	0.01899 (0.1577)	0.04273 ($4{\times}10^{-7}$)
25	61	31	4	0.07593 (0)	0.09556 (0.1770)	0.05226 (0.0905)	0.07367 ($1.8{\times}10^{-4}$)
64	155	78	4	0.07593 (0)	0.10295 (0.1923)	0.06647 (0.0323)	0.07582 ($2.0{\times}10^{-5}$)
12	31	16	8	0.01086 (0)	0.00978 (0.0140)	0.00137 (0.0240)	0.00079 ($1{\times}10^{-6}$)
25	61	31	8	0.01086 (0)	0.01295 (0.0151)	0.00486 (0.0228)	0.00552 ($4.3{\times}10^{-5}$)
64	155	78	8	0.01086 (0)	0.01625 (0.0189)	0.00755 (0.0113)	0.01065 ($2.4{\times}10^{-5}$)
64	155	78	12	0.00193 (0)	0.00480 (0.0064)	0.00100 (0.0066)	0.00171 ($1.3{\times}10^{-4}$)

Table 11: Six-stage IEEE `float16` conditioning ablation ($R_{\text{px}} = 25$, $W = 61$, $N = 20$ random 512×512 HDR scenes, `seed=0..19`, peak 48×): Stage 1 puts weights on H and unit basis on V; Stage 2 rescales $\|h_n\|_2 = \|v_n\|_2$; Stage 2b reorders Stage 2 channels smallest-$\|h_n\|_2\|v_n\|_2$-first *without* orthogonalization; Stage 2c sums Stage 2b's `float16` 1-D passes in a `float32` outer accumulator *without* orthogonalization; Stages 3 and 4 apply $\mathrm{GL}_R(\mathbb{R})$ orthogonalization summed largest-$|e_n|$-first vs. smallest-$|e_n|$-first. Rightmost column: Stage 3 → Stage 4 linear-light max error (/255, `seed=7`) at peaks 48× and 256×.

	20-scene mean ± std max 8-bit sRGB error (/255, peak 48×)						Linear (/255)
Design (R)	1. Unbal.	2. Pair-bal.	2b. Reord.	2c. `fp32`-acc.	3. Orth. (desc)	4. Orth. (asc)	48×; 256×
$2c$ (R=4)	0.12 ± 0.02	0.11 ± 0.01	0.11 ± 0.02	0.09 ± 0.02	0.20 ± 0.04	0.24 ± 0.08	0.24→**0.16**; 1.00→**0.60**
$4c$ (R=8)	0.58 ± 0.07	0.54 ± 0.06	0.64 ± 0.10	0.46 ± 0.05	0.25 ± 0.07	0.26 ± 0.10	0.36→**0.16**; 1.23→**0.69**
$6c$ (R=12)	7.80 ± 1.16	7.90 ± 0.92	8.18 ± 0.92	6.76 ± 1.10	$\mathbf{0.30 \pm 0.06}$	$\mathbf{0.31 \pm 0.13}$	0.40→**0.15**; 1.41→**0.60**
$1c_6$ (R=12)	`ovf` (20/20)	15.17 ± 1.39	15.59 ± 1.84	13.33 ± 1.30	$\mathbf{0.28 \pm 0.05}$	$\mathbf{0.27 \pm 0.10}$	0.40→**0.16**; 1.45→**0.62**

- *Direct ℓ_2 SVD vs. Chebyshev minimax and large-radius convergence ($R_{\text{px}} \geq 16$):* Direct ℓ_2 SVD degrades in both pass/stopband ripple (0.09556–0.10295 vs. 0.07593 at $R = 4$) and 17.7%–19.2% shell anisotropy (visible cross-streaks in Figure 3), while unconstrained discrete minimax lowers ripple (0.05226 at $R_{\text{px}} = 25$) *only by warping the transition rim* $1 < r < 1.2$ *anisotropically* ($\Delta_{\text{sh}} = 0.09052$). Once $R_{\text{px}} \geq 16$ and equal-radius shell isotropy is enforced, discrete minimax converges monotonically onto our continuous $1c/2c/4c/6c$ asymptotes (Figure 4a; 0.23245, 0.07582, 0.01065, and 0.00171 vs. 0.23245, 0.07593, 0.01086, and 0.00193 at $R_{\text{px}} = 64$).
- *Does warm-starting discrete minimax from continuous complex Gaussians speed up optimization? (Figure 4b):* The answer splits cleanly by regime. At *small* radii ($R_{\text{px}} = 6, R = 4$), warm-starting from $2c$ does *not* accelerate optimization: raised-cosine ℓ_2 SVD already starts 3.5× lower on the joint shell-isotropic objective $\max(\text{rip}, \|K - \mathcal{P}_{\text{iso}} K\|_\infty)$ (0.0218 vs. 0.0759) and stays ahead through 500 SVDs (0.0058 vs. 0.0106) because small stencils optimize discrete inter-shell gaps rather than continuous equioscillation. At *large* radii ($R_{\text{px}} = 25$), by contrast, Theorem 2.4 makes the intersection $\mathcal{V}_{\text{iso}} \cap \mathcal{M}_{\leq R}$ tangential to non-exponential-polynomial cold starts: warm-starting from $2c/4c/6c$ starts at $k = 0$ directly on the zero-anisotropy manifold $(0.0759, 0.0109, 0.00193)$, whereas raised-cosine SVD starts $1.3\times$ –$2.2\times$ higher $(0.0956, 0.0129, 0.00422$, needing ~ 120 SVDs at $R = 12$ just to match $6c$'s $k = 0$ error and still trailing at $k = 500$: 0.00109 vs. 0.00076), and hard-disc SVD (0.027 at $k = 500$) and random rank-R starts (> 1.0) stall in poor nonconvex local minima.

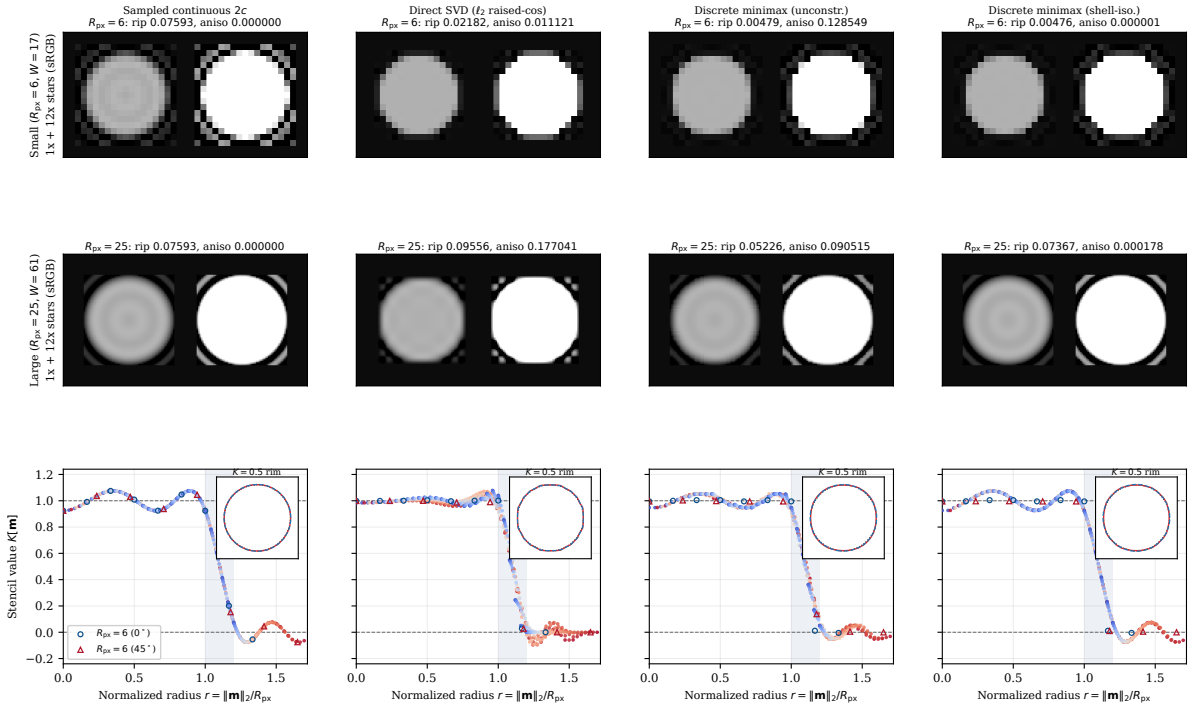


Figure 3: Visual and quantitative comparison at exact rank $R = 4$ between sampled continuous $2c$ Gaussians, direct ℓ_2 SVD of a raised-cosine disc, unconstrained discrete rank-4 minimax, and shell-isotropic discrete rank-4 minimax (`Applications/blur_2d.py`): top row, pixelated small stencil ($R_{\text{px}} = 6, W = 17$) sRGB bokeh on 1× (left) and 12× HDR (right) point stars; middle row, pixelated large stencil ($R_{\text{px}} = 25, W = 61$) sRGB bokeh (with 5-decimal ripple and 6-decimal shell anisotropy Δ_{sh}); bottom row, per-pixel radial scatter ($R_{\text{px}} = 25$ dots colored from 0° blue to 45° red; $R_{\text{px}} = 6$ open 0°/45° markers) and $K = 0.5$ rim contours ($R_{\text{px}} = 6$ solid red, $R_{\text{px}} = 25$ dashed blue, reference circle dotted black).

5.4 Worked application II: isotropic 3-D acoustic wave simulation

Consider the 3-D constant-speed scalar acoustic wave equation $\partial_{tt} p = c_0^2 \nabla^2 p$ discretized on $\mathbb{Z}^3$ with spacing Δx and time step $\Delta t = \nu \Delta x / c_0$ (Courant number $\nu > 0$) by the explicit two-step leapfrog update

$$p^{n+1}[\mathbf{m}] + p^{n-1}[\mathbf{m}] = (M * p^n)[\mathbf{m}], \qquad \mathbf{m} \in \mathbb{Z}^3. \tag{18}$$

Substituting a discrete plane wave $p^n[\mathbf{m}] = e^{i(\mathbf{k}\cdot\mathbf{m} - \omega_{\text{num}} n \Delta t)}$ with normalized wavenumber $\mathbf{k} = (k_x, k_y, k_z) \in [-\pi, \pi]^3$ turns the left-hand side of (18) into $2\cos(\omega_{\text{num}}\Delta t)\, p^n[\mathbf{m}]$ and the right-hand side into $\widehat{M}(\mathbf{k})\, p^n[\mathbf{m}]$, where $\widehat{M}(\mathbf{k}) = \sum_{\mathbf{m}} M[\mathbf{m}] e^{-i\mathbf{k}\cdot\mathbf{m}}$ is the discrete Fourier symbol of M. Since an exact continuous plane wave travels at speed c_0 ($\omega_{\text{exact}}\Delta t = c_0(\|\mathbf{k}\|_2/\Delta x)\Delta t = \nu\|\mathbf{k}\|_2$), matching the exact wave equation requires $\widehat{M}_{\text{exact}}(\mathbf{k}) = 2\cos(\nu\|\mathbf{k}\|_2)$ [67], and the numerical-to-exact phase-velocity ratio is $c_{\text{num}}(\mathbf{k})/c_0 = \arccos(\widehat{M}(\mathbf{k})/2)/(\nu\|\mathbf{k}\|_2)$. Whenever $|\widehat{M}(\mathbf{k})| \le 2$, both roots of $\zeta^2 - \widehat{M}(\mathbf{k})\zeta + 1 = 0$ have $|\zeta| = 1$ (spectral von Neumann stability); strict interior containment $\widehat{M}(\mathbf{k}) \in (-2, 2)$ for $\mathbf{k} \neq \mathbf{0}$ avoids repeated roots $\zeta = \pm 1$ (linear-in-n secular growth) and yields a conserved positive discrete energy on all nonzero modes. Standard 7-point FDTD ($\nu \le 1/\sqrt{3}$) suffers 19.88% axial phase-velocity error by $\|\mathbf{k}\|_2 = 0.80\pi$, motivating dispersion-relation-preserving and high-order room-acoustics FDTD designs [68, 40, 73, 30, 31]. Two structural facts explain why radial multi-Gaussians are natural here:

- **Inadequacy of additive axial stencils of any order:** Any purely axial scheme $\widehat{M}_{\text{ax}}(\mathbf{k}) =$

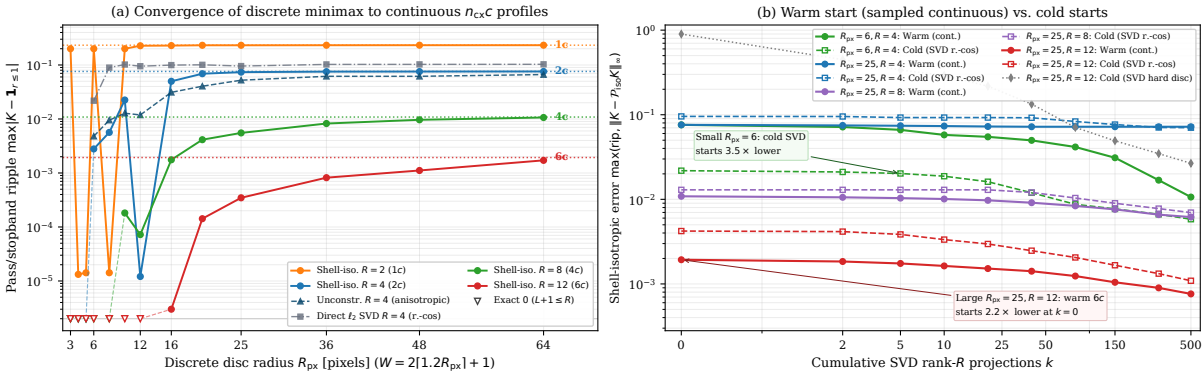


Figure 4: Discrete-to-continuous convergence and warm-starting (`Applications/blur_2d.py`): (a) pass/stopband Chebyshev ripple across $R_{\text{px}} \in [3, 64]$ for shell-isotropic discrete rank-R minimax ($R \in \{2, 4, 8, 12\}$, converging to the dotted continuous $1c/2c/4c/6c$ asymptotes once $R_{\text{px}} \geq 16$; open triangles ∇ mark exact 0 error when $L+1 \leq R$), alongside $R = 4$ unconstrained minimax and direct ℓ_2 SVD; (b) single-run trajectories of the shell-isotropic minimax objective $\max(\text{rip}, \|K - \mathcal{P}_{\text{iso}}K\|_\infty)$ vs. cumulative SVD projections $k \in [0, 500]$ comparing continuous-Gaussian warm starts against raised-cosine SVD and hard-disc SVD cold starts.

Table 12: GPU execution times in milliseconds (mean±std across $N = 10$ CUDA-Event batches of 15 calls = 150 timed launches per cell; 20–50 launches for `DirectBlur2D`) for 1920×1080 RGBA 2-D blur on an RTX 4070 Ti Super across $R_{\text{px}} \in \{8, 16, 32, 64\}$ ($W = 2L + 1$, $L = \lceil 1.2\, R_{\text{px}} \rceil$): tuned dense $W \times W$ Triton (`DirectBlur2D`, rank-independent), complex-plane-packed `cuFFT` at 7-smooth zero- and clamp-padded periods (`FFTBlur2D`, rank-independent), and two-pass register-windowed L2-strip Triton cascades on $2c/4c/6c$ (`SeparableBlur2D`; inter-run autotuning shift $\leq 6\%$, full metadata in `Applications/results_blur_2d.json`).

R_{px}	W	Direct W^2	`cuFFT` (zero)	`cuFFT` (clamp)	Sep. $R = 4$		Sep. $R = 8$		Sep. $R = 12$	
		`fp32`	`fp32`	`fp32`	`fp32`	`fp16`	`fp32`	`fp16`	`fp32`	`fp16`
8	21	0.516±.001	0.465±.0002	0.457±.001	**0.331**±.0003	0.417±.0004	0.665±.001	0.670±.001	0.860±.001	1.033±.032
16	41	1.750±.033	0.451±.0005	0.454±.001	**0.354**±.005	0.396±.001	0.719±.001	0.682±.0003	1.076±.005	1.143±.028
32	79	6.449±.132	**0.452**±.001	0.481±.001	0.528±.004	0.496±.001	1.121±.001	0.937±.005	1.661±.013	1.522±.012
64	155	24.38±.110	**0.473**±.002	0.555±.001	0.871±.003	0.825±.003	1.755±.007	1.672±.001	2.641±.007	2.594±.005

$2+A(k_x)+A(k_y)+A(k_z)$ ($A(0) = 0$) lacks mixed cross-terms $k_x^2 k_y^2$: even if $A(k) = 2(\cos(\nu k) - 1)$ has zero dispersion along the coordinate axes, along $(1, 1, 1)/\sqrt{3}$ it gives $2+6(\cos(\nu k/\sqrt{3}) - 1) = 2 - \nu^2 k^2 + \frac{1}{36}\nu^4 k^4 - \ldots \neq 2\cos(\nu k) = 2 - \nu^2 k^2 + \frac{1}{12}\nu^4 k^4 - \ldots$, leaving an irreducible $O(\nu^4 k^4)$ anisotropy gap while Brillouin-corner stability $\widehat{M}_{\text{ax}}(\pi, \pi, \pi) \geq -2$ forces $A(\pi) \geq -4/3$ ($\nu k_c \leq \arccos(1/3) \approx 0.392\pi$ when A tracks $2(\cos(\nu k) - 1)$).

- **Kowalczyk–van Walstijn IWB is a single real Gaussian:** In room acoustics, Kowalczyk and van Walstijn's 27-point Interpolated Wideband (`IWB`) stencil [40] ($a = \frac{1}{4}, b = \frac{1}{16}$, $\nu \leq 1$) has face, edge, and corner weights $\frac{1}{4}\nu^2, \frac{1}{8}\nu^2, \frac{1}{16}\nu^2$ and center weight $2 - \frac{7}{2}\nu^2$, which is *identically*

$$M_{\text{IWB}}[\mathbf{m}] = (2 - 4\nu^2)\delta_{\mathbf{0}}[\mathbf{m}] + \frac{\nu^2}{2}\, 2^{-(m_x^2 + m_y^2 + m_z^2)} \qquad (\mathbf{m} \in \{-1, 0, 1\}^3),$$

the ($n_{\text{re}} = 1, n_{\text{cx}} = 0, L = 1, R = 1$) single-real-Gaussian member of our radial family (plus a zero-pass center tap). Because a single positive real Gaussian cannot match higher radial derivatives of $2\cos(\nu\|\mathbf{k}\|_2)$, `IWB` at $\nu = 1$ is exact on axes but incurs 9.38% diagonal phase-velocity error by 0.80π (Figure 5a,d).

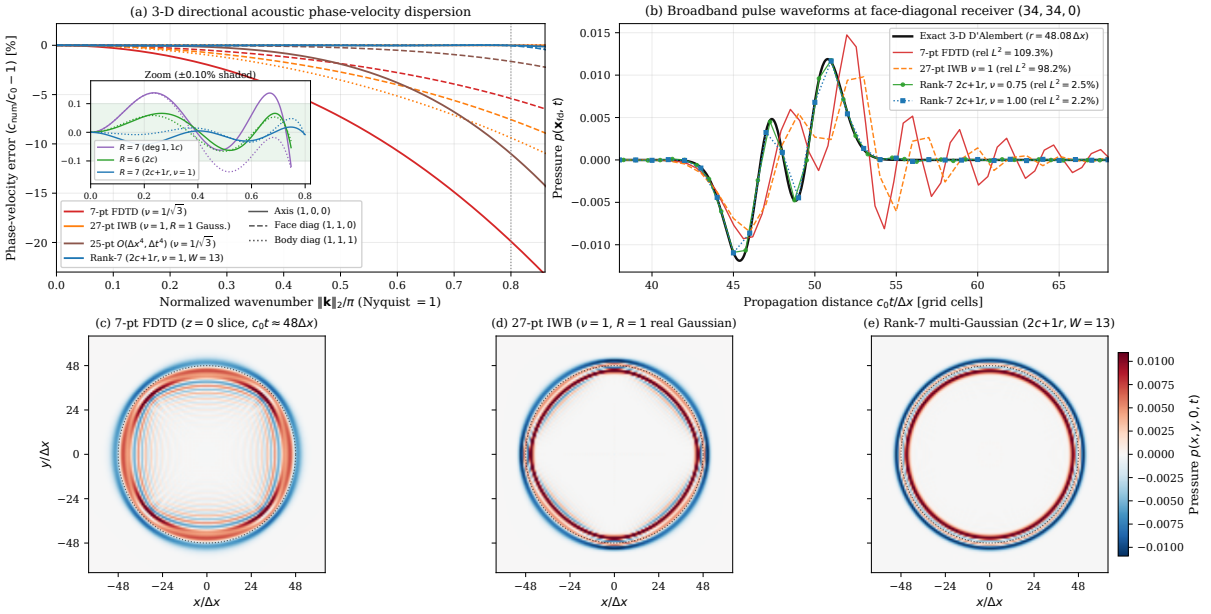


Figure 5: Three-dimensional acoustic wave propagation (`Applications/acoustics_3d.py`): (a) directional phase-velocity error $(c_{\text{num}}(\mathbf{k})/c_0 - 1)$ along $(1,0,0)$, $(1,1,0)$, and $(1,1,1)$ across $\|\mathbf{k}\|_2/\pi \in [0, 0.86]$, with inset zoom ($\pm 0.10\%$ shaded) for $R = 6, 7$; (b) broadband pulse waveforms ($\sigma_0 = 1.40\,\Delta x$) on a 256^3 grid at the face-diagonal receiver $(34, 34, 0)$ ($r = 48.08\,\Delta x$) vs. the exact 3-D D'Alembert solution; (c–e) mid-plane wavefield slices $p(x, y, 0, t)$ at $c_0 t \approx 48\,\Delta x$ (dotted circle: $r = 48\,\Delta x$).

Brillouin-octant Chebyshev optimization and exact Neumann walls. Generalizing `IWB` to wider stencils $\mathbf{m} \in [-L, L]^3$, we set $M[\mathbf{m}] = \alpha_0 \delta_{\mathbf{0}}[\mathbf{m}] + \varphi(m_x^2 + m_y^2 + m_z^2)$ (parameters in Table 16), where α_0 fixes $\widehat{M}(\mathbf{0}) = 2$ identically and φ has exact 3-way real rank $R = n_{\text{re}} + 3n_{\text{cx}}$ (Proposition 2.14) or $R = 7$ for $\deg P = 1$ (Proposition 2.15; verified to $\leq 5.6 \times 10^{-16}$ on $[-L, L]^3$). For fixed rates (searched via differential evolution and Nelder–Mead), minimizing $\max_{0<\|\mathbf{k}\|_2 \leq k_c} |\widehat{M}(\mathbf{k}) - 2\cos(\nu\|\mathbf{k}\|_2)|/(2\nu\|\mathbf{k}\|_2 \sin(\nu\|\mathbf{k}\|_2))$ subject to exact asymptotic sound speed $\partial_{k_x}^2 \widehat{M}(\mathbf{0}) = -2\nu^2$ and strict interior stability $-2 + 10^{-3} \leq \widehat{M}(\mathbf{k}) \leq 2 - \min(10^{-3}, \frac{1}{4}\nu^2\|\mathbf{k}\|_2^2)$ across 232,620 symmetry-inequivalent Brillouin-octant and spherical-shell wavevectors (including the exact sphere $\|\mathbf{k}\|_2 = k_c$) is solved via cutting-plane HiGHS linear programming, followed by multi-start continuous Nelder–Mead maximization of $|\delta c(\mathbf{k})/c_0|$ over $0 < \|\mathbf{k}\|_2 \leq k_c$. Moreover, because every 1-D radial factor $h[m] = h[-m]$ is even, half-sample symmetric index reflection at domain boundaries simultaneously diagonalizes all 3 directional passes in the 3-D `DCT-II` cosine basis—enforcing exact rigid-wall Neumann boundary conditions ($\partial_n p = 0$) without padding buffers or boundary instabilities ($\leq 7.0 \times 10^{-7}$ max `float32` relative energy drift across all three `Rank-7` schemes over 1,000 steps in a 64^3 room; 2.0×10^{-7} at $\nu = 0.75$ and 6.1×10^{-8} at $\nu = 1.00$ for $2c + 1r$).

Table 13 and Figure 5 summarize accuracy, mean ± std GPU timings, and VRAM footprints across all five baselines (`DirectStep3D` is a tuned dense W^3 reflecting leapfrog Triton kernel using +66–68 MiB for its z-padded buffer; `SeparableStep3D` sweeps x-slabs sized to L2 cache in three passes per slab—x fan-out to R channels, y, and z fused with the leapfrog update):

- **Within separable spatial stencils:** Replacing naive $(n_{\text{re}} + 4n_{\text{cx}})$ expansion by Proposition 2.14's and 2.15's exact rank R cuts `float32` GPU step time by 24.0% on `Rank-7` (1.51 ± 0.03 ms vs. 1.98 ± 0.02 ms, 4.7× faster than the tuned 13^3 direct Triton stencil; 15.6% on `Rank-4`, 23.5% on `Rank-6`, and 31.2% on `Rank-9`), alongside 21.2% on the 14-core CPU (283.6 ± 2.3 ms vs. 360.1 ± 0.6 ms on 128^3). On the 256^3 broadband spherical pulse test

Table 13: Illustrative 3-D acoustic propagators vs. four compact explicit FDTD baselines (7-pt, 13-pt $O(\Delta x^4, \Delta t^2)$, 27-pt IWB, and fourth-order 25-pt $O(\Delta x^4, \Delta t^4)$ [30]) and L2-chunked periodic k-space `cuFFT` on 256^3 voxels (parameters in Table 16). $|\delta c/c_0|$ is evaluated on each row's passband (0.80π for FDTD, 1.00π for `cuFFT`, parenthesized k_c for propagators); $k_{0.1\%}$, $k_{0.5\%}$, and Wave L^2 use identical definitions across all rows. Step times are mean ± std in ms over $N = 10$ CUDA-Event batches of 15 calls = 150 steps (6×2 steps for `DirectStep3D`; inter-run shift $\leq 3\%$); `+MiB` reports extra scratch VRAM beyond the leapfrog field arrays (128 MiB in-place / 192 MiB out-of-place on 256^3).

Scheme	ν	W	R (nv)	$\lvert\delta c/c_0\rvert$	$k_{0.1\%}$	$k_{0.5\%}$	Wave L^2	Direct	Naive	`fp32`	`fp16`	+MiB
7-pt FDTD	0.58	3	—	19.88%	0.060π	0.135π	109.3%	—	—	0.324±.0003	—	0
13-pt $O(\Delta x^4, \Delta t^2)$	0.50	5	—	8.74%	0.099π	0.222π	100.8%	—	—	0.324±.0001	—	0
27-pt IWB ($1r$)	1.00	3	1 (1)	9.38%	0.085π	0.191π	98.2%	—	—	0.324±.0001	—	0
25-pt $O(\Delta x^4, \Delta t^4)$	0.58	5	—	11.00%	0.236π	0.355π	78.2%	—	—	0.325±.0002	—	0
k-space `cuFFT`	1.00	256	—	0.000%	1.00π	1.00π	0.35%	—	—	1.44±.05	—	+101
Rank-4 ($1c$+$1r$, $.75\pi$)	0.75	9	4 (5)	0.293%	0.14π	0.76π	11.6%	2.68±.04	1.02±.02	0.857±.003	0.627±.009	+16
Rank-6 ($2c$, $.75\pi$)	0.75	11	6 (8)	0.066%	0.76π	0.80π	5.2%	4.63±.05	1.66±.02	1.27±.02	0.920±.016	+24
Rank-7 (deg 1 $1c$, $.75\pi$)	0.75	13	7 (9)	0.138%	0.16π	0.78π	7.1%	7.07±.01	1.98±.02	1.51±.03	1.09±.02	+28
Rank-7 ($2c$+$1r$, $.80\pi$)	0.75	13	7 (9)	0.045%	0.81π	0.85π	2.8%	7.07±.01	1.98±.02	1.51±.03	1.09±.02	+28
Rank-7 ($2c$+$1r$, $.80\pi$)	1.00	13	7 (9)	0.049%	0.81π	0.87π	2.5%	7.07±.01	1.98±.02	1.51±.03	1.09±.02	+28
Rank-9 ($3c$, $.85\pi$)	0.75	17	9 (12)	0.044%	0.86π	0.89π	2.0%	14.51±.03	3.21±.02	2.21±.02	1.57±.01	+36/18

($\sigma_0 = 1.40\,\Delta x$), all compact explicit stencils (7-pt, 13-pt, 27-pt IWB, and 25-pt $O(\Delta x^4, \Delta t^4)$) suffer 78.2%–109.3% waveform error from high-wavenumber dispersion, whereas `Rank-7` ($2c + 1r$) keeps $|\delta c/c_0|$ at 0.045% ($\nu = 0.75$) and 0.049% ($\nu = 1.00$, < 1 cent) up to 0.80π (2.47 points per wavelength at $k_{0.1\%} = 0.81\pi$), achieving 1.6%–2.8% broadband waveform error across all axes and diagonals (2.8% with `float16` slab intermediates at $\nu = 0.75$). Consistent with the small-stencil regime in Table 10, freeing the $(L + 1)$-tap even 1-D discrete factors from the continuous Gaussian formula under the same octant LP/Nelder–Mead objective substantially improves the smallest stencil (`Rank-4`, $W = 9, L + 1 = 5$: 0.293% → 0.108%) while yielding diminishing gains as L grows (`Rank-6`, $W = 11, L + 1 = 6$: 0.066% → 0.056%).

- **Fair accuracy/effort and memory-footprint comparison against 1-pass FDTD and `cuFFT`:** To compare methods fairly, we report *both* equal-grid cost on 256^3 and equal-accuracy grid scaling for a fixed physical domain and maximum resolved frequency $\omega_{\max}$. On the fixed 256^3 grid (64 MiB per `float32` field: 128 MiB in-place when p^{n+1} overwrites p^{n-1}, 192 MiB out-of-place), all four fused 1-pass FDTD kernels execute at 92.5% of peak GDDR6X memory bandwidth (0.324–0.325 ± 0.0003 ms/step, 51.7–51.8 GVox/s) with *zero* scratch VRAM—3.4× to 4.7× faster per step than `Rank-7` (1.09 ± 0.02 ms `fp16` / 1.51 ± 0.03 ms `fp32`, which adds +28 MiB of reusable L2-slab scratch, i.e. 156 MiB vs. 128 MiB in-place, vs. +101 MiB for `KSpaceStep3D`). When coarse dispersion suffices, 27-pt/25-pt FDTD [40, 30] is therefore faster and leaner. Conversely, when a target phase-velocity tolerance $\tau \in \{0.1\%, 0.5\%\}$ up to $\omega_{\max}$ is required, a scheme with effective τ-bandwidth k_τ needs grid spacing $\Delta x = c_0 k_\tau / \omega_{\max}$, scaling voxel count (and field memory!) as $(k_{\tau,\text{sep}}/k_{\tau,\text{FDTD}})^3$ and space-time work as $(\nu_{\text{FDTD}}/\nu_{\text{sep}})(k_{\tau,\text{sep}}/k_{\tau,\text{FDTD}})^4$: at $\nu = 1.00$, `Rank-7` ($k_{0.1\%} = 0.81\pi, k_{0.5\%} = 0.87\pi$) uses 865× less field memory and an estimated 1,800×–2,400× less runtime than 27-pt IWB ($k_{0.1\%} = 0.085\pi$) at $\tau = 0.1\%$ (94× less memory and 90×–130× less runtime at 0.5%), and against fourth-order 25-pt $O(\Delta x^4, \Delta t^4)$ ($\nu = 1/\sqrt{3}, k_{0.1\%} = 0.236\pi, k_{0.5\%} = 0.355\pi$) uses 40× less field memory and 52×–71× less estimated runtime at 0.1% (15× less memory and 13×–19× less runtime at 0.5%). On a *homogeneous periodic or rectangular box*, L2-chunked periodic k-space `cuFFT` [67] takes 1.44 ± 0.05 ms (83.3 ± 4.9 ms on the 14-core CPU) with zero dispersion up to π (0.35% pulse L^2 error), tying `Rank-7 fp32` (1.51 ms) in speed while trailing `Rank-7 fp16` (1.09 ms) and `Rank-4`/`Rank-6` (0.86/1.27 ms); local separable stencils

are thus primarily motivated for multi-GPU halo decomposition and *spatially varying media* $c(\mathbf{x})$ below.

- **Spatially varying sound speed $c(\mathbf{x})$ where 1-step `cuFFT` k-space is inapplicable (Figure 6):** In 3-D seismic velocity modeling (sedimentary depth gradients and high-velocity salt bodies [25]), transcranial and soft-tissue medical ultrasound, and ocean or atmospheric sound-speed channels, $\partial_{tt}p = c(\mathbf{x})^2\nabla^2 p$ has a spatially varying local Courant number $\nu(\mathbf{x}) = c(\mathbf{x})\Delta t/\Delta x$. Here exact 1-step `cuFFT` k-space propagation (`KSpaceStep3D`) fails because $2\cos(c(\mathbf{x})\Delta t\|\mathbf{k}\|_2)$ mixes $\mathbf{x}$ and $\mathbf{k}$ (requiring multi-FFT low-rank symbol expansions [25] while still imposing periodic wrap-around), whereas standard $O(\Delta t^2)$ Fourier-Laplacian pseudospectral leapfrog ($p^{n+1} = 2p^n - p^{n-1} + \nu(\mathbf{x})^2\mathcal{F}^{-1}(-\|\mathbf{k}\|_2^2\widehat{p^n})$) is unstable above $\nu_{\max} = \frac{2}{\sqrt{3\pi}} \approx 0.368$ and incurs 3.08% uncompensated temporal phase lead even at $\nu = 0.35$ (Figure 6a). By fixing *shared* exponential rates across space—two fixed-phase complex Gaussians plus two real Gaussians on $W = 13$ (exact rank $R = 3+3+1+1 = 8$, `acoustics_3d.py --hetero-only`)—all 8 shift-invariant separable 1-D cascades $\Delta_u p^n = U_u * p^n - \widehat{U}_u(\mathbf{0})p^n$ ($u = 1, \dots, 4$) become *independent* of $c(\mathbf{x})$, and $c(\mathbf{x})$ enters solely through local polynomial weights $\gamma_u(\nu(\mathbf{x})) = c_{u,1}\nu^2 + c_{u,2}\nu^4 + c_{u,3}\nu^6$ at accumulation:

$$p^{n+1}(\mathbf{x}) = 2p^n(\mathbf{x}) - p^{n-1}(\mathbf{x}) + \sum_{u=1}^{4} \gamma_u\big(\nu(\mathbf{x})\big)\,(\Delta_u p^n)(\mathbf{x}).$$

  Across an entire 2.3 : 1 Courant range $\nu \in [0.35, 0.80]$, this single shared-rate `Rank-8` basis keeps isotropic phase-velocity error $\le 0.185\%$ up to 0.75π with $\widehat{M}_\nu(\mathbf{k}) \in [-1.47, 2.00]$ (vs. 16.8%–23.8% for 27-pt IWB and 8.96%–10.30% for 25-pt $O(\Delta x^4, \Delta t^4)$ up to its $\nu \le 0.577$ limit; Figure 6a), and remains bounded over 1,000 reflecting-room steps ($\max|p| = 1.0000$, decaying to 0.0521 and 0.0642) on both a smooth 2 : 1 Gaussian lens and a sharp 2 : 1 spherical jump (whereas naive $\frac{1}{2}(\gamma_u\Delta_u p + \Delta_u(\gamma_u p))$ symmetrization diverges across jumps because $\Delta_u\gamma_u \neq 0$ violates constant-state preservation). On a 128^3 broadband refraction test ($\sigma_0 = 1.50\,\Delta x$, $c_{\max}t \in [0, 62\,\Delta x]$) through a low-velocity focusing lens $c(\mathbf{x})/c_{\max} = 1 - 0.45e^{-\|\mathbf{x}\|^2/(14\Delta x)^2}$ ($\nu(\mathbf{x}) \in [0.41, 0.75]$, 83 steps; Figure 6b–e) benchmarked against a 2×-refined (256^3) fourth-order spectral PDE reference, shared-rate `Rank-8` achieves 8.3% off-axis and 7.8% 3-D oblique waveform L^2 error (40.2% across the full singular caustic pulse at $(20, 0, 0)$, limited by spatial non-commutativity of the $W = 13$ stencil across steep $\nabla(c^2)$ curvature), compared with 65.2%–105.2% for 25-pt $O(\Delta x^4, \Delta t^4)$ (115 steps), 81.6%–91.0% for 27-pt IWB (83 steps), and 41.2%–73.7% for $O(\Delta t^2)$ `PS-FFT` (191 steps at $\nu_{\max} = 0.325$).

6 Limitations, Uncertainties, and Open Problems

To avoid overstating either the theoretical scope or the practical applicability of our results, we summarize six limitations, trade-offs, and remaining uncertainties:

1. **Continuous-domain exactness, discrete aliasing, and non-certified heuristic optimization:** Theorems 2.4–2.6 classify *exact* representations of *continuous* radial kernels on $\mathbb{R}^d$ (leaving $\varphi(s)$ for $s < 0$ unconstrained). Smooth non-exponential-polynomial targets ($e^{-\|x\|^4}$, Airy patterns $(J_1(r)/r)^2$, ideal circular low-pass filters) have infinite exact rank, while discontinuous kernels require separate arguments ($\mathbf{1}_{\{\|x\|=0\}}$ has rank 1; $\mathbf{1}_{r\le 1}$ has infinite rank). Sampling a continuous rank-R kernel on $\mathbb{Z}^d$ yields discrete rank *at most* R (strictly lower at aliased frequencies $\operatorname{Im}\lambda_u \in \pi\mathbb{Z}$ or when $W < R$). Moreover, while our inner Chebyshev weight solver is an exact linear program, outer optimization over nonlinear rates (α_u, ω_u) is nonconvex and heuristic: for 6c disc blur ($R = 12$), `blur_2d.py` reaches ripple ± 0.00193 whereas dedicated Remez exchange in [58] reaches ± 0.0014, and in

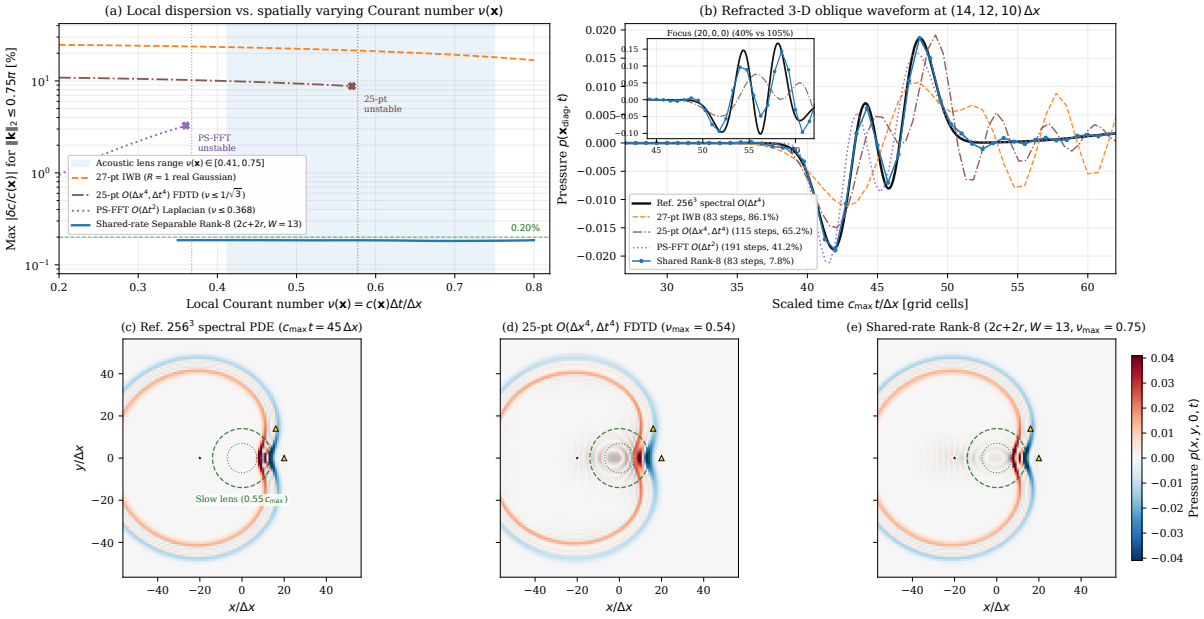


Figure 6: Three-dimensional wave propagation in a spatially varying medium $c(\mathbf{x})/c_{\max} = 1 - 0.45e^{-\|\mathbf{x}\|^2/(14\Delta x)^2}$ where 1-step `cuFFT` k-space propagation does not apply (`Applications/acoustics_3d.py --hetero-only`): (a) worst-case 3-D directional phase-velocity error up to 0.75π across local Courant numbers $\nu(\mathbf{x}) = c(\mathbf{x})\Delta t/\Delta x$ (shaded: lens range $[0.41, 0.75]$; crosses mark the stability limits of `PS-FFT` at 0.368 and 25-pt FDTD at $1/\sqrt{3}$); (b) refracted broadband waveforms ($\sigma_0 = 1.50\,\Delta x$, 128^3 grid) at the 3-D oblique receiver $(14, 12, 10)\,\Delta x$ and on-axis caustic focus $(20, 0, 0)\,\Delta x$ (inset) vs. the $2\times$-refined 256^3 fourth-order spectral PDE reference; (c–e) mid-plane wavefield slices $p(x, y, 0, t)$ at $c_{\max} t = 45\,\Delta x$ (dashed/dotted green circles: slow acoustic lens $r = 14\,\Delta x, 7\,\Delta x$; black star: source $(-20, 0, 0)$; yellow triangles: receivers).

3-D acoustics our stability check ($|\widehat{M}(\mathbf{k})| < 2$ across 232,620 Brillouin-octant/shell points plus multi-start continuous Nelder–Mead searches) is a dense numerical audit rather than a formal interval-arithmetic or sum-of-squares proof over all $\mathbf{k} \in [-\pi, \pi]^3 \setminus \{\mathbf{0}\}$.

2. **Visual and rendering limitations in 2-D lens blur:** Truncating 1-D passes to $[-S, S]$ produces a square $[-S, S]^2$ support ($r \le \sqrt{2}\,S$), and flat-top approximations built from oscillatory complex Gaussians inevitably exhibit small negative stopband ripples ($\pm\delta$). Around extreme HDR impulses ($10^3\times$–$10^5\times$ scene white), negative lobes can create dark ringing halos unless clamped or tone-mapped, whereas positive Monte Carlo disc gathering is strictly non-negative. Separable convolution is also inherently shift-invariant: real depth-of-field rendering with spatially varying circle-of-confusion $R_{\text{px}}(x, y)$ suffers foreground/background occlusion bleeding unless paired with depth-layer decomposition [27], and for purely shift-invariant blurs at $R \ge 8$ or large radii ($R_{\text{px}} \ge 32$), complex-packed 2-D `cuFFT` (0.45–0.56 ms on 1920×1080 RGBA, Table 12; layout conversion between interleaved RGBA and two complex planes not timed, and tensor-core `float16` matrix-multiply formulations not explored) is faster than direct FIR cascades.

3. **Numerical stability at large d, high degree k, or small border step h:** As quantified analytically in Section 2.4, Proposition 2.14's 2^{d-1} bound (12) is benign for $d \in \{2, 3, 4\}$ $(2, 4, 8)$ but grows exponentially for $d \gg 1$; high-degree syntheses (13) require saddle-point radius scaling $\rho \approx k/S^2$ (and Chebyshev nodes for real rates); and finite-h border-rank approximations hit an $\Theta(\varepsilon_{\text{mach}}^{1/(k+1)})$ relative-error floor ($\Theta(\varepsilon_{\text{mach}}^{1/4})$ in (7)). In $d = 2$, exact *a posteriori* $\mathrm{GL}_R(\mathbb{R})$ orthogonalization eliminates inter-component cancellation on rowSpan(K) (Table 11), whereas rigid $d \ge 3$ CP decompositions favor well-separated pure Gaussians

$(k = 0)$.

4. **Wide-stencil obstacles, boundaries, hardware sensitivity, and global FFTs:** While `DCT-II` half-sample reflection handles homogeneous rectangular rigid rooms cleanly, wide stencils ($L = 4$–8 grid cells in `Rank-4`–`Rank-9`) reach across thin walls or baffles and cannot be applied directly at staircased, curved, or frequency-dependent impedance boundaries without one-sided boundary closures whose passivity and stability for signed multi-Gaussian stencils are currently unknown. On simple homogeneous boxes without interior obstacles, L2-chunked periodic `cuFFT` k-space propagation (1.44 ms/step, Table 13) is dispersion-free and matches `Rank-7 fp32` (1.51 ms) in speed (though slower than `fp16`-intermediate `Rank-7` at 1.09 ms and `Rank-4`/`Rank-6`). Moreover, all GPU timings reflect one Ada Lovelace GPU (48 MB L2 cache, 672 GB/s GDDR6X) under Triton `3.9.0` / CUDA `13.2` with up to 3%–6% run-to-run autotuning variation: because `gpu_kernels.py` sizes 2-D strips and 3-D slabs explicitly to the 48 MB L2 cache, exact crossovers depend on L2 capacity, register file size, and kernel fusion (as illustrated by the 14-core CPU comparisons in Sections 5.3–5.4). Within the 3-pass-per-slab GPU kernel, replacing naive $n_{\mathrm{re}} + 4n_{\mathrm{cx}}$ by exact $n_{\mathrm{re}} + 3n_{\mathrm{cx}}$ reduces per-slab arithmetic and L2 traffic (24.0% step-time reduction on `Rank-7`), whereas on the unfused CPU baseline it reduces pass count directly (21.2% time reduction).

5. **Scope of formal verification and cross-literature connections:** Lean 4 guarantees that the 32 formal theorems in `KernelRankLean` follow logically from their definitions and standard classical axioms, while human inspection remains essential to verify that those definitions (`IsSumOfProducts`, `separableRank`, `IsRepresentation`) faithfully capture the intended mathematical objects, and none of the steerable pass-count arguments, Python/Triton GPU code, or numerical tables in Section 5 are formally verified. Likewise, while (11) links our exact rank formulas directly to Kronecker ($d = 2$), Winograd–Alder–Strassen ($d = 3$), and Bläser–Mayer–Shringi ($d \geq 4$) algebra-complexity lower bounds [75, 1, 10], special cases of the explicit root-of-unity functional syntheses (Propositions 2.14–2.15) may also appear under different terminology in other communities.

6. **Open problems — closed-form real border rank for general nonreal (d, k) and passive boundary/interface closures:** Theorem 2.10 and Table 4 resolve compact-open border rank over $\mathbb{C}$ in all dimensions, over $\mathbb{R}$ for all purely real-rate representations, and on $\cos(\|x\|^2)$ $(= 3)$ and $\|x\|^2 \cos(\|x\|^2)$ $(= 5 < B_3 = 6)$ in $\mathbb{R}^3$. Determining a closed-form formula for the exact real border rank of arbitrary nonreal higher-degree $(k \geq 2)$ or higher-dimension $(d \geq 4)$ representations Γ (compare border-rank degeneration bounds on local and higher-order algebras [79, 16]), incorporating $\nabla(c^2)$ commutator corrections across steep velocity gradients, and constructing provably passive boundary closures for wide signed multi-Gaussian stencils remain open.

7 Conclusions

Our theoretical, formalization, and experimental results reduce to three practical takeaways:

- **Theory (which round kernels split into 1-D passes, and at what exact cost):** A continuous radially symmetric kernel $K(x) = \varphi(\|x\|^2)$ on $\mathbb{R}^d$ $(d \geq 2)$ is an exact finite sum of 1-D coordinate products if and only if $\varphi(s)$ $(s \geq 0)$ is a sum of polynomials times real or complex exponentials (verified across 32 theorems in the Lean 4 proof assistant; continuous profiles that fade to zero at infinity can in turn be approximated uniformly across $\mathbb{R}^d$ by decaying complex-rate Gaussians with one shared width and equally spaced frequencies). Over $\mathbb{R}$, an alternating angle-sampling identity evaluates each complex-rate Gaussian in d real products rather than 2^{d-1} (3 vs. 4 in 3-D, a constant $d - 2$ overhead over complex

Table 14: Summary of the rank notions studied in this paper, illustrated on $K_3(x) = \|x\|^2 \cos(\|x\|^2)$ in $d = 3$ dimensions ($s = x_0^2 + x_1^2 + x_2^2$, single nonreal degree-1 component $\mathrm{Re}(se^{is})$).

Rank notion	Plain-language meaning (for $K : \mathbb{R}^d \to \mathbb{R}$)	General $d \geq 2$ formula / bound	On K_3
2-D slice / $\mathbb{C}$-border rank	Least complex terms in a uniform limit on bounded sets (equals 2-D slice rank)	$\sum_{\mu \in S}(\deg P_\mu + 1) = C_2(\Gamma)$	4
Real border rank	Least *real* product terms in a uniform limit on bounded sets	$\in [C_2(\Gamma),\ B_d(\Gamma)]$ ($= B_d$ if $\mathrm{Im}\,\lambda_u{=}0$)	5
Multi-Gaussian border	Least real terms in a uniform limit of *radial multi-Gaussian* kernels	$\leq B_d(\Gamma) = C_d(\Gamma) - (d-2)\sum_u k_u$	6
Exact complex $\mathrm{srank}_{\mathbb{C}}$	Least exact product terms $\sum_{n=1}^{R}\prod_{j=0}^{d-1} f_{n,j}(x_j)$ allowing $f_{n,j} : \mathbb{R} \to \mathbb{C}$	$\sum_{\mu \in S}((d-1)\deg P_\mu + 1)$	6
Exact real $\mathrm{srank}_{\mathbb{R}}$	Least exact product terms $\sum_{n=1}^{R}\prod_{j=0}^{d-1} f_{n,j}(x_j)$ with real $f_{n,j} : \mathbb{R} \to \mathbb{R}$	$C_d(\Gamma)$ ($+(d-2)$ per nonreal pair)	7

arithmetic across all polynomial degrees). Allowing uniform limits of lower-rank kernels on bounded regions (*border rank*) reveals a strict hierarchy in $d \geq 3$ dimensions (Table 14): over $\mathbb{C}$ (and over $\mathbb{R}$ when all rates are real), border rank drops all the way to the 2-D slice lower bound $C_2(\Gamma)$ obtained by setting $d-2$ coordinates to zero, whereas over $\mathbb{R}$ on $K_3(x) = \|x\|^2 \cos(\|x\|^2)$ in 3-D, exact real rank is 7, limits of radial Gaussians take 6 terms, and general real separable limits take 5—at the price of $O(h^{-1})$–$O(h^{-3})$ cancelling weights that favor pure Gaussians ($k = 0$) in finite precision.

- **Numerics (16-bit precision and discrete vs. continuous filter design):** In 2-D, individual complex-Gaussian components of high-accuracy disc blurs cancel by up to $595\times$–$1{,}110\times$ in amplitude; whereas magnitude reordering alone ($8.18 \pm 0.92/255$) and `float32` outer summation without orthogonalization ($6.76 \pm 1.10/255$) leave intra-pass `float16` rounding intact, rotating the 1-D filter basis into orthogonal terms and summing smallest-first cuts 20-scene `float16` sRGB error by 96% ($7.90 \pm 0.92/255 \to 0.31 \pm 0.13/255$ on a 12-term blur) with zero extra passes. Optimizing discrete rank-R stencils directly on a pixel grid beats continuous formulas only on tiny stencils where grid half-width L approaches R; once blur radius reaches 16 pixels (61×61 taps at radius 25), enforcing equal weights at equal Euclidean radii makes discrete optimization converge onto our continuous complex-Gaussian profiles (± 0.0758 vs. ± 0.0759 pass/stopband ripple at $R = 4$), and sampling the continuous profile gives an immediate isotropic starting point ($2.2\times$ lower initial error at $R = 12$ than truncating a smooth disc by SVD).
- **GPU execution (when separable cascades beat direct convolution, FDTD, and `cuFFT`):** On an RTX 4070 Ti Super eGPU (with all outputs checked against `float64` references), keeping intermediate channels in L2 cache makes separable FIR cascades fastest in two concrete regimes where global fast Fourier transforms (`cuFFT`) are slower or inapplicable: (i) 2-D blurs of small-to-medium radius at modest rank (0.331 ± 0.0003 ms at radius 8 px and 0.354 ± 0.005 ms at 16 px on 1920×1080 RGBA for $R = 4$, vs. 0.45–0.46 ms for `cuFFT`, while `cuFFT` wins for larger radii or $R \geq 8$), and (ii) wideband 3-D acoustic wave propagation, where exact 3-term-per-Gaussian decompositions cut 256^3 step time by 24% over naive 4-term expansion (1.51 ± 0.03 ms vs. 1.98 ± 0.02 ms in `float32`, 1.09 ± 0.02 ms with `float16` intermediates, +28 MiB slab scratch, and $< 0.05\%$ wave-speed error up to 0.80π; compact 25-point FDTD takes 0.325 ± 0.0002 ms/step on the same grid, but requires $40\times$ more field memory and $52\times$–$71\times$ more estimated runtime when refined to match 0.1% dispersion) and—by fixing shared Gaussian widths with local sound-speed weights—keep wave-speed error below 0.19% across $> 2:1$ sound-speed contrasts where 1-step `cuFFT` k-space propagation cannot be used.

Determining a closed-form formula for exact real border rank at arbitrary degree $k \geq 2$ and

Table 15: Explicit canonical coefficients (19) for representative 2-D circular disc-blur designs ($\Delta r = 0.2$, Table 9).

Design ($R = C_2$, C_3, $\pm\delta$)	u (q)	α_u	ω_u	$p_{u,q}$	$q_{u,q}$
1c (R=2, C_3=3, $\pm$0.2324)	1 (0)	0.862715	1.624792	+0.767553	−1.862917
2c (R=4, C_3=6, $\pm$0.0759)	1 (0)	0.894640	5.265923	+0.413699	+0.555222
	2 (0)	1.973219	1.554245	+0.510373	−4.611414
1c_2 (R=4, C_3=7, $\pm$0.1165)	1 (0)	2.878934	3.152485	+0.883521	−7.976371
	1 (1)	2.878934	3.152485	−16.607444	+7.931853
2c+1r (R=5, C_3=7, $\pm$0.0446)	1 (0)	0.962816	7.154625	+0.091824	−0.456790
	2 (0)	2.075279	3.285822	−3.434558	+0.588375
	3 (0)	2.494577	0	+4.387353	0
4c (R=8, C_3=12, $\pm$0.0109)	1 (0)	4.338835	1.553577	−5.754013	−46.161778
	2 (0)	3.838091	4.692138	+9.776587	+15.218676
	3 (0)	2.790788	8.181910	−3.042741	−0.293993
	4 (0)	1.345593	12.330086	+0.009312	−0.245279
6c (R=12, C_3=18, $\pm$0.0019)	1 (0)	5.143786	2.080541	−82.448179	−111.252655
	2 (0)	5.613157	6.153062	+113.961587	−58.149957
	3 (0)	5.982623	9.803250	+39.766911	+161.948149
	4 (0)	6.506533	11.060675	−71.545624	−94.857314
	5 (0)	3.871248	14.810010	+1.416029	+3.711177
	6 (0)	2.202364	19.033594	−0.152648	+0.108514

dimension $d \geq 4$, adding gradient corrections across steep sound-speed interfaces, and designing provably passive boundary closures for wide signed wave stencils remain open problems for future work.

A Explicit Filter and Propagator Coefficients

To allow standalone reproduction of the main designs in Section 5 from this PDF (complete JSON archives for all 18 blur variants and all 6 acoustic propagators are provided in `Applications/`), every radial profile in Tables 15–16 is tabulated in the single canonical convention of Definition 2.2:

$$\begin{aligned}\varphi(s) &= \sum_{u=1}^{m} \mathrm{Re}\left(\left(\sum_{q=0}^{k_u}(p_{u,q}+iq_{u,q})s^q\right)e^{(-\alpha_u+i\omega_u)s}\right)\\ &= \sum_{u=1}^{m}\sum_{q=0}^{k_u} s^q\big(p_{u,q}\cos(\omega_u s) - q_{u,q}\sin(\omega_u s)\big)e^{-\alpha_u s},\end{aligned} \tag{19}$$

where real-rate components satisfy $\omega_u = 0$ and $q_{u,q} = 0$. In 2-D circular disc blur (Table 15), $s = (x^2+y^2)/R_{\mathrm{px}}^2$ on $|x|, |y| \leq (1+\Delta r)R_{\mathrm{px}}$ ($\Delta r = 0.2$), normalized in 2-D discrete execution so $\sum_{x,y} K[x,y] = 1$. In 3-D acoustic leapfrog propagation (Table 16), $M[\mathbf{m}] = \alpha_0\delta_{\mathbf{0}}[\mathbf{m}] + \varphi(m_x^2 + m_y^2+m_z^2)$ on integer grid offsets $\mathbf{m} \in [-L, L]^3$ ($W = 2L+1$), where $\alpha_0 := 2-\sum_{\mathbf{m}\in[-L,L]^3}\varphi(\|\mathbf{m}\|^2)$ must be computed from the evaluated φ grid so $\widehat{M}(\mathbf{0}) = 2$ holds to machine precision (avoiding exponential constant-mode blow-up $\widehat{M}(\mathbf{0}) > 2$ and preserving spatially constant initial states; note $\zeta = 1$ is a double root of $\partial_{tt}\bar{p} = 0$, so nonzero initial velocity mean gives physical $a + bn$ motion). Table 16 lists the exact α_0 recomputed from the 9-decimal coefficients (within 2.2×10^{-8} of `float64` JSON), matching $\partial_{k_x}^2\widehat{M}(\mathbf{0}) = -2\nu^2$ to $\leq 8 \times 10^{-8}$ and keeping $\widehat{M}(\mathbf{k}) \in [-1.9990, 1.9946] \subset (-2, 2)$ on the verification grid.

Table 16: Explicit 9-decimal canonical coefficients (19) for all six 3-D acoustic leapfrog propagators in Table 13 (set $\alpha_0 := 2 - \sum_{\mathbf{m}\in[-L,L]^3} \varphi(\|\mathbf{m}\|^2)$; reference α_0 shown for verification).

Scheme (ν, W, R, α_0)	u (q)	α_u	ω_u	$p_{u,q}$	$q_{u,q}$
Rank-4 (1c+1r, ν=0.75, W=9, R=4, α_0= − 1.440427676)	1 (0)	0.439979390	0.502750129	+0.137876459	+0.024929224
	2 (0)	0.888101329	0	+0.492695925	0
Rank-6 (2c, ν=0.75, W=11, R=6, α_0= − 1.419560209)	1 (0)	0.606539114	0.240982835	+0.597581203	+0.180225818
	2 (0)	0.267859019	0.385016560	−0.007190817	+0.050272529
Rank-7 (deg 1 1c, ν=0.75, W=13, R=7, α_0= − 1.434649594)	1 (0)	0.429753880	0.229296784	+0.605530664	+0.525638174
	1 (1)	0.429753880	0.229296784	−0.061586407	−0.061209821
Rank-7 (2c+1r, ν=0.75, W=13, R=7, α_0= − 1.439243312)	1 (0)	0.381717837	0.311635272	+0.110431944	+0.263079509
	2 (0)	0.163094730	0.300649461	−0.015761751	−0.008519990
	3 (0)	0.599866965	0	+0.491386225	0
Rank-7 (2c+1r, ν=1.00, W=13, R=7, α_0= − 1.959018821)	1 (0)	0.321809087	0.262377242	+0.159170237	+0.129231489
	2 (0)	0.126172334	0.278598443	−0.010612383	+0.004174471
	3 (0)	0.438442015	0	+0.241253606	0
Rank-9 (3c, ν=0.75, W=17, R=9, α_0= − 1.447076570)	1 (0)	0.301382846	0.246772855	−0.286641369	+0.234406754
	2 (0)	0.398975160	0.094363260	+0.866865360	+0.865008109
	3 (0)	0.111967878	0.275076161	−0.002296910	−0.011770692

Code and Data Availability

All source code, machine-checked proofs, numerical optimization scripts, GPU benchmarks, JSON parameter/result archives, and figure-generation scripts for this work are openly available in the public domain (CC0 1.0 Universal) at `https://github.com/OlliNiemitalo/kernel-rank-lean`:

- `KernelRankLean/` and `Experiments/`: Lean 4 (`v4.36.0-rc2` with Mathlib `v4.36.0-rc2`) formalizations of all d-D and 2-D separable-rank, root-of-unity, differential-annihilator, border-rank, and alternative semigroup theorems;
- `Isabelle/`: supplementary Isabelle2026 (`HOL-Nonstandard_Analysis`) semigroup formalization `Semigroup_NSA.thy`;
- `Applications/`: Python, SciPy, PyTorch, and Triton optimization, GPU-kernel, and figure scripts (`blur_2d.py`, `acoustics_3d.py`, `gpu_kernels.py`, `generate_figures.py`) plus JSON result archives (`results_blur_2d.json`, `results_acoustics_3d.json`);
- `scripts/`: hermetic setup (`scripts/setup`), formal verification and axiom/oracle audit (`scripts/check`), isolated GPU Python wrapper (`scripts/python`), and PDF build (`scripts/build-paper`).

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the six-stage `separableChannels` ablation and memory/fairness comparisons, and Appendix A. Both the repository and this manuscript are released under CC0 1.0 Universal.

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